



Learning Management System for Rural India

Supporting education in rural communities in India

Bachelor thesis

Degree program:

BSc Computer Science

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Management Summary

Problem

India grew very rapidly in the last couple of decades. This growth brought wealth and opportunity with it, especially in the IT industry. Nowadays the children of the original “IT-Boomers” have grown up enjoying a good education and are given preference for the new IT jobs. The poorer and rural areas can no longer compete so have been left behind and are struggling to catch up.

Goal

We set out to create a mobile learning app to reach those left behind. The app should support the teaching of spoken English and aid in automating and organizing the curriculum. We partnered with the Sharada Educational Trust to tailor the app to their needs and to overcome their specific challenges, including reaching rural community teachers and college girls.

Requirements

The final product was created to be used with mobile phones in mind. The people in the rural communities mostly have no access to a computer or laptop, and often they do not even have an email address. So, we used many creative solutions, such as a date-of-birth and phone number to authenticate the users.

The app should be able to support user management, offer reusable courses, and provide a way to run a course with a batch of students and a tutor.

Results

The final product is an app “optimized for the minimum.” The end users do not have a background in or much experience with modern technology and usually have very old devices. So, of high importance was the creation of an easily understandable and navigable app that also does not overwhelm with an overload of information.

In May 2024 we carried out a soft launch at a rural government school in Cheeluru, India, which will be the first to use the app. At present the app is live and the Sharada Educational Trust is expanding the rollout to the whole of Karnataka (a state in India), and later throughout the country.

Conclusion and outlook

The project had many challenges, mainly based on cultural differences and the very inexperienced target user base. Nevertheless, a working first version was developed that will continue to be used and enhanced, perhaps even by another group of students doing their bachelor's thesis at BFH and eager to make a positive impact on the world.

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1 Introduction

With 1.4 billion people, 28 states and 8 territories, speaking over 19,500 languages and dialects, and 9 major religions with tens of thousands of variations, India is an amazing and very diverse country. Thanks to the so-called “IT Boom” in the late 1990’s and early 2000’s, massive numbers of Indians have been uplifted from poverty, but sadly considerable poverty and underdevelopment remains a challenge for India today.

This Chapter will look briefly at the situation in India and attempts to improve the situation by means of non-profit organizations such as the Sharada Educational Trust. The previous work will be reviewed, and goals of this Bachelor Thesis will be clearly stated.

1.1 The Situation in India

During the late 1990s and early 2000s India experienced the so-called “IT Boom.” This was a large-scale transformation of the Indian economy, driven by high need but also good possibilities to carry out high-tech work in what was (at that time) a lower-wage country. During these years, the IT industry exploded, creating around 3 million high tech jobs in the Business Process Outsourcing (BPO) sector alone, as well countless others that supported them. This enabled vast numbers of rural Indians to be uplifted from poverty. Even today the average IT “campuses” owned by companies such as InfoSys, Wipro, and Cognizant are comprised of many dozens of buildings, shopping and childcare, restaurants, and even golf courses and tennis courts, as shown for InfoSys in Figure 1, far more modern than anything found in Western countries.



Figure 1: Campus Infosys [1]

Sadly, today the situation in India is different. Because the high-tech industry has been established for over 20 years, many of the current staffing needs are filled by the children of the original “IT Boomers.” As such, this new generation usually has an advanced education and excellent knowledge of English, which is the lingua franca of IT. Because of this, the hiring barriers have been raised and it has become increasingly difficult for Indians living outside of large cities to attain the needed education and level of English proficiency that modern IT jobs now expect and require. The contrast between the modern IT world depicted by InfoSys and the rural school is evident, as shown in the next figure depicting the rural government school of Cheeluru.



Figure 2: Rural goverment school in cheeluru

1.2 Sharada Educational Trust

Registered in April 2022, Sharada Educational Trust was founded by Arvind and Anita Kamath as a Not-for-Profit Charitable Trust, with a clear mission to "Empower Lives Through Education by providing Quality Education to All".

The NGO primarily focuses on education of the under-privileged rural students and empowerment of rural women in the southern state of Karnataka in India, which has the modern IT city of Bangalore as its capital. To appreciate the size, note that Karnataka has a geographical area of 192K m², compared with 41Km² for Switzerland, as shown in Figure X. The population of Karnataka is 61 million, compared with 8.9 million in Switzerland. Therefore, the state of Karnataka is 4.6 times larger than Switzerland with 6.7 times more people; its capital Bangalore alone has roughly the population of Switzerland (8.4 million).

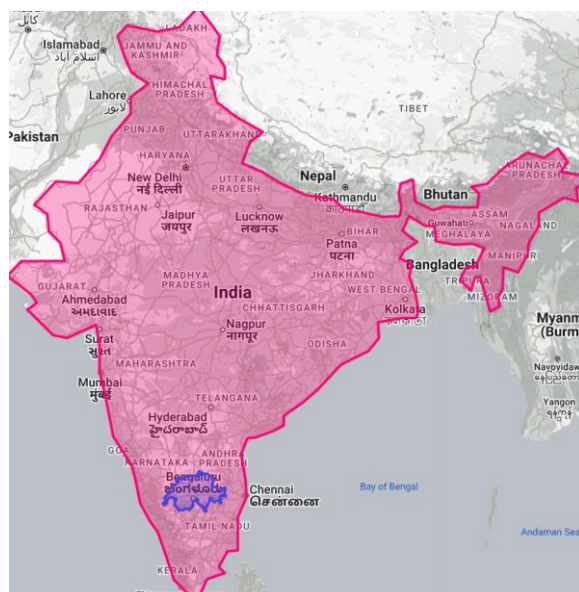


Figure 3 : India and Switzerland size [2]

The NGO strives to solve the following problems through its education focused programs (listed in the figure below):

- Teachers from rural government schools lack the quality and expertise to teach English, which affects the prospects of their students in the long run. Similarly, the less privileged College students, who hail from rural or semi-urban areas face the same problem, which affects them academically and professionally.
- "Financial Literacy" among rural kids empowers them to make informed choices to break the cycle of poverty and lays down the foundation for a financially secure and prosperous future. Unfortunately such kids are deprived of this education, since "Financial Literacy" as a subject is not taught in rural Govt. schools, unlike their counterparts in urban areas.

In the last 15 months of operation, Sharada Educational Trust has impacted more than 2,000 lives in 5 districts of Karnataka in India, through various programs detailed below. However the NGO now intends to reach the beneficiaries pan India, which is possible only through an EdTech based solution viz. Learning Management System supported by Mobile App, under the umbrella of Sharada Academy.

In this way, the Sharada Organization performs a very valuable service that directly or indirectly contributes to the UN Sustainable Development Goals (UN SDG), especially the following [3]:

- 1) No poverty
- 4) Quality education
- 5) Gender equality
- 8) Decent work and economic growth
- 9) Industry, innovation and infrastructure
- 10) Reduced inequalities
- 11) Sustainable cities and communities
- 17) Partnerships for the goals



Figure 4 : UN Sustainable Development Goals [3]

1.3 The need for a learning management system

In order to not only carry out an effective educational program but especially to ensure that it is scalable and can grow, a Learning Management System (LMS) application is required. The best-known LMS in the academic world is Moodle, or its industrial cousin Totara; but there are countless other LMS systems on the market.

In the previous work in Project 2 a MAKE-or-BUY analysis was carried out, and it was determined that a MAKE approach best met the needs of the Sharada Educational Trust. As a short review, here are the important factors that led to the decision to make the product instead of buying it, as well as a short explanation of the economic and risk factors from most important to least:

- ▶ **Unique Value Proposition (UVP), for Sharada to have their own LMS**
They show commitment to their case if they are willing to go the extra mile to create an LMS tailored to their mission.
- ▶ **Tailor made**
All the aspects of the product are tailor made for Sharada Trust and their needs. No functionalities of an existing product have to be neglected or bent to achieve the desired result.
- ▶ **Simplicity**
The product can focus on the parts and requirements most important to Sharada Trust. In contrast to general LMS only functionality really required by Sharada Trust are present. This makes the product leaner and simpler to use.
- ▶ **Economic**
Sharada Trust does not have the knowhow to host such a platform. In addition, it makes a lot of sense to create an application that is completed by students that work as volunteers (as part of their bachelor thesis).
- ▶ **Risks**
IT solutions change. This project and the technologies involved are no exception. This product should be available in unchanged form for a long time, since the product does not change and will probably be available to be hosted on older versions even if newer ones come out. This is a bigger risk if a third-party tool was used.

1.4 Previous Work

In the previous project (Modul BT13041 Project 2) the authors developed a minimal working version (MWV) of the LMS for Sharada Educational Trust. The MWV contains user management including role-based access control (RBAC), basic course management, and a basic enrolment management where batches of students including tutors are mapped to a course. The app is available as a website where the administration parts are optimized for a desktop to keep a better overview, and the student part is optimized for mobile devices, since not all students have access to a desktop. [4]

The product was just started but not deployed and there were no user tests so far. The goal of the bachelor thesis is to build on the foundation laid by the previous work. The whole project team from authors, thesis advisor and project partner will stay the same, which means the team is a well-oiled machine so to speak. [4]

1.5 Goals

The goal of the project, including the previous project [4], is to create an important foundational tool for the Sharada Educational Trust, tailored to their needs, but that can enable their scaling and growth to serve the needs of larger and larger communities. For the Trust it should be easy to create and manage users, keep their tutor led but also self-paced courses up to date and organize batches where students can participate in such a course. The idea of this bachelor thesis is to build upon the work started in Project 2, to implement additional, more complex features and create an environment that is running and easily manageable even when the development team no longer works on the project.

Feedback on student skill and progress is a very important part of learning. Therefore, the first additional feature will be a way to give homework or even create a little quiz. The assessment management will be coordinated by the tutor of a batch. The plan is to make them as simple as the rest of the app but elaborate enough to give different options for didactic reasons and so it does not get too boring and makes learning an interesting experience instead of being a chore.

In regard to the longevity and support of the project, first a complete version of the product will be built. In order to keep frustration low and confidence high the team has decided on a “success-driven” approach, investing much more time to ensure all features work in a productive environment. The idea is to have a productive version of the product hosted and free of major software defects, so no one at the Sharada Educational Trust must know anything about the hosting for the time being.

As additional goals, the following documentation will be created:

- ▶ a user manual will be essential, to keep time spent on support low for the Sharada Trust staff and to answer some questions before they arise.
- ▶ a system administrator manual. This manual contains all information about the data structure, hosting, and maintenance of the final product. It aims to achieve 2 goals: (1) if problems occur there is a documentation useful for resolving issues; (2) important documentation for future developers that may continue the work.
- ▶ Developer documentation (in e.g. markdown format), so that future developers have the basic information required to build and deploy the application

2 Organization

Appropriate attention to appropriate project management is essential for every Bachelor thesis. But in this case, with three students, an advisor, and a major stakeholder located in India, the design and implementation of an effective and high-quality project organization is absolutely essential.

In this Chapter the project organization is described. First, the project methodology is presented; this is not merely Scrum but rather tries to address each of the principles in the Agile Manifesto. Subsequent sections address communication (very important with the key stakeholder located in India), project phases, deliverables, the tools in use and finally risks and how they are managed.

2.1 Project Methodology

The project is focused on agile development based on two important factors. First, the initial Project 2 work had great results and a smooth working process in the team with agile. Second, an Internet-facing, customer-accessible platform has been created, so that the development team working within the BFH network can easily make deployments and work with the business stakeholders in India.

Based on the development team's (TA02) size and the parties involved, it was decided to appoint a Head of Project Management (PR03). This role oversees the project methodology, ensures agile development, and makes the time planning. In essence, it is also known as the "Agile Team Lead," since Scrum Master is usually applied to situations using the Scrum Methodology. For more information see 2.1.4 Project Roles and Teams.

Due to external responsibilities the development team (TA02) works on different days of the week but coordinate and synchronize all members meet in person every Thursday. It is not planned that the team then necessarily works at the same tasks. The plan is more to have an efficient, productive working environment and to maybe share problems to find solutions and give a feeling for the other members challenges and their part of the project and plan bigger tasks.

2.1.1 Agile development

The teamwork is based agile development on the 12 principles of the Agile Manifesto [5]. The following table shows the agile principles from the Manifesto and how it is ensured that they are met.

Id	Description	Measure Id	Measure
AP01	Our highest priority is to satisfy the customer through the early and continuous delivery of valuable software.	APM011	We have a minimal working version deployed at the start of the product.
		APM012	We will deploy a working version of the product whenever we have a new feature ready for user testing.
AP02	Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.	APM021	We are in constant exchange with the customer (see 2.2.1 Meetings) and will take feedback and implement small changes.
		APM022	This is a bachelor thesis. Therefore we have requirements that have to be met at the end of the project. To ensure the success of the bachelor thesis we can't accept major changes to them.
AP03	Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.	APM031	We try to deploy in feature intervals (see APM012). If a feature takes too long, we find a way so split it. We will deploy at least every other month.
AP04	Businesspeople and developers must work together daily throughout the project.	APM041	Since we don't work daily on the project we meet with the Project Partner (PR06) as often as reasonable: every week (see 2.2.1 Meetings). A WhatsApp channel ensures ad hoc requests are immediately fulfilled.
AP05	Build projects around motivated individuals. Give them the environment and support they need and trust them to get the job done.	APM051	The Development Team (TA02) is motivated through their thesis, degree, that they work on a project that will be used and working for an NGO.
		APM052	The Project Partner (PR06) is motivated because they get a tool to enhance their mission.
		APM053	The Thesis Advisor (PR05) is motivated to work for a good cause and can together with the Head of Project Management (PR03) support the Development Team (TA02) with the environment they need.
AP06	The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.	APM061	The Development Team (TA02) has weekly meetings with the Thesis Advisor (PR05) and the Project Partner (PR06).
		APM062	The Development Team (TA02) has a weekly in person workday.

Id	Description	Measure Id	Measure
AP07	Working software is the primary measure of progress.		Won't be met. It is still first and for most a bachelor thesis which many other measures like all the deliverables.
AP08	Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.		Won't be met. The foreseeable project ends with the end of the bachelor thesis.
AP09	Continuous attention to technical Excellence and good design enhances agility.	APM091	At the weekly in person meeting, we will make small code review of the most important things developed in the last week.
		APM092	The documentation will be proofread by multiple internal and external advisors.
AP10	Simplicity—the art of maximizing the amount of work not done—is essential.	APM101	Only implement things that fit in the narrative of the requirements.
		APM102	Contest every change to evaluate whether it is really needed.
AP11	The best architectures, requirements, and designs emerge from self-organizing teams.	APM111	The Development Team (TA02) is on their own with helpful correctional advice from the Thesis Advisor (PR06).
AP12	At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.	APM121	The Development Team (TA02) will hold a review and improvement section after each finished feature.

Table 1: Principles of agile development

2.1.2 Kanban

Kanban is employed as a lightweight work management system to visualize the progress, bring order to the workflow, and share the current state of the work. Kanban has been chosen over other alternatives (mostly Scrum) because it is lighter, and the team expects a continuous stream of small tasks resulting from feedback from the Project Partner (PR06). With Kanban there is no need to plan them into a sprint (or alike) and can just implement them on the fly but still integrate in the system. [6]

2.1.2.1 Kanban board

The Kanban board is the place to share all the current state of the tasks. In addition, it defines the basic workflow all tasks have to run through to be able to complete. The Kanban board has been set up in GitLab. [4]

Id	Name	Description
KL01	Open	For all tasks currently planned but not yet started.
KL02	In progress	For all tasks that are currently being worked on.
KL03	Blocked	For all tasks that currently can't be worked on.
KL04	Review	For all tasks that need reviewing by the project partner, project advisor or another team member. Also used for code review when in proofreading.
KL05	Closed	For all tasks that are already done.

Table 2: Kanban board lanes [4]

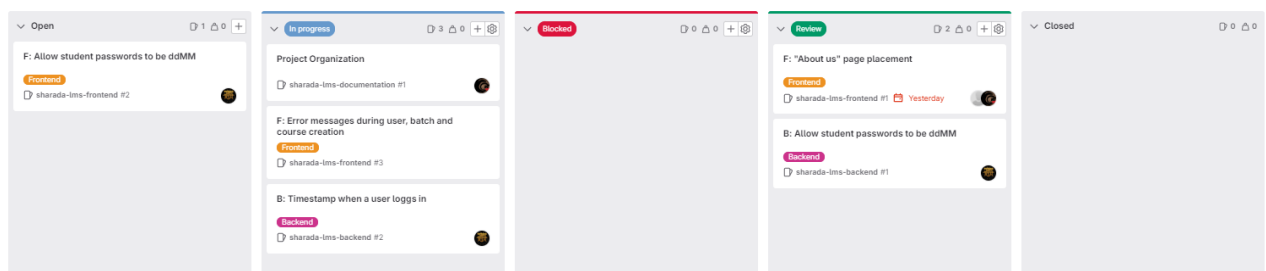


Figure 5: Kanban board

2.1.2.2 Kanban Cards

A Kanban card represents a task of sufficient size. Features are broken down onto different cards. One card is aimed at having about 4 to 16 working hours. GitHub issues are used as the Kanban cards.

2.1.2.3 WIP

The team strives to always focus on one feature, with maybe up to two urgent cards. With three members of the Development Team (TA02) and three cards, all lanes should never have more than 8 cards. Open (KL01) and Closed (KL05) are excluded from that.

2.1.3 GitHub issues

The GitHub issues are used to organize the development of the code by the developers and as Kanban cards. They are not used for tracking time since the learning established from the experience with the overall time evaluation in the previous project. [4]

2.1.4 Project Roles and Teams

The following diagram shows the different actors and teams involved in the project and the official communication channels.

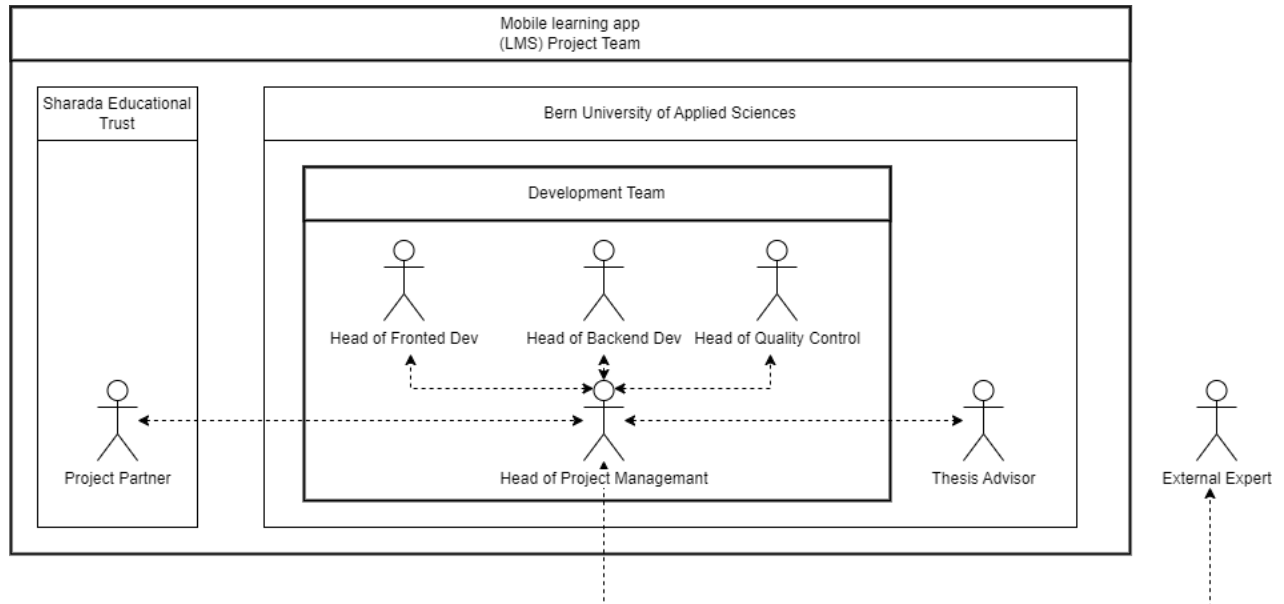


Figure 6: Project roles

In this project, every member of the development team took an active role in software development and coding.

However, because the scope of the project is much larger than a normal Bachelor thesis (namely, to develop a full-featured productive application including the needed deployment, quality, and related artifacts) it was necessary to assign focus areas to team members. External reviewers were not mentioned with their own project role but at the places where their review is discussed.

For an example of this, consider the role of testing. Because at least 5 (and sometimes more) people carried out application tests, it was necessary to include test management into the project, to create test cases, to assign test cases to testers, to collect the structured feedback, and finally to prioritize and assign the bugs for fixing. This was handled by Mr. Tobias Erpen (PR05).

The following chart describes the additional responsibilities and focus areas of the team members.

Id	Name	Additional Responsibilities	Person
PR01	Head of Frontend Development	▶ Development of the features in the frontend ▶ User experience (UX)	Ms. Alayne Hiltmann
PR02	Head of Backend Development	▶ Development of the features in the backend ▶ Unit test coverage	Mr. Lukas Vogel
PR03	Head of Project Management	▶ Scrum Master ▶ Project planning, meeting organization, etc. ▶ Requirements Engineering ▶ Integration and deployment	Mr. Tobias Erpen
PR04	Head of Quality Control	▶ Oversee and adapt code reviews ▶ Concepting tests	Mr. Tobias Erpen

Id	Name	Additional Responsibilities	Person
		▶ Coordinate tests and fixes	
PR05	Thesis Advisor	▶ Support the project team in questions regarding organization with the BFH ▶ Support the development team (TA02) with expertise in software development	Dr. Kenneth Ritley
PR06	Project Partner	▶ Provide the features needed and desired quality ▶ Prioritize the software defects and bugfix priorities ▶ Test and provide feedback for improvement	Mr. Arvind Kamath
PR07	External Expert	▶ Evaluate and grade the project from an outside perspective	Mr. Alain Joray

Table 3: Project roles description

The following chart describes the teams and parties involved.

Id	Name	Description
TA01	Mobile learning app (LMS) Project Team	Includes all those who actively and directly contribute to the product of the project.
TA02	Development Team	Includes all those who actively and directly are developing the product.
PA01	Bern University of Applied Sciences	The university at which the members of the development team do this bachelor thesis. Is the client in the regard of the project happening.
PA02	Sharada Educational Trust	The NGO that serves as the client regarding them being the end user of the product.

Table 4: Team and parties description

2.2 Communication

Communication is an important part of every project. To ensure smooth communication between all the parties involved the head of Project Management (Project Role PR03) was responsible for handling all the communications concerning the Development Team (TA02) in any way.

Most communications outside of the Development Team (TA02) will be through electronic means. The following table shows the main channels for communication.

Id	Name	Purpose
CH01	WhatsApp-Chat	For small questions and to share information fast with the Project Team (TA01).
CH02	email	To setup meetings and share information that has to be more traceable like meeting protocols or contracts.
CH03	Online Meeting via MS Teams	For discussion of the Development Team (TA02) with or without the Thesis Advisor (PR05) and the Expert (PR07).
CH04	Online Meetings via Zoom	For discussions in the Project Team (TA01).

Table 5 : Communication channels

Since the Project Team (TA01) can be distributed over the globe at times it is important to consider the relevant time zones. To avoid confusion all times will be given in 24-hour time including time zone. The following table shows the relevant time zones and should aid in planning meetings.

Time Zone	Abbreviation	Reference	Current Period
Coordinated Universal Time	UTC	UTC + 0:00	
Central European Time	CET	UTC + 1:00	29.10.2023 – 31.03.2024
Central European Summer Time	CEST	UTC + 2:00	31.03.2024 – 27.10.2024
Indian Standard Time	IST	UTC + 5:30	
Central Daylight Time (Texas)	CDT	UTC – 5:00	05.11.2023 to 10.03.2024
Central Standard Time (Texas)	CST	UTC – 6:00	10.03.2024 to 03.11.2024

Table 6: Relevant time zones

2.2.1 Meetings

To ensure an agile approach and to make sure all team members are adequately synchronized, a short weekly meeting with the Thesis Advisor (PR05) was established as well as one with the Project Partner (PR06). The meetings always follow each other to keep organizational effort lower. If there is nothing to present or discuss, the regular meeting is skipped. The following tables show the regular meetings and their topics. The full protocols can be found in the Appendix.

Id	Planned Date	Topics discussed
M01	19.02.2024	Briefing about the structure of the project
M02	29.02.2024	<i>Cancelled because of absence of the Thesis Advisor</i>
M03	07.03.2024	<i>Cancelled because of absence of the Thesis Advisor</i>
M04	14.03.2024	How good we are in the time plane considering we are three
M05	21.03.2024	Testing imagination
M06	28.03.2024	<i>Cancelled because of absence of the Thesis Advisor</i>
M07	04.04.2024	Organizing after easter and user documentation
M08	11.04.2024	Test results and debriefing after demo
M09	18.04.2024	<i>Cancelled because we did not have anything to discuss</i>
M10	25.04.2024	<i>Cancelled because we did not have anything to discuss</i>
M11	02.05.2024	Cancelled because we had another meeting M36
M12	09.05.2024	Preparing the ride to India
M13	16.05.2024	<i>Cancelled because the whole Project Team (TA01) was together in India</i>
M14	23.05.2024	<i>Cancelled because of absence of the Thesis Advisor</i>
M15	30.05.2024	Coordinating finalization of the project
M16	06.06.2024	Coordinating finalization of the project

Table 7: Regular meetings with the Thesis Advisor

Id	Planned Date	Topics discussed
M17	20.02.2024	Testing feedback from Arvind
M18	29.02.2024	The “About us” page and planning going forward, with the Gnatt diagram.
M19	07.03.2024	The Requirements were discussed
M20	14.03.2024	Future Plans
M21	21.03.2024	Testing feedback
M22	28.03.2024	Catch up before easter
M23	04.04.2024	Test results and fixing QNA format
M24	11.04.2024	Test results and two week outlook
M25	18.04.2024	Demo for testing
M26	25.04.2024	Demo for Students
M27	02.05.2024	Demo for Tutors
M28	09.05.2024	Preparing the app and expectations for the India trip
M29	16.05.2024	<i>Cancelled because the whole Project Team (TA01) was together in India</i>
M30	23.05.2024	<i>Cancelled because we did not have anything to discuss</i>
M31	30.05.2024	Prepare final deployment and answer last questions
M32	06.06.2024	Coordinating finalization of the project

Table 8: Regular meetings with the project partner

2.2.1.1 Additional Meetings

There were some additional meetings, including coordinated setup of the deployment environment, expert meetings and emergency meetings. These meetings were not planned from the start but scheduled when needed. The following list shows all the additional meetings and their topics. The full protocol can be found in the 0 Appendix.

Id	Planned Date	Topics discussed
M33	21.02.2024	Payment and Infomaniak hosting (Jelastic)
M34	26.03.2024	Expert Meeting
M35	16.04.2024	Organization India field trip
M36	02.05.2024	Introduction to India
M37	15.05.2024	Field test in the rural village in India: Cheeluru
M38	06.06.2024	External Code review
M39	06.06.2024	Mail flights refunding

Table 9: Ad hoc meetings

2.3 Project Phases and Milestones

The project phases are based on the features that are to be developed. To always ensure a working product but improving product it was decided to make the features the focus point. As expected from an agile approach, the team returns to the concept board for every feature instead of trying to build a complete concept at the beginning. This is done to ensure the commitment to the agile process, specifically the measure APM031 and to a lesser extend APM111. So, it was decided to structure the Bachelor thesis into the following project phases.

Id	Name	Description	Closing Milestone
PP01	Project Initialization	Cleaning up the carryovers from the previous project. Preparing the organization and creating feature overarching concepts like security, testing and color concepts.	MS02
PP02	Feature 1: Assessment	The concepting, implementation, testing and deployment of the assessment functionality	MS03
PP03	Feature 2: Certification	The concepting, implementation, testing and deployment of the certification functionality	MS04
PP04	Feature 3: Smaller functionalities	The concepting, implementation, testing and deployment of smaller functionalities found during the project	MS05
PP05	Testing	Coordinated and free user tests with actual users to improve the product and a final test run to verify the requirements are met. Includes continuous bug fixing.	MS06, MS07, MS08, MS09
PP06	Additional Deliverables	Creating all additional deliverables needed in the bachelor thesis (see 2.5 Deliverables)	MS10, MS13
PP07	Presentation and Demo	Preparing the presentation and Defence including test runs.	MS12

Table 10: Project phases

The following table shows the different milestones. They serve as a guide through the project and communicate to the stakeholders when certain things are planned to be done. They are mostly ordered according to the Gantt Diagrams in the following section. The only exception is India Visited (MS13) this was added to the project once it was clear that the funding for the field trip will be given and that the Project Team (TA01) will meet in India.

Id	Name	Description
MS01	Contract signed	The contract required by the BFH for bachelor thesis concerning rights over the code, is signed by all needed parties.
MS02	Project Initialization finalized	All preparations are done, and the development of the new features can begin.
MS03	Feature 1: Assessment online	The assessment feature is complete and deployed and ready for user feedback.
MS04	Feature 2: Certification online	The certification feature is complete and deployed and ready for user feedback.
MS05	Feature 3: Smaller functionalities online	The chat feature is complete and deployed and ready for user feedback.
MS06	Testing feedback Feature 1 arrived	The feedback from the customer for the enhancements of Feature 1: Assessment (PP02) arrived.
MS07	Testing feedback Feature 2 arrived	The feedback from the customer for the enhancements of Feature 2: Certification (PP03) arrived.
MS08	Testing feedback Feature 3 arrived	The feedback from the customer for the enhancements of Feature 3: Smaller functionalities (PP04) arrived.
MS09	Testing complete	All user test runs, and final test run are done. The window for feedback considered in the bachelor thesis is closed. All results are documented
MS10	Documents final version delivered	The final version of the documentations is reached, and all deliverables are delivered.
MS11	Presentation held	The presentation and exposition of the project is done.
MS12	Defense held	The defense is held.
MS13	India visited	Visited stakeholders in India and done user tests with them.

Table 11: Milestones

2.3.1 Software Development Gantt

The following figure shows the project schedule plan for the software development, expressed as a Gantt chart.

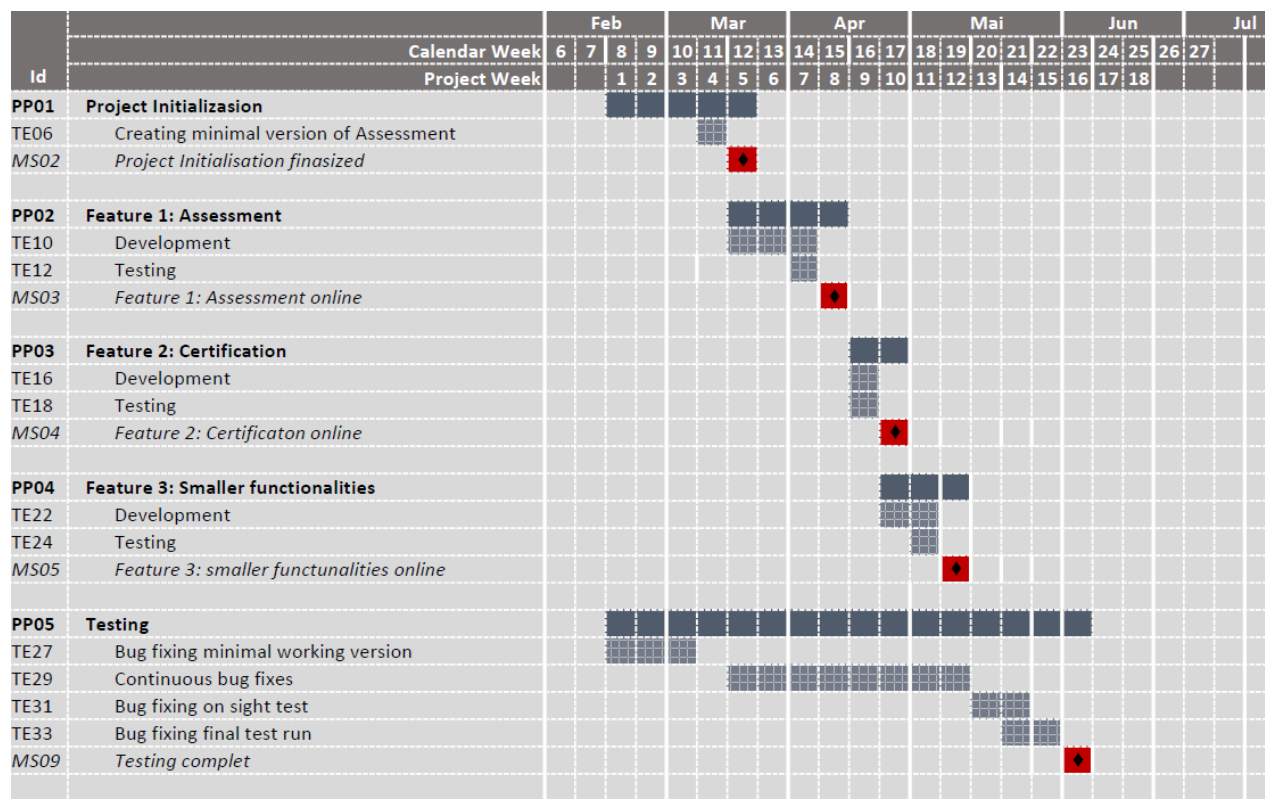


Figure 7: Project schedule plan for software development

2.3.2 Documentation and Software Quality Control Gantt

The following figure shows the project schedule plan for the documentation and quality control.

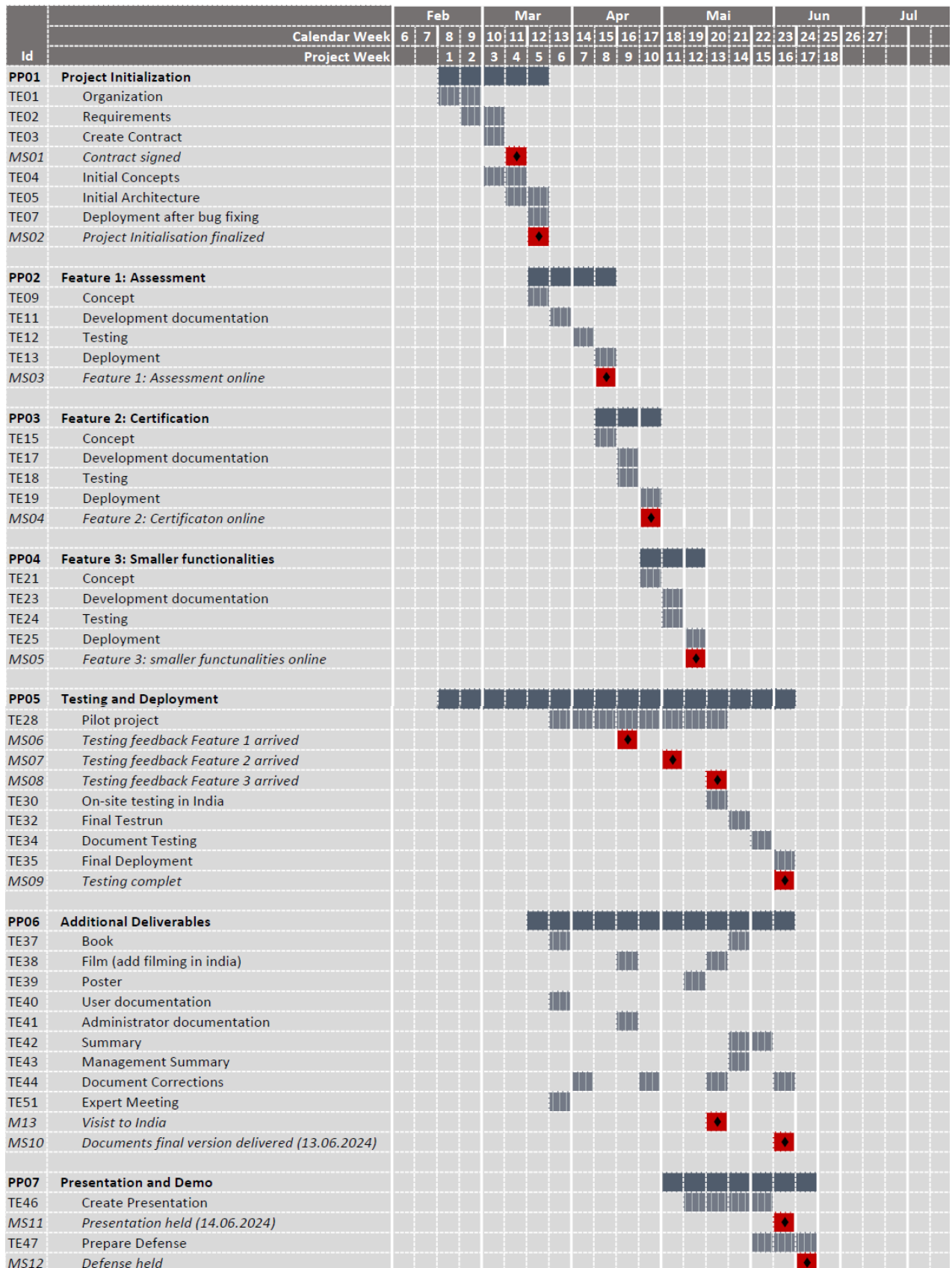


Figure 8: Project schedule plan for documentation and software quality

2.4 Time Estimate

The project is a bachelor thesis and therefore has the scope of 12 ECTS per student. With 30 working hours per ECTS and a team of 3 students the time budget for this project comes to 1080 working hours. In the following is listed how the working hours are planned to be spent separated by phase.

2.4.1 Estimate by phase

The following tables show the effort expected to use for the given phase. At the end they were completed with the time invested.

Id	Activity	Effort Estimated [person-hours]	Time invested [h]
PP01	Project Initialization		
TE01	Organization	20.00	25.75
TE02	Requirements	20.00	4.00
TE03	Create contract	4.00	6.25
TE04	Initial Concepts	20.00	21.75
TE05	Initial Architecture	15.00	3.50
TE06	Creating minimal version of Assessment	40.00	116.75
TE07	Deployment after bug fixing	4.00	16.50
TE08	Reserve	15.00	18.50
	Total	138.00	213.00

Table 12: Time estimate project initialization

Id	Activity	Time Estimated [h]	Time invested [h]
PP02	Feature 1: Assessment		
TE09	Concept	20.00	22.75
TE10	Development	70.00	105.00
TE11	Development documentation	10.00	29.75
TE12	Testing	14.00	11.50
TE13	Deployment	6.00	3.50
TE14	Reserve	10.00	0.50
	Total	130.00	173.00

Table 13: Time estimate feature 1: assessment

Id	Activity	Time Estimated [h]	Time invested [h]
PP03	Feature 2: Certification		
TE15	Concept	12.00	4.50
TE16	Development	24.00	35.00
TE17	Development documentation	6.00	2.00
TE18	Testing	6.00	8.75
TE19	Deployment	6.00	2.00
TE20	Reserve	6.00	0.00
	Total	60.00	52.25

Table 14: Time estimate feature 2: certification

Id	Activity	Time Estimated [h]	Time invested [h]
PP04	Feature 3: Smaller functionalities		
TE21	Concept	20.00	10.00
TE22	Development	56.00	74.50
TE23	Development documentation	6.00	23.00
TE24	Testing	12.00	8.75
TE25	Deployment	6.00	5.00
TE26	Reserve	10.00	2.00
	Total	110.00	123.25

Table 15: Time estimate smaller functionalities

Id	Activity	Time Estimated [h]	Time invested [h]
PP05	Testing		
TE27	Bug fixing minimal working version	60.00	32.25
TE28	Pilot project	4.00	0.00
TE29	Continuous bug fixes	80.00	84.50
TE30	On-Sight Testing	20.00	12.00
TE31	Bug fixing on sight test	20.00	2.00
TE32	Final Test run	12.00	8.50
TE33	Bug fixing final test run	30.00	18.25
TE34	Document testing	6.00	10.00
TE35	Final deployment	4.00	5.50
TE36	Reserve	25.00	7.00
	Total	261.00	180.00

Table 16: Time estimate testing

Id	Activity	Time Estimated [h]	Time invested [h]
PP06	Additional Deliverables		
TE37	Book	8.00	9.00
TE38	Film	16.00	20.00
TE39	Poster	8.00	0.75
TE40	User documentation	16.00	8.00
TE41	Administrator documentation	16.00	2.25
TE42	Summary	12.00	14.50
TE43	Management Summary	4.00	0.00
TE44	Document Corrections	20.00	27.00
TE45	Reserve	10.00	21.50
	Total	110.00	103.00

Table 17: Time estimate additional deliverables

Id	Activity	Time Estimated [h]	Time invested [h]
PP07	Presentation and Demo		
TE46	Create Presentation	15.00	30.75
TE47	Prepare Defence	15.00	6.75
TE48	Reserve	10.00	7.00
	Total	40.00	44.50

Table 18: Time estimate presentation and demo

2.4.2 Meetings

The following table shows the effort expected to be used for meetings and how much time invested.

Id	Activity	Time Estimated [h]	Time invested [h]
TE49	Thesis advisor meeting	30.00	17.00
TE50	Project partner meeting	30.00	32.00
TE51	Expert meeting	6.00	3.00
TE52	Code reviews	30.00	24.50
TE53	Organization Development Team	40.00	36.00
TE54	Reserve	15.00	23.00
TE55	Representation & Publicity	0.00	66.00
	Total	151.00	201.50

Table 19: Time estimate meetings

2.4.3 Time Tracking

As mentioned previously, in the previous project time was tracked with GitHub but the team was disappointed with the extraction and processing of that data. To ensure comprehensive and easily evaluable time tracking, a simple time-tracking tool was set up using Microsoft Excel. The Excel file is split into one sheet per student and contains the date, the category and the time.

The total time invested by the Development Team (TA02) (actual) is 1090.50 hours, which exceeds both the estimated time (1080 hours) and the ECTS requirement (1080 hours). Note that this does not include the time invested by the Thesis Advisor (PR05), Project Partner (PR06), and other people involved with e.g. testing. A detailed assessment of these results is provided at the end of this report in Section 7.1.5 Working hours.

2.5 Deliverables

This Section lists all items that have to be delivered at the end of the project. The success of the project will be measured by the number of deliverables delivered and how complete they are (quality).

Id	Description	Corresponding Requirements
D01	A working LMS enhanced by all the functional requirements	FR01, FR02, FR03, FR04, FR05, FR06, FR07, FR08, NR01, NR02
D02	A project documentation i.e. the bachelor thesis document	NR03
D03	User documentation	NR04
D04	System administrator documentation	NR05
D05	A book entry to be published on the BFHs website and in the bachelor thesis booklet	NR06, NR07
D06	A video describing the project	NR08
D07	A poster summarizing the project for the exhibition	NR09

Table 20: Deliverables

2.6 Tools

In this section all tools used in the project and their purpose are described. The tools for this chapter are software that enables project management and collaboration. They differ from the 4.1 Technology Stack, which are specialized tools not just used to create but also to run the product. To increase readability and understanding they were separated into communication, documentation, development, and hosting tools.

2.6.1 Collaboration

The following table shows all tools used for communication inside the development team (TA02) as well as the project team (TA01). These tools are mostly based on 2.2 Communication.

Id	Tool	Purpose
TL01	WhatsApp	As an easy-to-use communication channel (CH01) for small questions and informal information.
TL02	MS Teams	Used as the communication channel (CH03) for meetings in the development team (TA02) with or without the thesis advisor (PR05) and expert (PR07).
TL03	Zoom	Used as the communication channel (CH04) for meetings with the project partner (PR06).
TL04	Miro	As a place to easily dump and share information in the project team (TA01). This information consists of data models, mockups, ideas, notes and more.

Table 21: Communication tools

2.6.2 Documentation

The following table shows all tools used to create and share the documentation and other deliverables.

Id	Tool	Purpose
TL05	MS Office	To document the project and create most of the other deliverables (D02, D03, D04, D06, D07). Additionally used to track the time.
TL06	MS SharePoint	Used as a platform to store and share documents to work on simultaneously in the development team (TA02) and share documents with reviewers.
TL07	draw.io	Used to create diagrams like the 2.7.3 Heat map.
TL14	ChatGPT	Used to rephrase sentences, get translations of phrases and quick opinion on the structuring of the document. It was not used to gather information.

Table 22: Documentation tools

2.6.3 Development

The following table shows all tools used to develop, share, store and review code.

Id	Tool	Purpose
TL08	IntelliJ IDEA	Mostly used to develop the backend code but also by one development team (TA02) member to develop the frontend.
TL09	WebStorm	Used to develop frontend code.
TL10	MS Visual Studio Code	Used to verify the frontend code.
TL11	h2	Used as the database for automated testing since it is an in-memory database.
TL12	GitLab	Used as the versioning tool for frontend and backend. Used as a data storage and backup for code and documentation. Used to as the kanban board.

Table 23: Development tools

2.6.4 Hosting

Id	Tool	Purpose
TL13	Jelastic Dashboard	Used to deploy and manage the productive version.

Table 24 : Hosting tools

2.7 Risk Management

This Section describes all the relevant risks that can occur during the project. This is absolute vital as a first step in the project planning process, so that possible disrupters can be identified and planned for. First listed are all the contexts and definitions necessary for the risk assessment. This is followed by all risks, including assessment and mitigations. Finally, all risks are graphically shown in a heat map, in order to assist with the comprehension of their impact.

2.7.1 Context for the risk assessment

The following tables explain the abbreviations used in the risk assessment.

2.7.1.1 Damage

Id	Description
D1	No damage
D2	Smaller changes in the time plan
D3	Notable changes in the time plan or cancelation of minor features or bug fixes
D4	Cancellation of major features
D5	Abortion of the project

Table 25: Risk assessment damage

2.7.1.2 Likelihood

Id	Description
L1	Unimaginable
L2	Very unlikely
L3	Unlikely
L4	Likely to occur but not necessarily expected to be certain
L5	Expected to occur ones or multiple times

Table 26: Risk assessments likelihoods

2.7.1.3 Risk color scheme

Name	Color (heat map)	Consequence
Low	Green	The risk can be accepted
Medium	Yellow	Take mitigation measures or monitor the risk closely
High	Red	Intolerable: take mitigation measures

Table 27: Risk assessment risks

2.7.1.4 Mitigation

Name	Description
Avoidance	Take actions to nullify the risk's probability or impact.
Reduction	Take actions to reduce the risk's probability and/or impact to a at least Medium risk.
Transfer	Transfer the risk to a third party.
Acceptance	Acknowledge the risk and accept that the counter measures are more expensive than the benefits in reduction.

Table 28: Risk assessment mitigations

2.7.2 Risk assessment

Id	Tite	Impact	Damage		Risk	Mitigation	Strategy	Damage		Risk	Mitigation
			Damage	Likelihood				Damage	Likelihood		
RA01	Document loss	The documents must be recreated. The thesis likely will not be of sufficient quality in the given time frame.	D4	L3	Medium	Reduction	We employ the GitLab Service from the BFH as backup after every version.	D4	L2	Low	
RA02	Source Code loss	The code must be recreated. The thesis won't be of sufficient quality.	D5	L3	High	Reduction	We employ the GitLab Service from the BFH for development and as a backup	D5	L1	Medium	Acceptance
RA03	Key staff temporarily unable to continue	Some meetings will be dropped. Urgent work shifted in time or by how done.	D2	L5	Medium	Acceptance	Monitoring and act quickly when it comes to it.				
RA04	Key staff permanently unable to continue	Development Team (TA02): recalculate time estimates possibly canceling features. Thesis Advisor (PR05): New one is assigned by the BFH. Project Partner (PR06): Continue to the best ability. Expert (PR07): Exchange the Expert and arrange new meetings.	D3	L2	Low	Acceptance					
RA05	Damage to infrastructure	Data needs to be recovered or the hosting setup needs to be redone.	D3	L2	Low	Acceptance					
RA06	Unsuitable hosting platform	The hosting platform needs to be changed.	D3	L1	Low	Acceptance					

Table 29: Risk assessment

2.7.3 Heat map

The heat map shows all the risks from the 2.7.2 Risk assessment. The black circles show the final results where the gray ones with the dashed lines show the assessment before the mitigation measures.

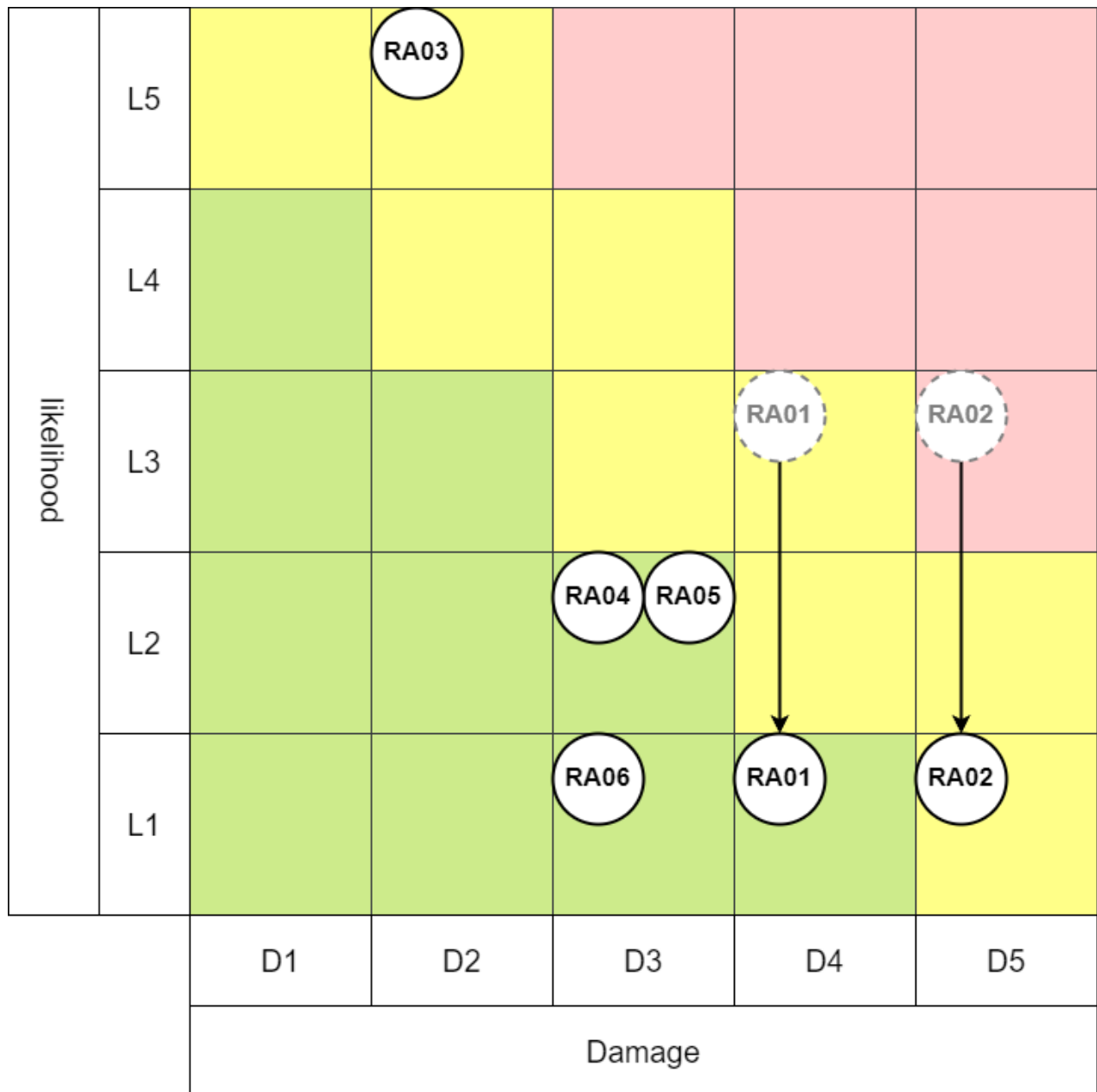


Figure 9: Risk heat map

As shown here, after mitigation the most significant risks are RA02 (source code loss) and RA03 (resources) so these will be monitored closely during project execution.

3 Requirements

In this Chapter the requirements are described. They are split into first functional and then non-functional requirements. Most of the requirements map to the final product composed of the Deliverables D01 – D04. The other requirements are given by the fact that this project is a bachelor thesis. To be regarded as an all-around successful project, all the requirements have to be met.

3.1 Agile approach to requirements & “feature freeze”

The initial requirements were gathered at the start of the project. These included requirements identified in the Project 2 module but were then re-examined and extended with new requirements.

This “agile approach” was continued on a weekly basis. Inputs from the Stakeholder were discussed with him and the project team. The results of this discussion often meant that the Stakeholder could not achieve all of the desired requirements.

Around Project Week 12, a “requirements freeze” was announced, in which the development team would accept no further requirements. This was absolutely essential in order to guarantee enough time to address all of the software defects and to ensure a “production-ready” quality. However, because this requirement freeze could be announced so late in the project was a reflection of the very good working system including quality of implementation and fast feedback loops.

3.2 Functional Requirements

The functional requirements show what features are required to be implemented. The following table shows the functional requirements and their respective sub requirements. The functional requirements FR01-FR05 were implemented in the previous project and are therefore not further elaborated here [4]. They are included for completeness since they still have to be working at the end of this project.

Id	Description
FR01	The system includes a user management
FR02	The system includes a course management
FR03	The system includes a batch management. A batch links students, tutor, and a course.
FR04	The system includes a way for a student to participate in a course.
FR05	The system includes a way to track whether students were present at a date.
FR06	The system includes an assessment functionality. Relates to Feature 1: Assessment (PP02)
SR01	An administrator or the course manager can create an assessment.
SR02	There is a way for a student of a batch to write an assessment.
SR03	A student can see the result of their assessments and redo the assessment if they did not pass.
SR04	The student cannot continue the course if they did not take or passed the assessment.
FR07	The system includes a way to automatically create certificates for students. Relates to Feature 2: Certification (PP03)
SR05	A course can be marked as completed.
SR06	A student can get an automatically generated certificate for all their successfully passed batches.

Table 30: Functional requirements

3.3 Non-Functional Requirements

The non-functional requirements are comprised of all the requirements that are not directly a feature. The following table shows the non-functional requirements and their respective sub requirements.

Id	Description
NR01	The system is well tested.
SR08	All implemented functionalities in the backend are tested via unit tests.
SR09	All endpoints in the backend are tested via integration tests.
SR10	All implemented features (FR06, FR07, FR08) are tested by the project partner (PR06).
SR11	A field test with actual users was conducted.
SR12	A final test run was conducted.
SR13	All major bugs were fixed.
NR02	The system is deployed.
SR14	The system is deployed after every new feature for testing by the project partner.
SR15	The system in its final form is deployed for the project partner to use.
NR03	The project documentation contains all necessary parts to comprehend how the project went.
SR16	The project documentation contains a management summary and gives an introduction to the project.
SR17	The project documentation shows the organization including project methodology, communications, phases, milestones, time estimates and a risk management.
SR18	The project documentation splits all requirements into understandable parts.
SR19	The project documentation contains all the necessary concepts.
SR20	The project documentation explains the architecture of the final product.
SR21	The project documentation shows the most important parts of the implementation.
SR22	The project documentation documents the test runs and its results.
SR23	The project documentation compares the times estimated and worked, checks whether the requirements were met and contains a review over the whole project.
NR04	A user documentation was created to answer questions when the project team (TA01) is no longer available.
SR24	The user documentation explains all functionalities.
SR25	The user documentation shows the workflow with matching imagery.
NR05	An administrator documentation was created to answer questions when the project team (TA01) is no longer available.
SR26	The administrator documentation contains the most important parts of the architecture.
SR27	The administrator documentation contains the current hosting setup.
NR06	The book entry contains all necessary information.
SR28	The book entry contains an abstract of the project.
SR29	The book entry contains an image of all members of the development team (TA02).
SR30	The book entry used the corporate format of the BFH.
NR07	The book entry ran through the correct process of approval.
NR08	The video contains the necessary information and format.
SR31	The video is 60-90 seconds long

Id	Description
SR32	The video uses the given BFH intro and outro
SR33	The video contains problem/task, solution and results
SR34	The video was shot employing a script
NR09	The poster contains all necessary information and format
SR35	The poster is A1 format
SR36	The poster is max. 5 GB large and in the format .pptx or pdf
SR37	The poster illustrates the most important parts of the thesis

Table 31: Non-functional requirements

3.1 Emergent Requirements

Emergent requirements are an important category of requirements. These are new requirements that are only able to be identified after they “emerge” naturally from productive use of the application. While they cannot (by definition) be predicted in advance, the extensive attention to testing and a “soft rollout” will hopefully minimize or even eliminate these requirements from developing after the go-live into production.

Given the format of the bachelor's thesis this won't be regarded as necessary in order for the project to be successful.

3.2 Use Cases

In this Section all use cases are listed. First are the use cases from the previous project as a reference (mostly for testing). This is followed by all new use cases added in this project, in detail split by the feature they belong to. The following figure visualizes the use cases and the respective actors.

3.2.1 Use cases from the previous project

The following table lists the use cases created and implemented in the previous project and to what requirement they belong to. The detailed use cases can be found in the Appendix A: User Cases. They were changed and reordered to fit this project.

Id	Name	Related Requirements
UC01	Create User	FR01
UC02	Manage User	FR01
UC03	Create Course	FR02
UC04	Manage Course	FR02
UC05	Create Batch	FR03
UC06	Manage Batch	FR03
UC07	List enrolled and tutoring courses	FR04, FR05
UC08	Detailed course	FR04, FR05
UC09	Manage attendance	FR05

Table 32: Use cases previous project

3.2.2 Use cases Assessment-Feature

The following table lists the use cases for the assessment feature (FR06) and to what requirement they belong to. The detailed use cases can be found in the Appendix A: Use Cases.

Id	Name	Related Requirements
UC10	Manage Assessment	SR01
UC11	Fill out an Assessment	SR02, SR04
UC12	View assessment results	SR03

Table 33: Use cases assessment

3.2.3 Use cases Certificate-Feature

The following table lists the use cases for the certificate feature (FR07) and to what requirement they belong to. The detailed use cases can be found in the Appendix A: Use Cases.

Id	Name	Related Requirements
UC13	Complete a Course	SR05
UC14	Get a Certificate	SR06

Table 34: Use cases certificate

4 Architecture

In this Chapter all architectural designs are mentioned, including the parts that are designed before implementing the code/environment. What is described here is the final state of the architecture, as small changes were made and some new features were added, or to overcome other difficulties.

This Chapter is structured in the following way. First, the chosen technology stack is described. Next, the data model of the application and the resulting class structure are presented. The system architecture is described in detail, including the REST APIs used for communication between the front-end and the back-end. Finally, details about the deployment and cloud hosting environment are given.

4.1 Technology Stack

The technologies are the software and frameworks used to create and to run the product. This is not to be confused with the 2.6 Tools which are not used to run the product. The technology stack was chosen in order based on the developers' experience, cost to save money for Sharada and if possible widespread technologies to enhance the possibility of enhancing it after the bachelor thesis. The following table shows the technologies that were employed for the project and describes them.

Id	Technology	Purpose	Version
TL01	Angular	Used as the front-end framework. In use to structure the project and increase the speed of creating the frontend. Used in combination with TypeScript.	16.2.12
TL02	Spring Boot	Used as the backend framework. Used to structure the project and increase the speed of creating the backend, including most notably JDBC and Hibernate for the database connection.	3.2.0
TL03	PostgreSQL	Used as the database for local testing, test and productive environment but not for the unit tests.	14.2
TL04	Apache Tomcat	Used as the webserver to host the frontend for the test and the productive environment	11.0.0-M14
TL05	Jelastic/Spring Boot container	Used to host the backend for test and productive environment.	Openjdk-21

Table 35: Technology stack

4.2 Class Structure

The following diagram shows the professional classes and how they are linked. As this work follows the concept of “Object First,” the entity objects to reproduce the reality inside the system are first defined. When starting the application, Hibernate will automatically create tables with relations. However, the resulting database structure does not vary much from the diagram shown in Figure 10. It is also not exactly the way it is represented through the API; for more information on that see the 4.6 API Description.

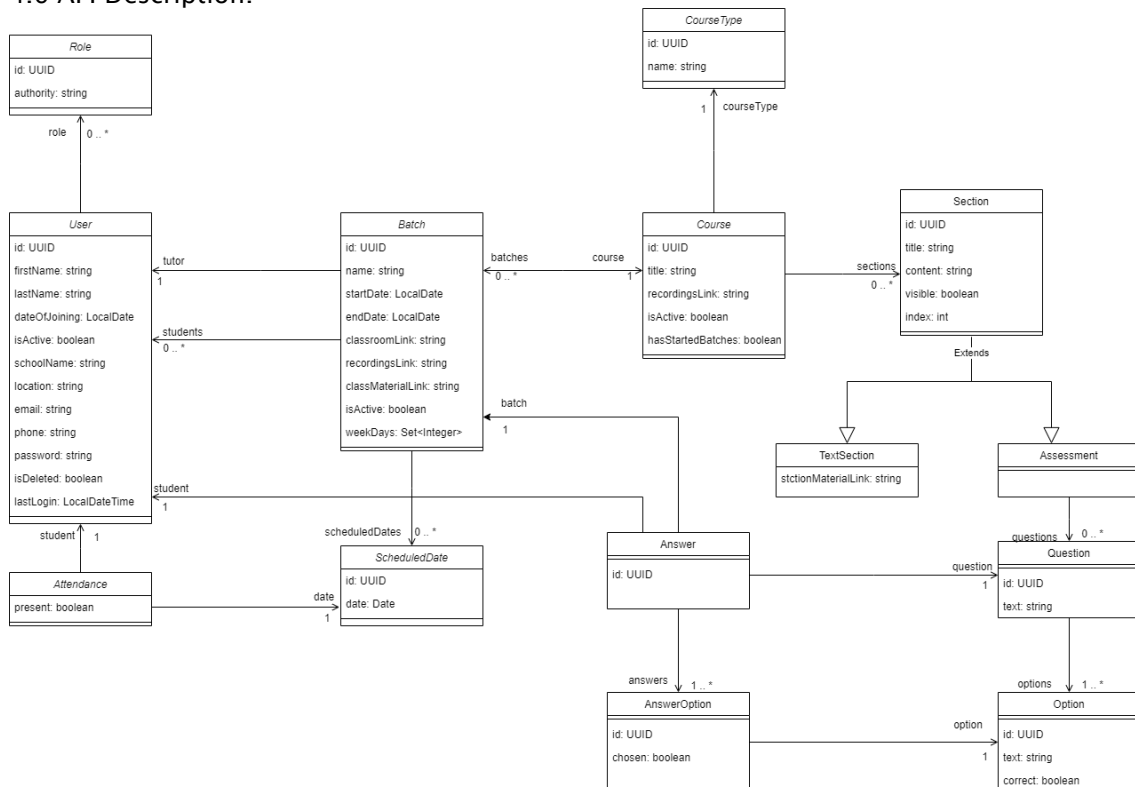


Figure 10: Entity class diagram

The most essential entities are **User**, **Course** and **Batch**. Each user can have multiple roles. For example, a tutor can also be a student allowing tutors to teach other tutors. Additionally, users with the “admin” role have unrestricted permission whereas course managers are able to only configure courses.

Courses are categorized with a **CourseType**. Until now, there is “self paced” and “tutor led” but in the future, more course types can be specified. Courses which are led by a tutor can have scheduled dates which are seen as lessons. When a lesson has started, a tutor must note if every student is attending the lecture.

A batch is the connecting part between users (students and tutors) and courses. Therefore, it embodies an instance of a course by containing multiple students and a tutor in addition to start and end dates.

In order to get a better understanding of this diagram, the following diagram depicts one of the important parts, sections and how they are arranged.

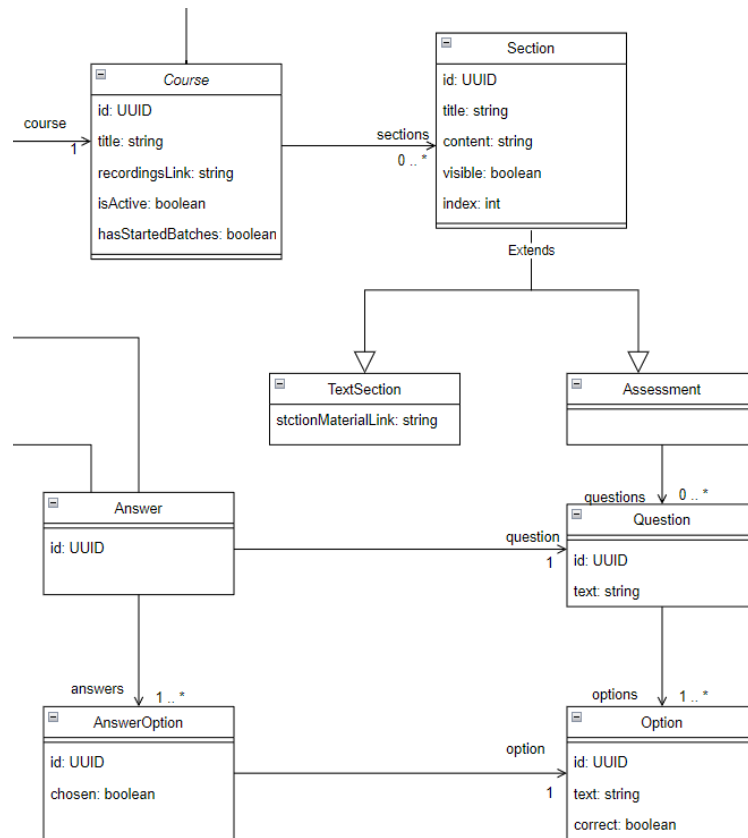


Figure 11: Section in detail

The first thing to understand is that courses have **Sections** which can either be **Text Sections** or **Assessments**. They are derived from the section class because both can be seen as components of a course but with different purpose. A text section can also be looked at as a lesson. During the project, we have seen that tutors would like to add text sections and assessments alternately. Like this, the idea of how a course is structured becomes very clear. A student can take a lesson and when they feel ready, they take the assessment. When they succeed, they are allowed to access the second lesson and so on.

The assessments were designed as a List of multiple-choice questions. This means that every question has a List of Options which either can be correct or wrong. On the answer side, for every option of a question, there must be an AnswerOption. These are grouped as a List inside an Answer, which is mapped to a question. The mapping of an answer to a batch is necessary, because a student might have to retake a course and therefore, they would be in a different batch but in the same course.

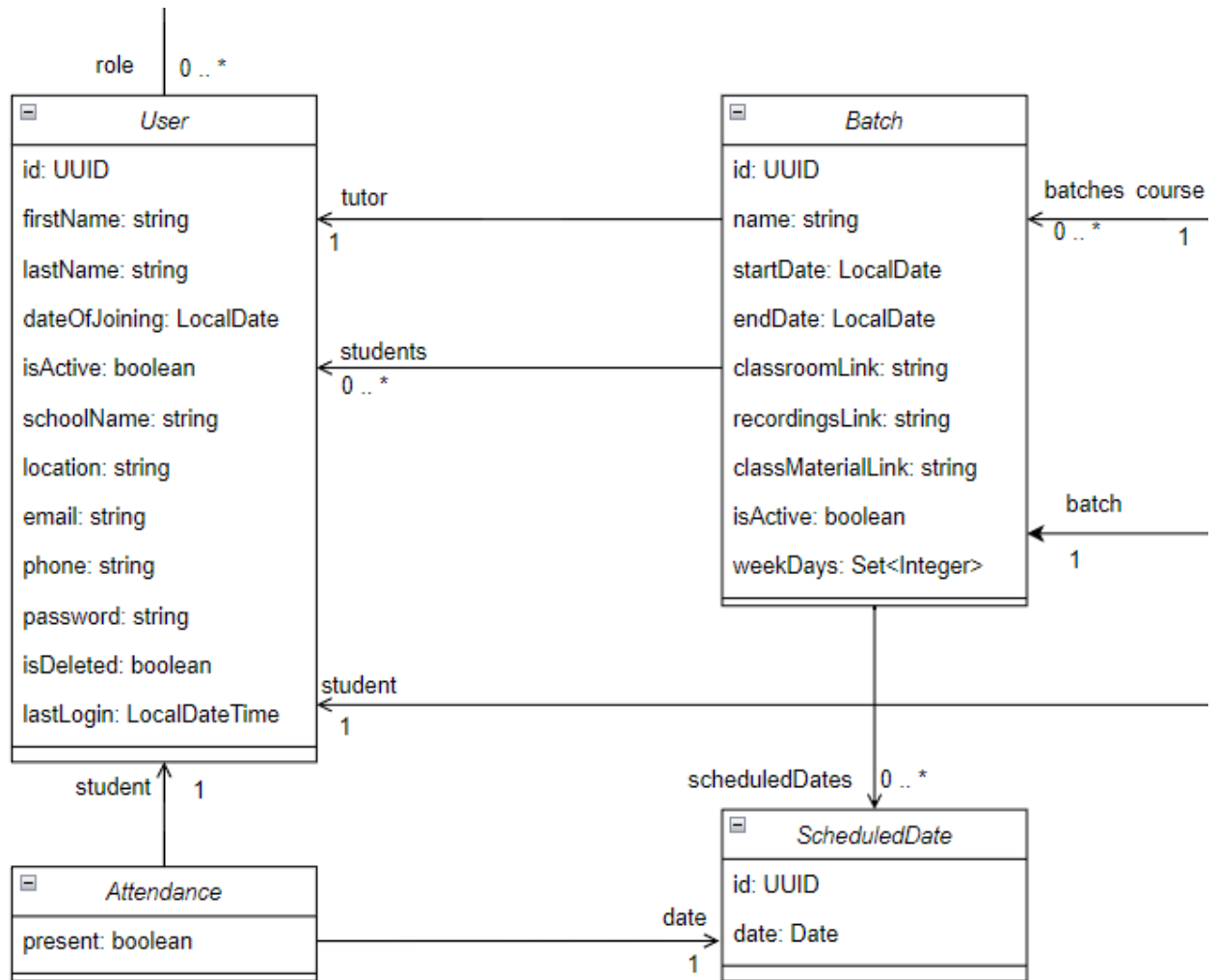


Figure 12: Attendance in detail

In order to fulfil the requirement of the stakeholder (namely, that tutors should check whether students are present to a lesson or not) Figure 12 shows how this was designed. When a batch has a tutor lead course, it will have scheduled dates. For every date and every student an attendance object is present. When a lesson takes place, a tutor can change the present state of the attendance object.

As a final note, we stress the generic nature of this design, in order to completely separate content from implementation. In this way, the application can be used to present teaching materials on any topic at all (English-as-a-second-language, financial literacy, business culture, etc.). The project partner (PRO6) feels that this is a critical success factor for the application to be adopted in future by other NGOs that have their own educational goals.

4.3 System Architecture

This Section describes the system architecture used to build the Sharada LMS application. The following diagram shows a container diagram of the productive Sharada LMS environment. To provide maximum clarity, the diagrams in this section are created according to the c4model [7] which is increasingly favored for agile development projects.

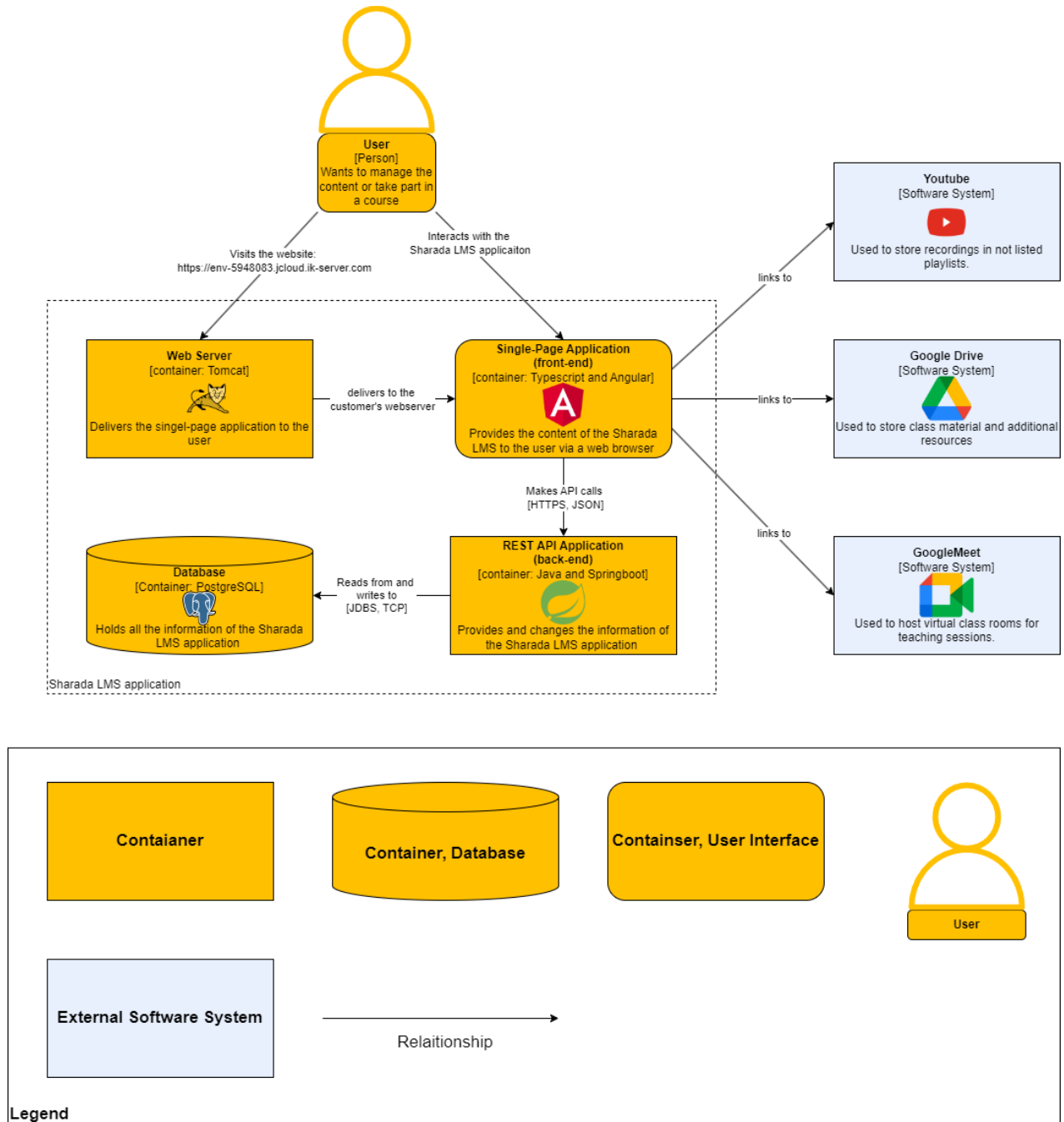


Figure 13: Container diagram for the Sharada lms application

4.4 Security-by-design (SbD)

Security-by-design tries to prevent vulnerabilities in the software by pre-emptively thinking about potential problems and solving them with design during the creation of the fundamental concepts. The following chapters show how which parts were handled.

4.4.1 Role-Based Access Control (RBAC)

The product includes many resources. The technical description can be found in the 4.6 API Description. To keep order among the user and data the following roles were used. The roles were defined back in the previous project and adapted in this work based on the new features [4]. The roles are not further split into permissions to keep it simpler for the end user, reduce sources of potential problems and focus on implementing features that bring more value to the project partner.

Id	Name	Description	Tasks	Related Use Cases
R01	Admin	Administrator for non-technical things.	Onboarding new staff and students.	UC01
			Granting and revoking roles to other users.	UC01, UC02
			Deciding which courses are offered, creating and deleting them.	UC03, UC04
			Organizing batches with dates, students, and tutors.	UC05, UC06
			Answering questions about the system.	Todo add UC
R02	Course Manager	Responsible for maintaining the content of the courses.	External content (i.e. presentations) and the linking to those places.	UC04
			Maintaining the content inside the system including the assessments.	UC04, todo add UC
R03	Tutor	Managing the execution of a batch assigned to them.	Teaching to the students in the batch.	UC07, UC08
			Keeping track of the attendances.	UC09
			Correcting assessments.	Todo add UC
			Grading the students.	Todo add UC
			Answering questions about the course	Todo add UC
R04	Student	Take part in their courses	Consulting the internal and external content.	UC08
			Solving assessments.	Todo add UC
			Viewing the result of their assessments.	ToDo add UC
			Getting their certificates.	ToDo add UC
			Asking questions to the admins or their tutors.	Todo add UC

Table 36: Security roles

The following table shows the relationship between the resources and the roles. The access rights table was created in the previous project [4] and updated in this project resulting in the table below. To show the relationship CRUD (create, read, update, and delete) are used. Additionally, the * symbol is used if a role can only access a resource related to them (i.e. a student can only access courses they take part in).

Resource	Admin	Course manager	Tutor	Student
Login	R	R	R	R
Roles	R	R	R	R
User	CRUD	R*	R*	R*
Course Type	R	R		
Course	CRUD	CRUD		
Batch	CRUD			
Scheduled Dates	CRUD			
Attendances			R*U*	
User Course View			R*	R*
Assessment	CRUD	CRUD		
Assessment Answer			R*U*	R*C*U*
Certificate				R*
Chat	C*R*	C*R*	C*R*	C*R*

Table 37: Access rights

4.4.2 Passwords

Although multifactor-authentication and third-party authentication are nowadays the gold standard for web-applications, this adds a degree of complexity that would challenge the intended users (rural students and teachers in India). Therefore, this application implements its own user management with no third-party authentication, meaning the passwords must be saved in the database. To ensure that the passwords are not readable by an attacker or the DB administrator they are saved as hashes using the BCrypt algorithm [8]. BCrypt automatically adds a salt to the password encoding. This prevents an attacker from reverse engineering the passwords using rainbow tables with pre-generated hashes and the according passwords. In addition, it employs a strength which is the number of times the password is rehashed to slow down brute force.

4.4.2.1 Password Strength

Since the project partner (PR06) and most of the end users are in rural India and have a not-so-versed understanding of information technologies, some compromises for the password strength were essential. For the students passwords are allowed with a length of 4 to allow the format ddMM; this will be their date-of-birth which is easy to remember and will greatly reduce support and training efforts. According to feedback from the project partner (PR06), the users with the student role are not experienced with passwords, since their lives do not revolve around computers. In terms of the system, the student role is very limited to what they can see and therefore it is a tolerable risk. All users who have another role (in addition or not) have to have a password with the minimal strength of 8.

4.4.3 Security Scan

In keeping with the fundamental principle of Security-by-Design (SbD), to protect against common security issues and potential problems, a static application security testing (SAST) scan is used. For the project the SAST collection provided by GitLab [9] is used [9]. This collection includes scans using SemGrep for Java and TypeScript, which are our most important languages for the project. The configurations for the scans are done by GitLab. A sample SAST vulnerability report is shown in the next figure.

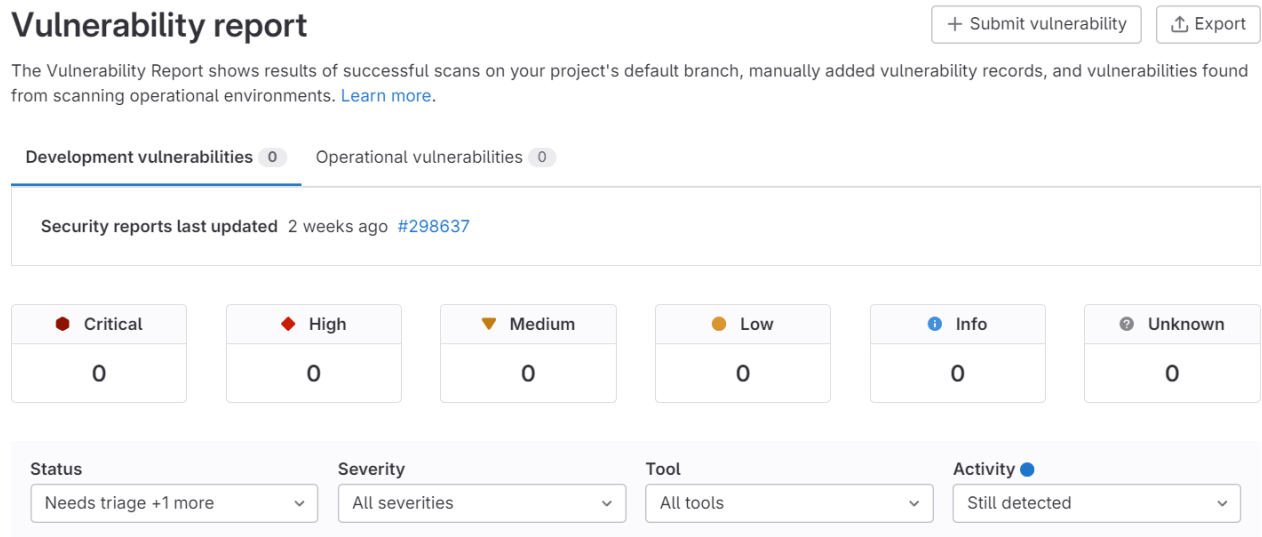


Figure 14: Vulnerability report

4.5 Hosting & Deployment

Hosting is done with Jelastic Cloud service offered by the Infomaniak company in Switzerland [10]. This provider was chosen because it has an attractive pricing, and the team already had some experience with it. The hosting was chosen and tested during the previous project [4]. The costs for this hosting are born by the advisor (PR05), as his personal volunteer contribution to the Sharada Educational Trust.

The deployment is initially done on the test environment. Once there is confirmation from the test, the deployment was carried out on the productive environment. The environment structure is described by the following C4 diagram.

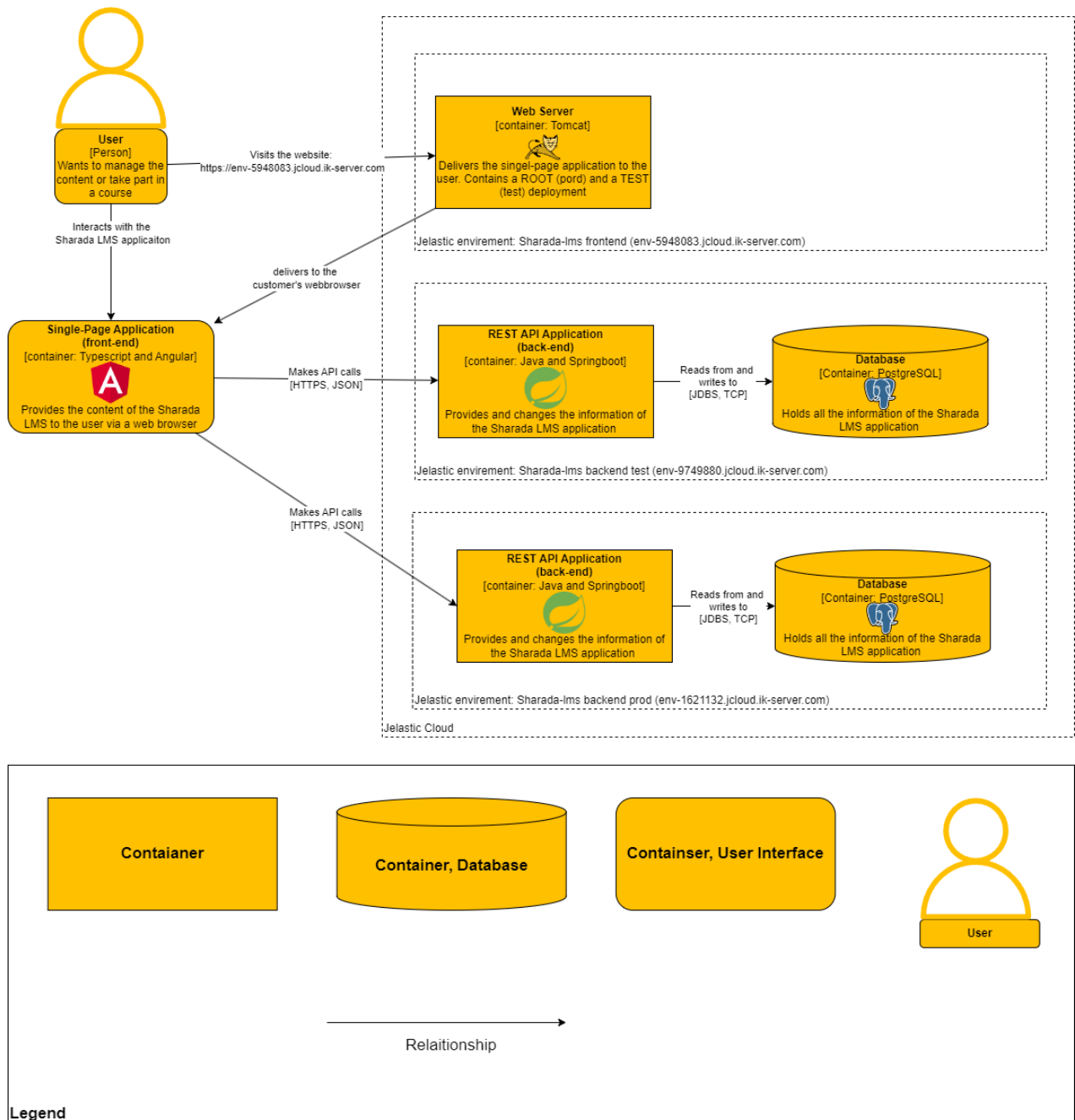


Figure 15: Deployment container diagram

The implemented version is deployed to the Jelastic cloud. The following diagram shows the Jelastic Cloud dashboard, as the implemented version of the previous diagram.

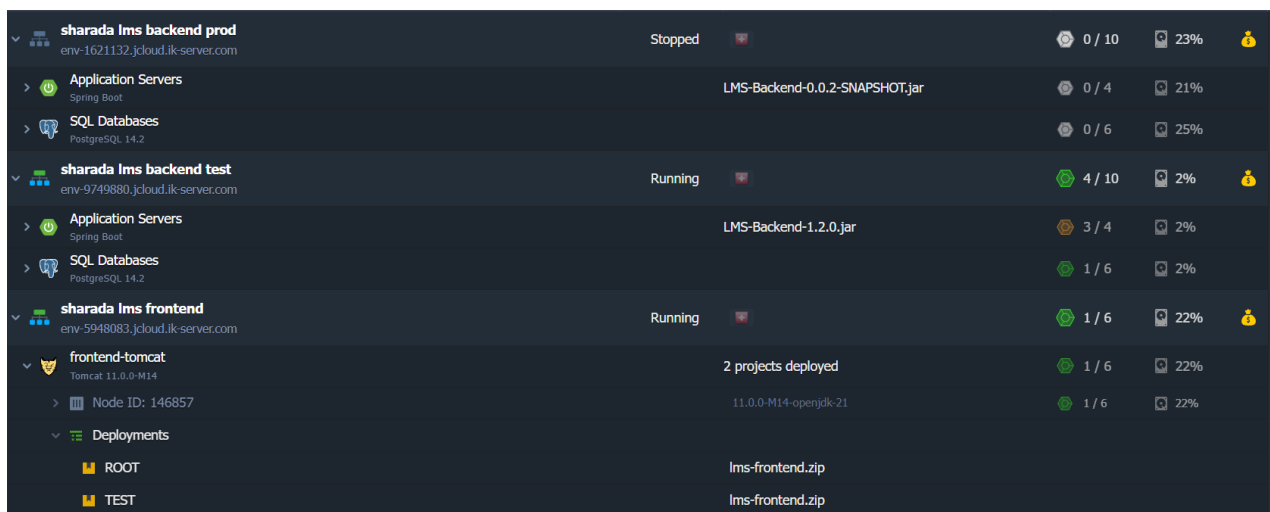


Figure 16: Deployment in the Jelastic cloud

4.6 API Description

The description of the API has been done in 2 different ways. Mainly an Excel sheet served as a basis for discussions and adjustments. For a better overview and to follow industry standards also included is a Swagger documentation. For every endpoint, annotations “Operation” and “ApiResponse” have been added to provide information about the API. To access the swagger GUI documentation, use the following URL:

<https://ENVIRONMENT/api-docs/swagger-ui/index.html>

Note: for “ENVIRONMENT”, enter the one you want to use:

Name	Environment
Local (default spring boot port)	localhost:8080 (without “https://”)
Test	env-9749880.jcloud.ik-server.com
Prod	env-1621132.jcloud.ik-server.com

Table 38: API description environments

To access the swagger YAML documentation, use the following URL:

<https://ENVIRONMENT/api-docs>

5 Implementation

In this Chapter the implementation of the project is explained. First, how the Git repositories GitLab are structures its documents; second, the look and feel of the project is provided; and lastly, the structure and most important bits of the front- and backend are discussed.

5.1 GitLab

This Section describes the ways GitLab was used and how the git repositories were organized.

5.1.1 Grouping and Projects

The whole project is version controlled in the BFH's GitLab. To fit it all in one, a Group named "Sharada-LMS Group" was created. It contains 3 different repositories: one for backend, frontend and documentation. The following figure shows the group overview in GitLab.

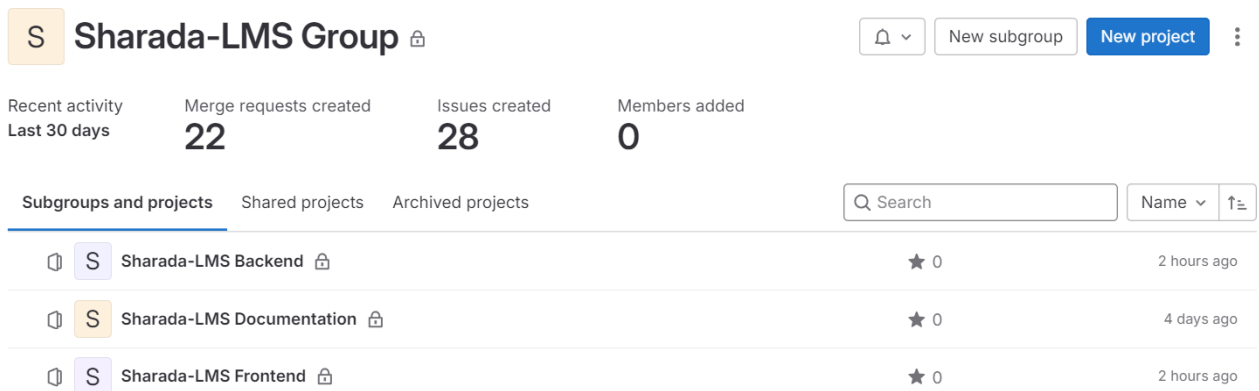


Figure 17: GitLab group Sharada-LMS Group

5.1.2 Pipelines

Both the "Sharada-LMS Frontend" and "Sharada-LMS Backend" have pipelines which must run through before a merge request can be approved. Both have a pipeline for 4.4.3 Security Scan. But only the backend has automated tests and therefore a pipeline to execute them. The following screenshots show the pipeline jobs first for the frontend and after for the backend.

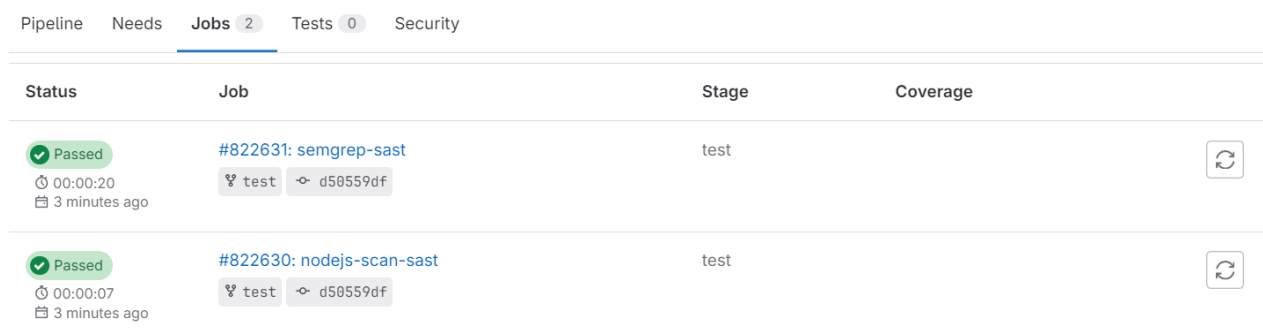


Figure 18: Pipeline jobs frontend

Pipeline	Needs	Jobs 2	Tests 0	Security
----------	-------	--------	---------	----------

Status	Job	Stage	Coverage
✓ Passed 🕒 00:01:54 📅 3 hours ago	#822449: backend-test 🔍 test 🔗 021972d9	test	
✓ Passed 🕒 00:00:22 📅 3 hours ago	#822448: semgrep-sast 🔍 test 🔗 021972d9	test	

Figure 19: Pipeline jobs backend

5.1.3 Branching

The branching has 3 main parts: production represented by the master branch; test represented by the test branch; and development represented by the other branches. The following figure shows the workflow and interaction between the different kinds of branches.

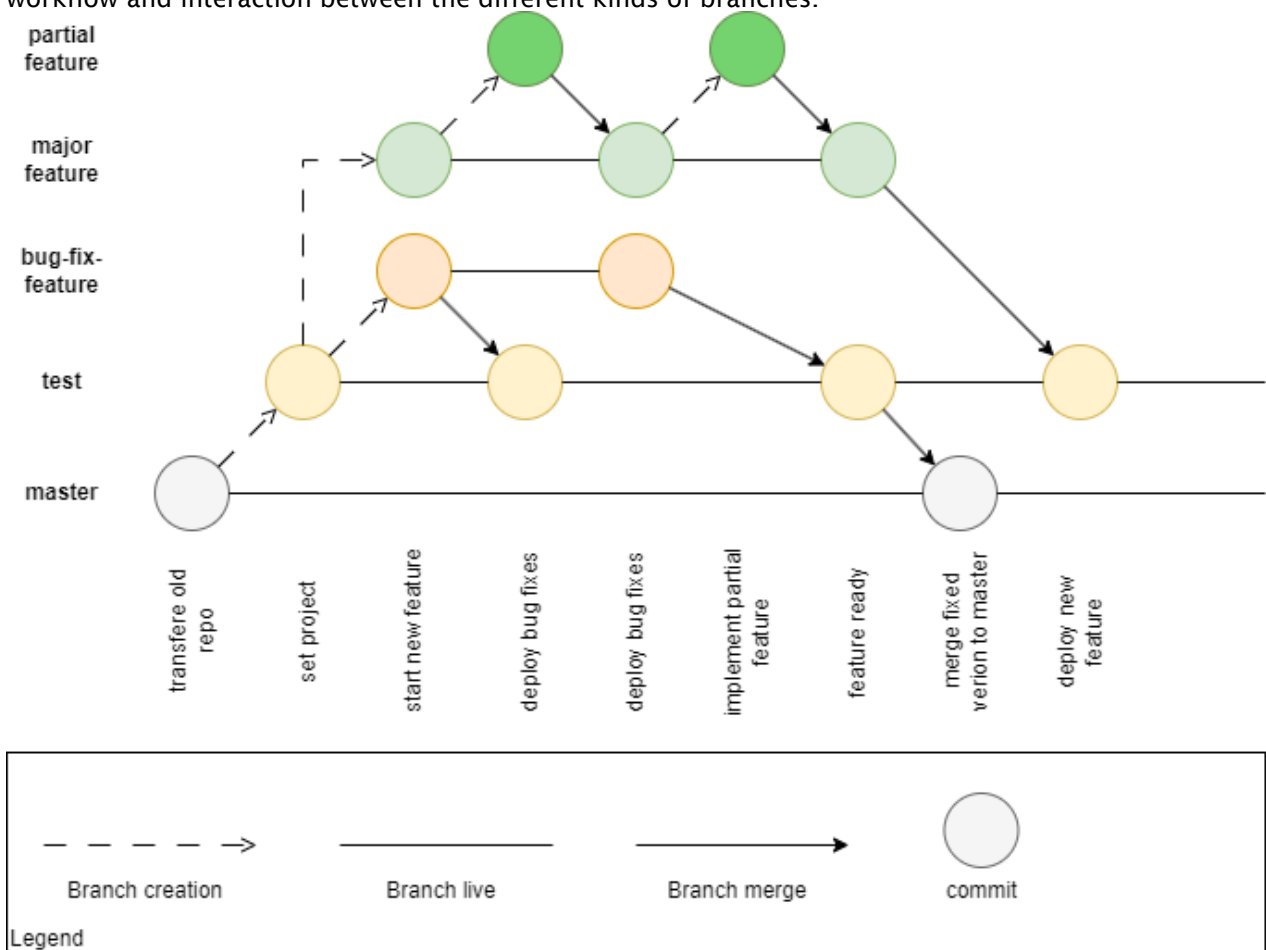


Figure 20: Branching workflow

5.2 Look & Feel

5.2.1.1 Intent

The intent of the Look-&-Feel is to create a harmonious application that is usable by the stakeholders, which in this case are not just children from rural students in India, but in fact children with older models of mobile devices. Therefore, this project has aimed to create a product for an audience that can appeal to non-tech-savvy to having very little access to modern computer technology in general. In addition, one must consider that the end users live in India, a culture that is far from what the Development Team (TA02) knows and that also has an impact on usability.

The goal therefore is to create a product which is light in content and information (to not lose the audience) and is simple to navigate and understandable – and for the student part of the application even with as little English as necessary.

5.2.2 Visual Design

The following sections include all the specifications on the visual design elements. It is very important to note that design aspects may not in general have a great importance for typical Bachelor theses. But for the present work, an application is developed that must be tightly integrated into an organization that has an existing branding in place.

5.2.2.1 Style

The style Material Design is used because it is an intuitive design for the end user, it is not coupled to a platform, and it encourages responsive design which is essential to reach the students. In addition, there is good support for Material Design in Angular.

5.2.2.2 Colors

The color seam uses the main colors of the Sharada Educational Trust: Carrot Orange and Navy Blue. To better visualize parts of the application different shades of these colors are used. The following table shows the different colors and their application.

Color	Name	Usage
#ED9121	Carrot Orange	This is the main color of the application. It is used to emphasize backgrounds, interactable elements and the title on the login page. Most notable elements are: ▶ Header ▶ Navigation ▶ Buttons
#FDF2E4	Old Lace	Old Lace is used as a lighter color of the Carrot Orange. It is used to bring listed elements up from the background. Most notable elements are: ▶ Tile backgrounds
#000080	Navy Blue	This is the main accent color of the application. It shows up in counterparts to the main color. This includes disabled elements headers and secondary backgrounds. Most notable elements are: ▶ Footer ▶ Disabled elements ▶ Page Titles
#FFFFFF	Withe	The background color for the whole application is used to make the rest pop out. In addition, it is used as the font color for elements with a colored background. Most notable elements are: ▶ Application background ▶ Font
#000000	Black	The main text color of the application. Most notable elements are: ▶ Text of filled out input fields ▶ Text blocks
# 646464	Granite Gray	Used as the secondary color for text. Most notable elements are: ▶ Placeholder and title text of input fields

Table 39: Colors used

5.2.2.3 Typography

The application uses the same font for all its text. All fonts are Sans-Serif fonts and were chosen because of their readability. They should enhance the understandability and not further any visually pleasing effect. The following table shows the different fonts in order of priority.

Name	Example
Roboto	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam pellentesque ante eget lorem feugiat efficitur. Sed nec massa tincidunt, volutpat arcu vitae, faucibus elit. Vestibulum consectetur ligula vitae quam tristique.
Helvetica Neue	Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam pellentesque ante eget lorem feugiat efficitur. Sed nec massa tincidunt, volutpat arcu vitae, faucibus elit. Vestibulum consectetur ligula vitae quam tristique.
Undefined Sans-Serif	-

Table 40: Typography

5.2.2.4 Icons

In trying to design the user interface as user friendly as possible for users with limited technological knowledge and a different cultural and economic background, the team has tried to abstain from using icons and opt for buttons with text. Icons are only used when space demands it, or if the icon is easily enough understood. The following table shows the different icons that are in use:

Icon	Name	Usage
	arrow_back	Indicates navigation to the previous page.
	arrow_forward	Indicates navigation to the next page.
	arrow_forward_ios	Indicates navigation to the detail view.
	edit	Indicates navigation to the edit page.
	visibility	Indicates navigation to the detail view of a not editable element.
	content_copy	Indicates navigation to the edit page with a copy of the element.
	delete	Indicates the action to delete the resource.
	add	Indicates the action to add another element of the same type.
	check	Indicates that something is okay or already done.
	menu	Indicates the action to open the navigation menu
	close	Indicates the action to close something

Table 41: Icon explanation

5.2.3 Layout and User Interface

Buttons to change pages and the navigation are at the top where users will see them first to simplify changing from one page to another. Save buttons are placed at the bottom of a page therefore guiding users down across all input elements to assist in not forgetting or overlooking elements which are not directly visible.

5.2.3.1 Content placement

The main area of the body is placed centrally on the page. This is used primarily due to two reasons. Firstly, the central user interface translated easily onto a phone screen with minimal layout changes. Secondly, this forces the placement of further input fields or actions below others instead of next to them. In this way a user only sees a limited number of interactive elements at once and is therefore less likely to be overwhelmed, especially when not used to working with technology. This also allows elements to take up enough space to be also usable on a smaller screen.

5.2.3.2 Navigation

The main navigation is accessible at all times via a hamburger icon in the header. Depending on the roles a user possesses, different pages are visible and accessible. Besides the Swagger API description, only the About Us and Login page are accessible without being logged in. The order of the navigation links was provided by the project partner (P06).

5.2.3.3 Buttons

The buttons for accepting or denying a form are oriented on «as read in English» since it is an English platform. This means that the reader starts in the top left and reads line by line to the bottom right. This results in the the following button placements:

- ▶ Return buttons are in the top-left corner. To make returning in case of a missclick easy
- ▶ Action buttons for a form are at the bottom of the form. The main action button in the bottom-right corner. The other buttons are in the same row to the left of the main button.
- ▶ When creating new content saving is the main action.
- ▶ When removing content the cancel button is the main action, to prevent accidental removal. Reading is much more frustrating than opening the delete dialog twice.

5.3 Frontend Implementation

The frontend is the client-side of the application which is directly engaged by users. Its main focus is to be a user-friendly interface for users to interact with the application and the presentation of data. Angular [11] was used as the Framework because of the two libraries: Angular Material [12] for the prefabricated components and Bootstrap [13] for the styling, designing the layout and UI components.

Additionally for the drop-in upload the library ngx-file-drop was added. This chapter describes the implementation process and structure of the application.

5.3.1 Framework and Libraries

Angular is an OpenSource TypeScript based single page application (SPA) framework. [14] Angular Material is a Design and UI component library designed to work seamlessly with the Angular Framework. It offers a variety of pre-designed templates to ease the design process. Those two build the foundation for the frontend part of the application.

Bootstrap provides a library used for designing the layout with HTML elements in a grid-based system. This eases the development for different screen sizes.

5.3.2 Structure

The frontend application is split up into different components which are representative of the various pages and reusable sections. A Component is made up of three connected files. Typescript for the logic and handling data, HTML for the layout and a SCSS file for local styling. Components that are used within one another are placed within a tree structure. Additionally, components which are closely related are grouped in the same directory.

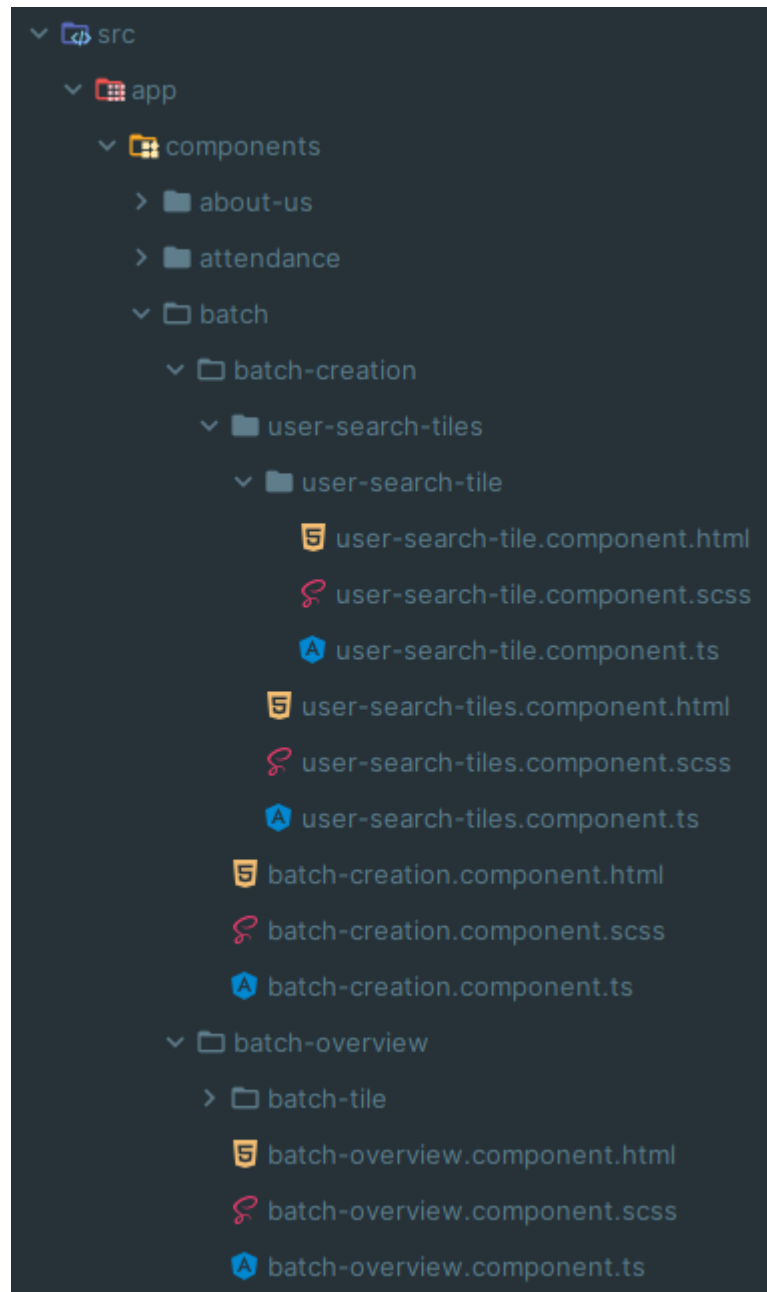


Figure 21: Class Structure

5.3.3 Services and Error Handling

Service classes are designed to retrieve data from an API and provide it to the frontend once ready. The service classes in the frontend often mirror the setup of the controller classes from the backend. Therefore, the service classes are split up depending on the endpoint and with that according to the data type as well.

The URLs are defined as static strings with the variable parameters enclosed within curly brackets. With this set up the placeholders are exchanged with the help of a designated method, so that the URL string does not have to be put together manually.

```
private _getAllActiveUserCourses :string = environment.BACKEND_URL + '/api/user/{userId}/batch';

private _getUserCourseByCourseId :string = environment.BACKEND_URL + '/api/user/{userId}/batch/{batchId}';

no usages  Alayne +1

constructor(private http: HttpClient,
  private cookieService: CookieService) { }

1+ usages  Gurkenblut +1

getAllActiveUserBatches(userId: string): Observable<UserCourseModel[]> {
  let token :string = this.cookieService.get('token');
  let headers :HttpHeaders = new HttpHeaders( headers: {
    Authorization: 'Bearer ' + token
  });
  let url :string = UrlUtils.replaceParams(this._getAllActiveUserCourses, params: {userId})
  return this.http.get<UserCourseModel[]>(url, options: {headers});
}
```

Figure 22: Service class with Get Method

Additionally, to authenticate the user a bearer token must be added to the headers, which was received during the login response, to authenticate the user in the backend.

The data is emitted as an Observable. Observables are streams of data that can be observed over time. [15]

This stream of data is accessed within the Typescript file, usually within the `ngOnInit` method, via calling `subscribe` on the service method. The `ngOnInit` method is part of Angulars Lifecycle Hook and is called once after Angular has initialized all the data-bound properties. It is used to handle any additional initialization tasks. [16] In this application the main use is to retrieve data from the backend and set additional dynamic values depending on the result.

```

ngOnInit(): void {
    this.userId = this.cookieService.get('userId');
    this.userCourseService.getAllActiveUserBatches(this.userId).subscribe( observerOrNext: {
        next: (courses : UserCourseModel[] ) :void => {
            this.userCourses = courses;
            this.displayedUserCourses = courses;
        },
        error: (error) :void => {
            if (error.status === HttpStatusCode.NotFound) {
                this.snackbar.openSnackBar( message: "Courses were not found", action: "OK", duration: 5000)
            }
            if (error.status === HttpStatusCode.BadRequest) {
                this.snackbar.openSnackBar( message: "You can only get your courses", action: "OK", duration: 5000)
            }
            if (error.status === HttpStatusCode.InternalServerError) {
                this.snackbar.openInternalServerErrorSnackbar()
            }
            else {
                this.snackbar.openGenericErrorSnackbar(error)
                console.error(error)
            }
        }
    })
}
}

```

Figure 23: ngOnInit Method within Component Typescript File

After the subscribe method the http response can take two distinct paths. If the response is without any errors, it moves through the “next” part. In this case the data received from the backend is set to two attributes in the class. Otherwise, the error path will intercept the message. In this case the code tests if the http error code which was sent with the response is one known to our application and then shows the user a snackbar with helpful information on what went wrong.

If the error code is not known to our system beforehand a generic error snackbar is shown to the user and the full error is logged into the console. In the snackbar class is the generic empty method to create an individual snackbar as well as predefined methods for known error cases such as http 500 Internal Server Error. In the case of the generic snackbar method the code first filters through a map of known error codes with associated error messages to display these for the user. If none of the error messages are known to the system, it defers to the snackbar for unknown errors, printing a generic message with the error code.

```

openSnackBar(message: string, action: string, duration?: number) :void {
    this.matSnackBar.open(message, action, config: {
        duration: duration ? duration : 5000,
        horizontalPosition: 'center',
        verticalPosition: 'top',
    });
}

1+ usages  tobias
openUncompletedFormSnackBar() :void {
    this.matSnackBar.open( message: "Warning: Some part of the form are missing or incorrectly filled in.", action: "OK", config: {
        duration: 5000,
        horizontalPosition: 'center',
        verticalPosition: 'top',
    });
}

1+ usages  Alayne +1
openInternalServerErrorSnackBar() :void {
    this.matSnackBar.open(this.genericErrorMessages[500], action: "OK", config: {
        duration: 5000,
        horizontalPosition: 'center',
        verticalPosition: 'top',
    });
}

```

Figure 24: SnackBar methods

5.3.4 Reactive Forms

Reactive forms are used to manage the user input when creating or updating a record via the frontend. Angular provides two ways to handle user input through forms: Template-driven forms and Reactive forms. The application uses reactive forms for their more robust design, reusability and scalability. Reactive forms are a model-driven approach to handling inputs and validating them directly in the input field.

A singular input field is bound to a FormControl by its designated given name. Multiple FormControls are grouped into a FormGroup. A FormGroup defines a fixed set of controls that can be managed together. Most of the forms within the application are grouped as FormGroups.

```

userForm :FormGroup< = this.formBuilder.group( controls: {
    firstName: this.formBuilder.control<string>({ formState: {value: null, disabled: false}, validatorOrOpts: [Validators.required]},
    lastName: this.formBuilder.control<string>({ formState: {value: null, disabled: false}, validatorOrOpts: [Validators.required]},
    roles: this.formBuilder.array<Role>({ controls: []}),
    school: this.formBuilder.control<string>({ formState: {value: null, disabled: false}},
    email: this.formBuilder.control<string>({ formState: {value: null, disabled: false}},
    password: this.formBuilder.control<string>({ formState: {value: null, disabled: false}},
    phoneNumber: this.formBuilder.control<string>({ formState: {value: null, disabled: false}, validatorOrOpts: [Validators.required, phoneValidator()]}),
    location: this.formBuilder.control<string>({ formState: {value: null, disabled: false}},
    date: this.formBuilder.control<string>({ formState: {value: null, disabled: true}},
    lastLogin: this.formBuilder.control<string>({ formState: {value: null, disabled: true}},
    active: this.formBuilder.control<boolean>({ formState: {value: true, disabled: false}},
    },
    options: {
        validators: PasswordValidator(),
        updateOn: "change"
    })
}

```

Figure 25: FormGroup for the user

The FormControl and FormGroup are initialized with the FormBuilder class and are given attributes such as an initial value, if they are disabled and an array of validators. The FormGroup is initialized directly within the class to allow access to the FormControl via their name. Otherwise, if initialized within a method such as the ngOnInit method the individual FormControl must be accessed by the get method via inputting the name as a string. This method is more prone to spelling mistakes.

```
setData(user: User) :void {
    this.userForm.controls.firstName.patchValue(user.firstName);
    this.userForm.controls.lastName.patchValue(user.lastName);
    this.userForm.controls.school.patchValue(user.schoolName);
    this.userForm.controls.email.patchValue(user.email);
    this.userForm.controls.phoneNumber.patchValue(user.phone);
    this.userForm.controls.password.patchValue(user.password);
    this.userForm.controls.location.patchValue(user.location);
    this.userForm.controls.date.patchValue(user.dateOfJoining);
    this.userForm.controls.lastLogin.patchValue(UserCreationComponent.formatDateTime(user.lastLogin));
    this.userForm.controls.active.patchValue(user.active);
    this.setRoles();
}
```

Figure 26: Method to set data for a FormGroup

FormGroups can contain other FormGroup to allow a nested structure and enable the option for FormGroup with dynamic amounts of data. This varying amounts of FormGroup or FormControl are contained in a structure called FormArray.

FormArrays initialize as an empty list; when a certain action is performed, they get a new FormGroup added into the array. The individual FormGroup are accessed via their index.

```
1+ usages  Alayne
addQuestion() :void {
    this.questions.push(this.newQuestion())
}

1+ usages  Alayne
newQuestion(): FormGroup {
    return this.formBuilder.group( controls: {
        id: [null],
        text: this.formBuilder.control( formState: {value: null, disabled: false}, validatorOrOpts: [Validators.required]),
        options: this.formBuilder.array( controls: [this.newOption()])
    })
}
```

Figure 27: Adding new Question FormGroup to a FormArray

These FormArrays allow the users to create dynamic amounts of questions with dynamic amounts of options for each question. When accessing deeper nested FormControl within multiple FormArrays the indexes of the preceding FormGroup are needed. Similarly, how data is accessed in a multi-dimensional array. These dynamic FormGroup can also be removed from the FormArray by giving their indexes to the removeAt method.

5.3.5 Observables and BehaviorSubjects

Data needs to be transferred between different components. With components that are connected via the HTML data can easily be transferred between the parent and child components with the keywords @Input and @Output. When data needs to be transferred between components that are not connected via HTML, the data constructs BehaviorSubject and Observable can be used.

The BehaviorSubject and Observable are situated within a service class of the same data type. BehaviorSubjects provide and maintain a “current value” to subscribers which will immediately receive the last value which was emitted by the BehaviorSubject. [17] To access the value present within the BehaviorSubject it needs to be transformed into an Observable, which can be retrieved with the subscribe method.

```
private readonly _copyCourse :BehaviorSubject<Course> = new BehaviorSubject<Course>( _value: null);  
copyCourse$ :Observable<Course> = this._copyCourse.asObservable();
```

Figure 28: Initialization of a BehaviorSubject and Observable

The data is retrieved from the service by calling the subscribe method on the observable attribute and the further processing is done akin to the subscribe methods for http service calls. To update the data present within the BehaviorSubject and Observable another set method is needed. The method is called before the router navigates to the other page and sets the data in the BehaviorSubject by calling next on it.

```
setCourse(course: Course) :void {  
    this._copyCourse.next(course);  
}
```

Figure 29: Setting Data in the BehaviorSubject

5.3.6 Screenshots

See the Appendix D: Screenshots of the frontend application for Mockups and further screenshots. This section provides a few highlights to inform the reader about what the key screens look like.

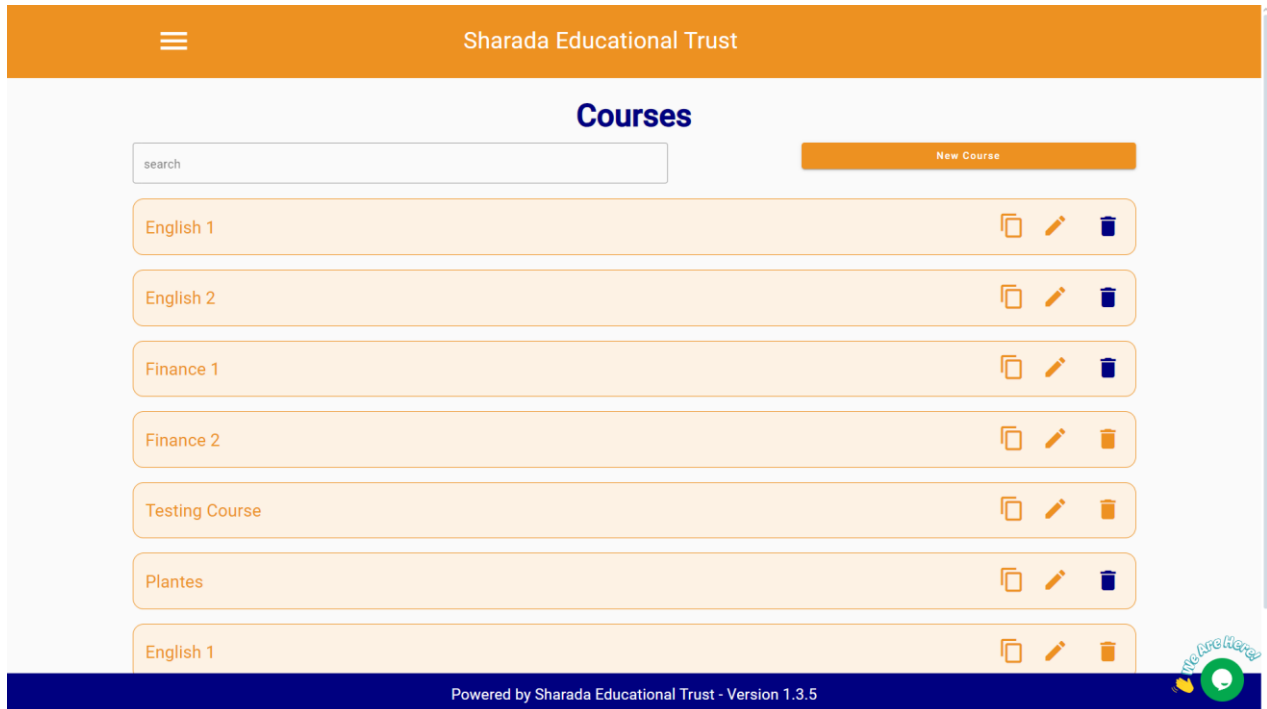


Figure 30: Course Overview Screenshot

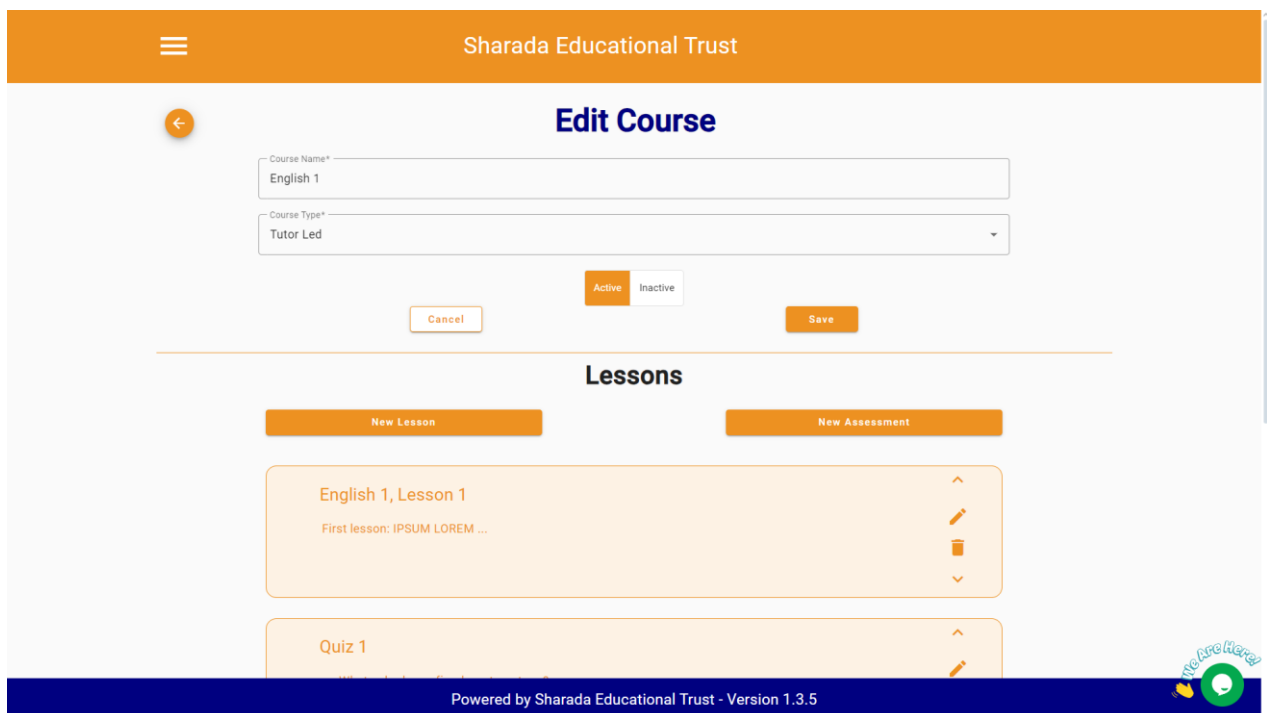


Figure 31: Course Creation Screenshot

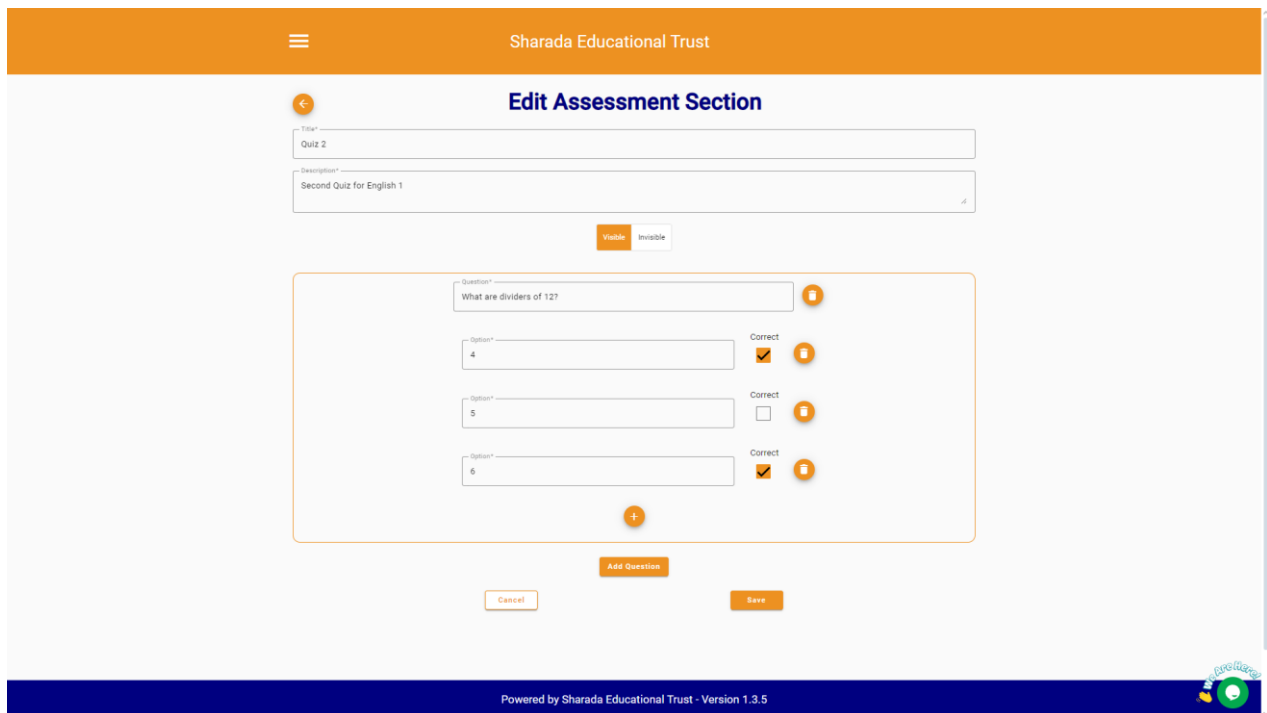


Figure 32: Assessment Creation Screenshot

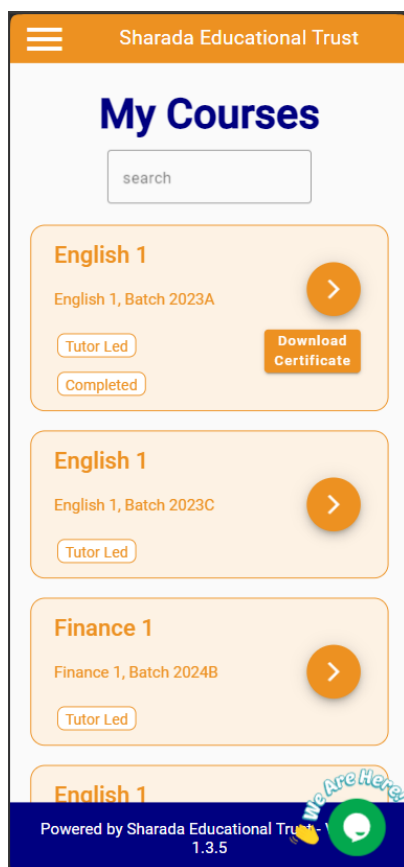


Figure 33: Course Details Screenshot

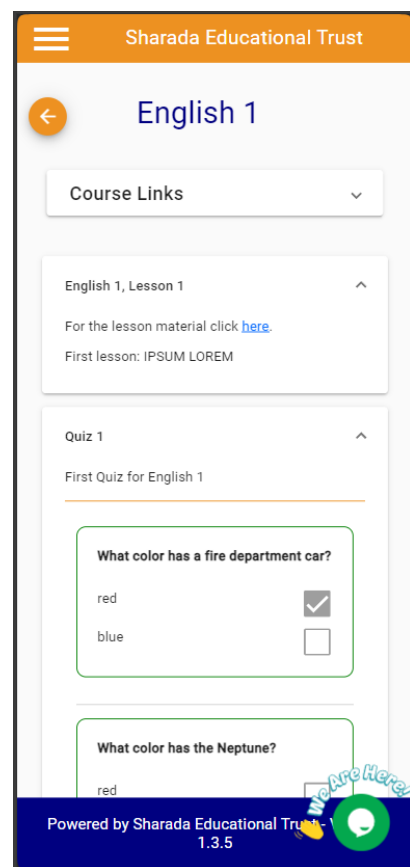


Figure 34: Student Course Overview Screenshot

5.4 Backend Implementation

The Backend is implemented with Spring Boot using Java as the programming language. For the communication between backend and database the Jakarta Persistence API (JPA) was used, which enables seamless data manipulation and retrieval; in this way, the backend can efficiently perform operations such as creating, reading, updating and deleting data in the database.

5.4.1 Framework

Spring Boot is an open-source Java framework used for production-grade Spring-based applications with minimal effort. As an extension of the Spring Java platform, it follows the convention-over-configuration paradigm. [18]

For the implementation of JPA, Hibernate is a ORM (Object-Relational Mapping) framework for Java that is often used in combination with Spring Boot. It enables developers to handle database operations with Java objects by abstracting the interaction with the database.

5.4.2 Structure

Inside the source folder, there are 2 separate application environments: test and main.

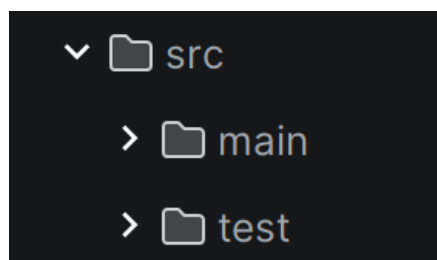


Figure 35: Environment distribution

Inside the main folder, the application logic is located. Inside the test folder all the unit and integration tests are defined.

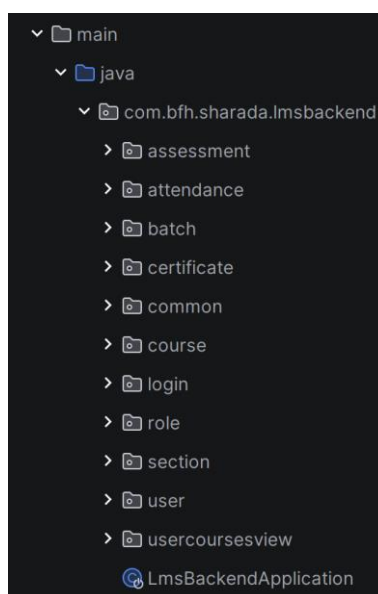


Figure 36: Java structure

Typically for Spring Boot, endpoints are arranged as folders. In this case, each folder also represents a resource needed by the frontend.

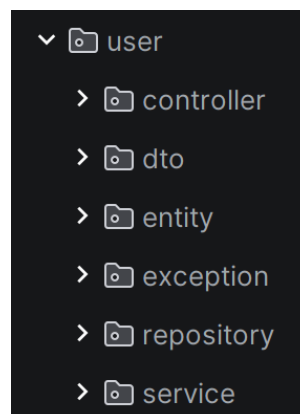


Figure 37: Endpoint components

Each endpoint consists of a variety of components. Most of the endpoints contain all of the components, but not all do. The following is a short table to describe each component with its intent:

Component	Intent
Controller	Embodies all parts which are responsible for receiving and sending HTTP messages. That means it is in charge for the REST communication.
DTO	Stands for data transfer object. It contains special objects which are only used for the REST communication. Sometimes the frontend needs data different of the form it is persisted in the backend. Therefore, DTOs are there.
Entity	This folder contains the classes of the objects which are the main business logic and backbone. Those classes specify relationships between each other. In addition, derives Hibernate the database structure from them.
Exception	If processes of handling requests need custom exceptions, they are stored here.
Repository	Inside this folder, interfaces for the communication with the database are located.
Service	A service is responsible to gather all information from other services or the repository and perform order from the controller. Business logic takes place here.

Table 42: Description of backend components

5.4.3 Update list of properties

The implementation of this topic was more complex than expected in order to increase the maintainability. This example shows the implementation of updating questions within an assessment.

```
53 public List<Question> updateAllQuestions(List<Question> questions, List<Question> questionsFromDB) throws DuplicatedQuestionIdException {
54
55     List<Question> questionsToDelete = new ArrayList<>(List.copyOf(questionsFromDB));
56     if (!questionsToDelete.isEmpty()) {
57         questionsToDelete.removeAll(questions);
58         deleteAllQuestions(questionsToDelete);
59     }
60
61     List<Question> questionsToAdd = questions.stream()
62         .filter(question -> question.getId() == null)
63         .toList();
64
65     List<Question> questionsToUpdate = questions.stream()
66         .filter(question -> question.getId() != null)
67         .toList();
68
69     Optional<UUID> duplicatedId = this.findDuplicatedId(questionsToUpdate);
70     if (duplicatedId.isPresent()) {
71         throw new DuplicatedQuestionIdException(duplicatedId.get());
72     }
73
74     List<Question> addedQuestions = addAllQuestions(questionsToAdd);
75     List<Question> updatedQuestions = questionsToUpdate.stream()
76         .map(this::updateQuestion)
77         .toList();
78
79     return Stream.concat(addedQuestions.stream(), updatedQuestions.stream()).toList();
80 }
```

Figure 38: Method to update all questions of an assessment

The method exists inside the `QuestionService.java` file. The key point about this method is that two lists of questions are needed as parameters: the questions which will be saved at the end and the ones which are already persisted in the database.

In the first block it is determined which questions are no longer needed. With overriding the equals and hash functions of the `Question` class, it is possible to remove the unwanted questions from the list which will be deleted; after that, they will be deleted.

```
32
33 @Override
34 public boolean equals(Object o) {
35     if (o == null) return false;
36     if (this == o) return true;
37     if (getClass() != o.getClass()) return false;
38     Question question = (Question) o;
39     return Objects.equals(id, question.id);
40 }
41
42 ± vogel4
43
44 @Override
45 public int hashCode() { return Objects.hash(id); }
```

Figure 39: Overridden equals and hash method

The second step of the method is to determine which are the questions to be saved. These are the new ones and the ones which will be updated. Therefore, the parameter “questions” is filtered and the results are saved to two new lists.

In order to detect errors beforehand, it needs to be checked whether an id of a question occurs two times. This may be possible when using the backend API without the frontend.

Finally, these two lists of questions must be saved, and after that the two results lists must be concatenated and returned as the result.

5.4.4 Evaluate assessments

Since the entity layer of the system does not specify the “passed” property which is seen in the API Description 4.6, it must be defined when gathering data to answer requests. This will always occur when a ResponseAssessmentDto is created.

After a new Assessment is saved, a method is called which gathers all information to build the data transfer object (DTO).

The first property inside the DTO – Layer is “answered correctly” of the ResponseAnswerOptionDto.

```
5 public record ResponseAnswerOptionDto(  
    no usages  
6     UUID optionId,  
    no usages  
7     boolean chosen,  
    1 usage  
8     boolean correctlyChosen
```

Figure 40: ResponseAnswerOptionDto

When building this object, an Option and its corresponding AnswerOption are compared. That way it is determined if the question was correctly answered or not.

The next step in the DTO – Layer is the ResponseQuestionDto with the property “answeredCorrect.”

```
8 public record ResponseQuestionDto(  
    no usages  
9     UUID questionId,  
    no usages  
10    String text,  
    no usages  
11    List<Option> options,  
    3 usages  
12    UUID answerId,  
    4 usages  
13    boolean answeredCorrect,  
    no usages  
14    List<ResponseAnswerOptionDto> answers
```

Figure 41: ResponseQuestionDto

This is defined after all AnswerOptions of the Answer were built, by going through each of them and checking if some options were not correctly chosen. The result is then set in the property.

```
7 public record ResponseAssessmentDto(  
    6 usages  
8     boolean passed,  
    11 usages  
9     List<ResponseQuestionDto> questions
```

Figure 42: ResponseAssessmentDto

In the last layer of this DTO there is the “passed” property. This is calculated with the method “userHasPassedAssesment” inside the AssessmentService.

```

122 @ private boolean userHasPassedAssessment(List<ResponseQuestionDto> responseQuestionDtos) {
123     double numberOfQuestions = responseQuestionDtos.size();
124     double amountCorrectlyAnswered = responseQuestionDtos.stream().filter(ResponseQuestionDto::answeredCorrect).count();
125     double percentageScore = amountCorrectlyAnswered / numberOfQuestions * 100;
126     return percentageScore >= MINIMAL_PERCENTAGE_TO_PASS_ASSESSMENTS;
127 }

```

Figure 43: Method which defines if a user has passed an assessment

5.4.5 Approach to clean code

In order to follow a clean code policy, much value has been set on the naming on variables. (In fact, this topic received high praise in the source code review, described elsewhere in this document.) As an example, first the number of questions is determined. Then the amount of correctly answered questions is identified. One divided by the other and multiplied by 100 will end with the percentage value of correctly answered questions. The variable “MINIMAL_PERCENTAGE_TO_PASS_ASSESSMENTS” defines the borderline how high the portion of questions must be in order to pass the assessment. It is defined inside the application.properties file.

After all these calculations, the DTO is sent as the answer body of a request.

5.4.6 Authentication and authorization

Mainly authentication and authorization are implemented with two parts. One of them is the JWTAuthorizationFilter. It is implemented as a OncePerRequestFilter. It checks whether an incoming request has an JWT-Token inside the Authorization header or not. When a token is available, it checks the validity of it.

The second part is the SecurityConfig which embodies 4.4.1 Role-Based Access Control (RBAC). There, the security guidelines for different endpoints inside the project are defined. With the use of the HttpSecurity configuration, access patterns can be linked to the corresponding permissions.

5.4.7 Casting Section from the frontend into Subclasses in the backend

Due to the class structure of sections in the backend, issues arise during data exchange with the frontend; these are explained in detail in what follows.

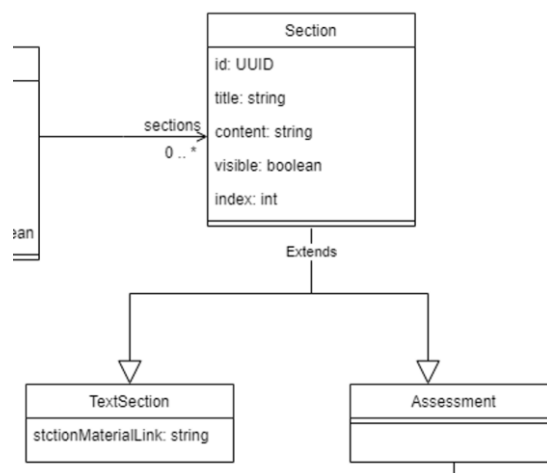


Figure 44: Class structure of sections

There are two classes which inherit from the class Section: TextSection and Assessment. When deserialization from JSON is carried out, it must be ensured that the object is converted correctly into

either a TextSection or an Assessment. This can be done with the FasterXML/jackson library and its annotations.

```
13 @JsonIgnoreProperties(ignoreUnknown = true)
14 @JsonTypeInfo(use = JsonTypeInfo.Id.NAME, property = "sectionType", visible = true)
15 @JsonSubTypes({
16     @JsonSubTypes.Type(value = TextSection.class, name = "text"),
17     @JsonSubTypes.Type(value = AssessmentSection.class, name = "assessment"))})
18 @Entity
19 @Inheritance(strategy = InheritanceType.TABLE_PER_CLASS)
20 public abstract class Section {
```

Figure 45: Annotations of the Section entity class

For example, a TextSection after conversion into JSON looks like this:

```
1 {
2   "sectionType": "text",
3   "id": "ac49c6e2-f98a-4956-941e-d9050977980f",
4   "title": "New posted Section 111",
5   "content": "This is some large content of a textSection",
6   "visible": true,
7   "index": 4,
8   "sectionMaterialLink": "www.course.material.in/1",
9   "sectionVideoLink": "www.section.video.in/1",
10  "additionalResourceLink": "www.additional.material.in/1"
11 }
```

Figure 46: TextSection after serialization into JSON

Here it is shown that when a TextSection is serialized into JSON, a field “sectionType” with the value “text” is added to the object. With this property it is much easier to display sections dynamically by the frontend. The property can also be used to correctly deserialize JSON into java objects. More precisely, when the “sectionType” value is “text” it will be converted to a TextSection and when it is “assessment”, into Assessment.

5.4.8 Handling password problems when updating users

When updating a user, problems may occur with his password. In order to circumvent these issues, two separate methods are used to update a user’s data.

```
102 @Transactional
103 @
104 public User updateUser(User user) throws
105     UserNotFoundException,
106     EmailAlreadyExistsException,
107     PhoneAlreadyExistsException,
108     PasswordRequirementsNotMetException {
109     User userFromDb = findUser(user.getId());
110
111     if (user.getEmail() != null &&
112         !userFromDb.getEmail().equals(user.getEmail()) &&
113         userRepository.existsByEmail(user.getEmail()))
114         throw new EmailAlreadyExistsException(user.getEmail());
115
116     user.setPhone(user.getPhone().replace( target: " ", replacement: ""));
117     if (!userFromDb.getPhone().equals(user.getPhone()) &&
118         userRepository.existsByPhone(user.getPhone()))
119         throw new PhoneAlreadyExistsException(user.getPhone());
120
121     user.setPassword(userFromDb.getPassword());
122     user.setDateOfJoining(userFromDb.getDateOfJoining());
123     return userRepository.save(user);
124 }
```

Figure 47: Method to update user data

The Annotation “@Transactional” is provided by the Spring Boot Project. It is necessary at this point because it makes sure that if something goes wrong, the whole context is rolled back; otherwise, partial and uncomplete changes could occur. First, data of the persisted user is fetched and compared to the data of the user parameter. When the email addresses differ from each other, a check if the new email already exists is mandatory before updating because of the uniqueness of the field. The same procedure is executed with the phone number. The key point about this method is, that the password is never updated which can be seen on line 120 where the password is set with the one from the database. Updating a user’s password happens in the other method to ensure, that it will not be changed by mistake.

The second reason why @Transactional was annotated is the following. In the login controller the lastLogin field of the user is updated. Then the updated user must be saved with the UserService. There, the user will be fetched again from the database, but it will still be the same context. However, without the annotation, it will still be the same user with the updated data. Hence, it is not possible to compare the two users. This behaviour can lead to bugs which are very hard to find and must be considered during further development.

```
126 public User changePassword(UUID id, String newPassword) throws PasswordRequirementsNotMetException {
127     User user = findUser(id);
128     user.setPassword(newPassword);
129
130     if (!meetsPasswordCriteria(user))
131         throw new PasswordRequirementsNotMetException();
132
133     user.setPassword(encoder.encode(newPassword));
134
135     return userRepository.save(user);
136 }
```

Figure 48: Method to update passwords

After fetching data of the persisted user with his id, the new password is set without encoding. Hence, the method which checks if the password meets the criteria to be strong enough can receive the user object as parameter. After the check, the password gets encoded and set again. Finally, the user with the updated password is saved directly calling the save method of the repository.

6 Software Quality

This Chapter describes how the quality of the product is ensured: first, how the quality of the code is ensured, and then what the concept for testing is and finally, what the results are.

6.1 Code Quality

This Section addresses how the quality of the code is enhanced, mainly via code reviews carried out by the team.

6.1.1 Code Reviews inside the development team

As outlined in the section 2.1.1 Agile development for the measure APM091 the code quality was enhanced by carrying out code reviews. In the project team (TA02) every week a code review meeting was planned to cover whenever someone had a question, an interesting thing someone learned, or whenever some more complicated code was encountered.

Here is an example from the code review from 18.04.2024. The function in question finds out whether a student has completed and therefore passed a course. The first code block shows the code before the review.

```
public boolean hasCompleted(Batch batch, UUID userId) {
    int amountOfAllSections = batch.getCourse().getSections().size();
    List<Section> sections = getAvailableSections(batch, userId);
    int amountOfAvailableSections = sections.size();
    if (amountOfAllSections != amountOfAvailableSections)
        return false;

    boolean hasFailedAnyAssessment = sections.stream()
        .filter(section -> section instanceof AssessmentSection)
        .map(section -> assessmentService.getAssessment(batch.getId(),
                                                    section.getId(), userId))
        .anyMatch(responseAssessmentDto -> !responseAssessmentDto.passed());
    return hasFailedAnyAssessment;
}
```

The next code block shows the code after the code review. Besides that the function now works (the last line in the block should be returning `!hasFailedAnyAssessment`;) it is now much more readable, does not contain a not in the return line and is a bit shorter.

```
public boolean hasCompleted(Batch batch, UUID userId) {
    List<ResponseAssessmentDto> assessments = batch.getCourse().getSections()
        .stream()
        .filter(section -> section instanceof AssessmentSection)
        .map(section -> assessmentService.getAssessment(batch.getId(),
                                                    section.getId(), userId))
        .toList();

    boolean courseHasAssessments = !assessments.isEmpty();
    boolean studentHasPassedAllAssessments = assessments.stream()
        .allMatch(ResponseAssessmentDto::passed);
    return courseHasAssessments && studentHasPassedAllAssessments;
}
```

The whole team strongly agreed that these code reviews contributed greatly to the quality of the code and the overall product.

6.1.2 External Code Review on 06.06.2024

An external code review was done at the end of the project. The review was carried out by Mr. Hugues Clavadetscher, a software architect working in the BFH CyberLab team who has experience from real-world projects with Angular and Spring Boot. This section outlines the main takeaways and feedback from the external code review. The protocol of the code review can be found in the Appendix E: Meeting protocols.

In preparation for the code review the external reviewer was added to the GitLab repository group and he had a look at the project and noted what could be improved. During the code review he presented the points demonstrated them in the GitLab and clarifying questions from both sides were posed and answered. The following list shows the most important takeaway points from the core review:

- ▶ It would be nice to (re-)visit the concepts and tools: DDD, SOLID, LIFT, nx/store and Cypress.io.
- ▶ The frontend could be improved by grouping the components that build a unit into a module. This is to improve reusability and the overall readability of the structure.
- ▶ It would be nice to separate the services and the front-end interaction in the form of the Snackbar. A solution would be to put a logger interface between the two.
- ▶ It would be nice to not have the static data (that will be added on every new setup of a data source) in a Spring runner instead of in a SQL file.
- ▶ It was noted that the backend was very well and comprehensively tested with automated tests.
- ▶ Next time it would be nice to have certain endpoints with the /admin prefix to separate the functionalities or permission to execute an endpoint. i.e. a user can get their own user only and an admin can get all users. So, the GET /user/{id} would stay for the user and become GET /admin/user/{id} if you want to get any user.
- ▶ Have some speaking minimal README.MD not just the generated.

The reviewer was quite confident the project was solid and well-built, and he described it with the following feedback:

- ▶ Good project
- ▶ Good work
- ▶ Easy to understand
- ▶ Good collaboration - we can see the integration, naming is good, mapping between frontend and backend
- ▶ Would be eager to have the students join his ongoing projects

6.2 Test Concept

This section contains the test concept heavily relying on the 3 Requirements and 3.2 Use Cases. It shows the objectives of the testing and defines how to go about them in ensuring a good quality product at the end.

As a guide to understanding how the different types of tests (described below) contribute to checking the quality of the requirements-collection, design and build process, the V-model is commonly used, as shown in the following diagram.

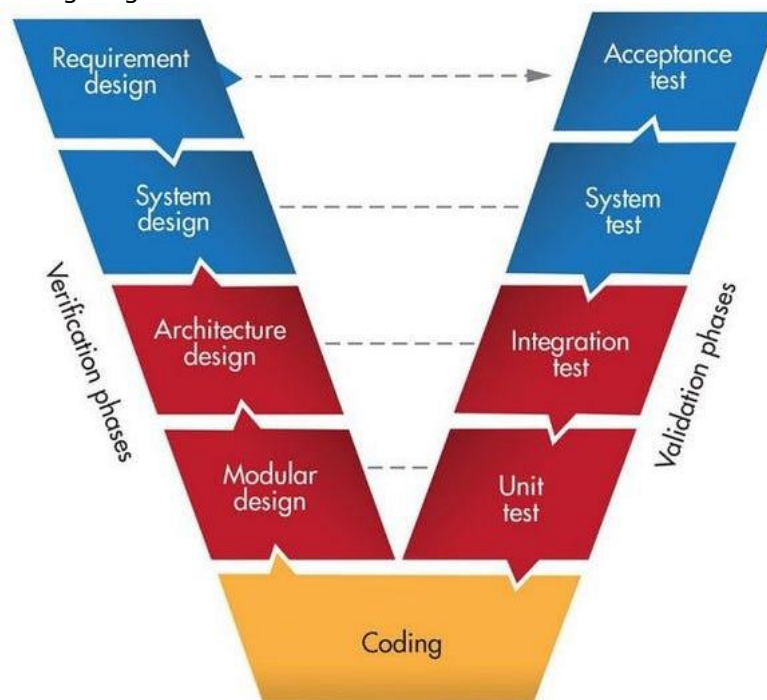


Figure 49: V-model [19]

6.2.1 Test objectives

The objective of testing in this project is to ensure that the functionality is met and working as planned. By extension, it is also used to show that the 3.2 Functional Requirements are met by showing all the related use cases are working. Therefore, in the end it is used to prove the success of one part of the project.

6.2.2 Testing Scope including Regression Testing

It is important to carry out regression testing of all the functionalities implemented in the previous project as well as new tests of the new features in this project. Such regression testing is essential because the testing in the previous project was only carried out by the developers and as an additional means to ensure that nothing broke during this project.

The test concept does not provide for usability testing nor for performance testing. Although dedicated usability tests are not practical because of the time consumption required, nevertheless feedback on usability was gathered, mostly from our Project Partner (PR06). Similarly, performance testing is out of scope because the user base is not expected to be in the range (100's of users) where performance bottlenecks would be visible, and therefore for the Bachelor work the performance is not an issue. The time gained by not testing usability and performance will be invested in bringing value to the product through additional features.

6.2.3 Testing Methods

The testing methods are mostly split into two parts: automated backend tests and manual user tests. The following table shows the different test strategies that are used.

Id	Name	Description
TM01	Unit tests	Unit tests are used to ensure that smaller contained units are working. This includes: ▶ the by us defined functions of repositories ▶ the service classes containing the business logic ▶ helper classes like not simple converter, validators and scheduled classes
TM02	Integration Tests	The integration tests are used to test the full backend via the APIs. This includes: ▶ the rest controllers ▶ indirect all parts of the backend that are called via the controllers
TM03	Explorative user tests	Explorative tests are used to find bugs and problems by the users not directly related to the requirements. Thies tests will mostly be done by our project partner (PR06). Thies tests are close to and the substitute to the system tests from the V-model.
TM04	Guided user tests	The guided tests are used to verify that the requirements are met. These tests are executed by various potential users or just normal testers with a protocol. This tests will be used in the final test run as a substitute for the acceptance tests from the V-model.

Table 43: Test strategies

6.2.4 Test Scenarios

A test scenario combines testing with a story. It groups different test cases together to help the tester realize the workflow behind the different tests. This does not necessarily mean that the tester takes the role of one person. In order to keep the tests contained and flowing a tester must switch between roles. The following table shows the different test scenarios.

Id	Name	Story	Test Cases
TS01	Get a student on a course	We want to offer a new course to a new student. To get there we need to set up the user and course with the content, which work independently. Then we can combine them in a batch. Then the user can see their new course and start working on it.	TC01, TC02, TC03, TC04
TS02	Assess the progress of a student	We want to check whether the user learned something. If we don't already have one, we need an assessment. Once we have one, the students can fill it out. Additionally, the student's availability has to be tracked via the attendance management.	TC05, TC06, TC07, TC08

Id	Name	Story	Test Cases
TS03	Clarify a question	The student has a question about the last session and needs some help from their admin. So, the student starts a chat with the tutor and the admin answers to it at the best of their knowledge.	TC09, TC10
TS04	Completing a course	The student wants to complete the course they have started. They have to meet all the requirements, before the student can get their certificate.	TC11, TC12
TS05	Explorative testing	The project partner (PR06) wants to get their hands on the product and test it, to see whether it fits their imagination by learning to use the tool by trial and error.	TC13
TS06	Automated Tests	The developer wants to see whether they broke something during development and whether the new features work on a technical level. The developer executes the automated test to find problems and fix it before they merge that.	TC14

Table 44: Test scenarios

6.2.5 Test Cases

The test cases are mostly derived from the 3.2 Use Cases. The test cases for automated testing (TS05) relying on TM01 and TM02 are given their own test cases, not directly derived from the use cases but from the non-functional requirement “the system is well tested” (NF01). The test cases can be found in the Appendix B: Test Cases.

The description of the test cases is kept to an instruction level with which a user could work. This is done to test not only the functionality but also to check the usability. The following tables shows the relationship between the test cases and use cases they test.

Test Case	Description	Related Use Case	Related Requirement
TC01	Onboard a new student and tutor.	UC01, UC02	FR01
TC02	Adding a new course	UC03, UC04	FR02
TC03	Combining the new users with the new course	UC05, UC06	FR03
TC04	Viewing new course	UC07, UC08	FR04, FR05
TC05	Add an assessment to the new course	UC04, UC10	FR02, SR01
TC06	Fill out the new assessment	UC11	SR02, SR04
TC07	Retake an assessment	UC12	SR03
TC08	Mark the student as attending	UC09	FR05
TC09	Asking a question	-	
TC10	Answering a question	-	
TC11	Complete a course	UC13	SR05
TC12	Generate a certificate	UC14	SR06

Test Case	Description	Related Use Case	Related Requirement
TC13	Free field testing in the application	-	
TC14	Automated testing with unit and integration test	-	

Table 45: Use cases link to test case

6.2.6 Test Data

For all the guided user tests (TM03) in the scenarios TS01 to TS04 the data should be created by the tester. The only pre-requisite for this is an admin account. The data created in the first tests is reused in the later tests to simulate one big story. The data entered by the tester can largely be chosen by them to have a wide range of tested data. And user/testers tend to come up with the best data to break a system even though it should work.

For the automated tests, static data is used that is always reloaded into the in-memory database during execution. This test data is a minimal set of data designed specifically for automated tests and does not need to be real-world related.

6.2.7 Test Environment

The test server will be hosted on the same containers as the productive version. For more information see 4.3 System Architecture.

To have a functional test environment for a tester the following preconditions must be present:

- ▶ A connection to the test server
- ▶ A browser to access the tool
- ▶ The test handbook

6.2.8 Defect Management

Defect management describes how to handle a software defect if found during the testing. After every release of a new feature there is some time planned to fix bugs that were found; the prioritization of the defect fixes is the responsibility of the Project Partner (PR06), and ensuring they are carried out is the responsibility of the Head of Quality Control (PR04). Every software defect not found during development and prior to testing has to go through the following stages.

Id	Description	Responsible
DMS01	Identify and describe the bug.	Finder
DMS02	Prioritize the bug.	Head of Quality Control (PR04)., Project Partner (PR06)
DMS03	Reproduce the bug. Search and agree on a solution	Head of Quality Control (PR04).
DMS04	Allocating resources and person to solve it.	Head of Quality Control (PR04).
DMS05	Implement the solution.	Solver
DMS06	Deploy to relevant test environment	Head of Quality Control (PR04).
DMS06	Retesting and validate	Finder

Table 46: Defect management stages

Although tools such as GitLab can be used to track software defects, these were too unwieldy for this project, especially with the key business stakeholder in India and not with easy access to the GitLab

system. Therefore, this process was managed using a shared Excel file where the whole project team (TA01) has access. There it is also possible to track the status of the defect resolution.

Testing Feedback - SET Mobile App												
Arvind Kamath												
S.no.	Date	Finder	Feature / Menu / Screen	Details	Comments / Feedback / Issue / Bug	Priority (H/M/L)	Notes/Solution	Assigned to (Name)	Implementation	Retesting feedback	Status	
171	167	9/5	Arvind	User id	Email id Test	High		Tobi				-
172	168	7/5	Arvind	Course end	Certificate template	Closed						-
173	169	7/5	Arvind	Course	Additional Links to each Lesson							-
174	170	7/5	Arvind	Users	User data upload							-
175	171	9/5	Arvind	Course	Assessment Quiz related		Can not reproduce it					-
176	172	9/5	Arvind	Course	Certificate Alignment	Medium	Won't be done in this project. The problem is that the course name is too long					-
177	173	9/5	Arvind	About Us	Screen			Tobi	Changed		Closed	-
178	174	11/5	Arvind	Course	After Login		Can't reproduce the source of the problem		An error will be thrown if the source of the problem is to arise again		Closed	-

Figure 50: Test feedback sheet

Special note: at the conclusion of the project, the test feedback sheet contained nearly 200 separate entries. This shows the high degree to which the testers provided written feedback.

6.2.9 Defect Classification Levels and Reports

The defect classification levels and reports only apply to the final test runs. For this the tester will fill out whether the test succeeded and if not, what went wrong. All the summarized results can be found in the section 6.3.4 Final Test run.

The classification levels for evaluating the software defects are severity and frequency. In most software projects there is one set of concerns, namely the concerns of the business stakeholder. In this project there is an additional set of concerns, namely the concerns of the Bachelor thesis. This is important to note, because a “showstopper” defect from the business stakeholder perspective may not necessarily be a “showstopper” defect from the Bachelor thesis perspective. The following table shows the levels adopted in this project.

Id	Name	Description
TRM01	Very High	A showstopper that prohibits the application from being used in production and/or prevents completion of the Bachelor thesis. When a mature feature consistently is not working correctly. No workaround available.
TRM02	High	A showstopper for production, but not a showstopper for the Bachelor thesis. When a mature feature rarely works correctly, or one part of a mature feature or of a minor feature is consistently not working correctly. No workarounds available.
TRM03	Medium	When one part of a mature feature or a minor feature is rarely not working correctly. No workarounds available, but the application is still usable.
TRM04	Low	If there is a suitable workaround for the defect.
TRM05	Very Low	Some text labels are not perfectly labelled, or elements are not correctly placed. Will only be remedied if they can be done so quickly.
TRM06	Additional	Some things that could be added to improve the functionalities. Will be recorded but not acted upon.

Table 47: Testing evaluation priorities

6.3 Test Results

In this Section all results from the tests are summarized. This includes the field tests carried out at a government school in the rural Indian village of Cheeluru, the backends automated tests and the final test run.

6.3.1 Field-Testing

On the 15th of May 2024, a field test was performed at a rural government school in Cheeluru. After a ceremonial and warm welcome, the following was executed:

1. Explanation of the main components of the app and how they work together.
2. Answering questions and uncertainties about the app.
3. Guiding teachers and students through the app and the provided course materials.
4. Gather feedback from the users.

The following table shows the issues that were found. The protocol for the field-testing can be found in Appendix E: Meeting Protocols.

Id	Name	Description
I01	Students must go through the lesson before taking the assessment.	Teachers want that there is an obstacle for users who go directly to the assessment without reading the lesson material. Some sort of check, whether the material was read has to be implemented.
I02	Assessment must only be passed when answered the questions 100% correctly.	Certificates could be downloaded without answering all questions right. This is by design because of a misunderstanding. The assessment is passed, when 80% of the questions were answered correctly. Change it to 100%
I03	Server error when fetching user course views.	With some users, a server error occurred when they logged in and were redirected to the course overview, without seeing their courses. Debug and fix the bug.
I04	Minimal OS version of devices used	The question was raised whether we know what the minimum OS system of the mobile device is. With an Android 7 Device, the app worked with some problems in displaying content. Check with different OS versions of Android and Apple whether the app works correctly. (Also with the platform where the lesson material is accessed)
I06	Optimizing the resource materials for mobile devices	As the students are using mobile devices to use the app, it is crucial that the resource material is also optimized for this format

Table 48: Issues gathered during field test

As a supplement here are some impressions of the field test:



Figure 51: Field tests: explaining the app to the audience



Figure 52: Field tests: the audience

Special note: the field testing took around four hours, during which time the students and teachers participated actively, asked questions, and also gave insight into how the application would be used. It was amazing to see the concentration and attention given by the children during this time frame. Based on the behavior of European children in these age groups, this dedication was highly unexpected but was a critical success factor for the tests.

6.3.2 Explorative Tests

The explorative tests described in this Section are based on the test scenario Explorative Tests (TS05) and cover the test case TC13. A working product was available at the beginning of the project because of the previous project [4]. Therefore, the exploration tests started with the project and ended with the visit to India. They were then followed by with highly structured acceptance tests.

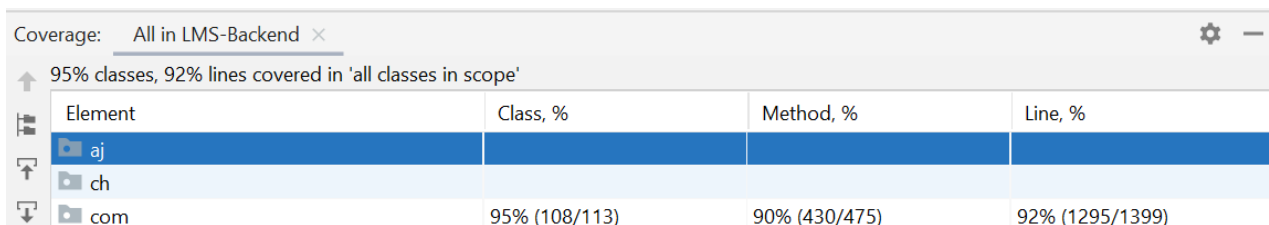
The results of the explorative tests were reported on a separate sheet. Everything from bugs, questions, ideas for new features came together. The list was later split in two: the newly found issues and the new features. The summarized and ordered lists are listed in the 7.4 Next Steps.

Out of all the 184 reported issues 179 have been resolved or deemed new features. This was a lot of feedback that was essential to making the app work as good as possible. Of all the 11 new features that were identified two were implemented in the project phase Feature 3: Smaller functionalities (PP04).

The number of new features shows that there is still an appetite for further development. The features were prioritized with the project partner (PR06). This way it was possible to make have so few open issues. Not all issues were of the same size and the biggest were made into new features. Nevertheless only 2.7% of all the reported issues are still open and none of them are showstoppers. This is a very good rate in general but especially considering the degree of feedback obtained.

6.3.3 Automated Tests

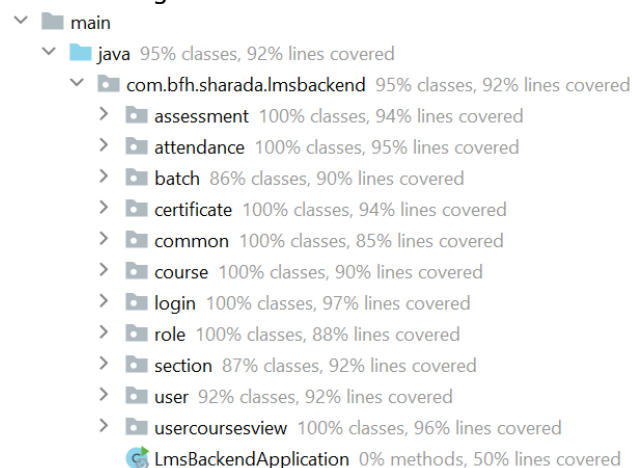
The automated tests described in this Section are based on the test scenario Automated Tests (TS06) and represented by the test case TC14 in the test concept. They include unit tests as well as automated integration tests. The overall coverage of the code is 1295 lines out of 1399 which comes to an 92% coverage. By industry standards this is a very good value.



Coverage: All in LMS-Backend ×			
↑ 95% classes, 92% lines covered in 'all classes in scope'			
Element	Class, %	Method, %	Line, %
aj			
ch			
com	95% (108/113)	90% (430/475)	92% (1295/1399)

Figure 53: Automated tests: overview test coverage

The detailed coverage looks like in the following figure. The missing percentages are mostly due to containing different system and configuration classes.



main	
java	95% classes, 92% lines covered
com.bfh.sharada.lmsbackend	95% classes, 92% lines covered
assessment	100% classes, 94% lines covered
attendance	100% classes, 95% lines covered
batch	86% classes, 90% lines covered
certificate	100% classes, 94% lines covered
common	100% classes, 85% lines covered
course	100% classes, 90% lines covered
login	100% classes, 97% lines covered
role	100% classes, 88% lines covered
section	87% classes, 92% lines covered
user	92% classes, 92% lines covered
usercoursesview	100% classes, 96% lines covered
LmsBackendApplication	0% methods, 50% lines covered

Figure 54: Automated tests: detailed test coverage

6.3.3.1 Local test run

The next figures show an execution of all the automated tests on a local machine with all the test classes executed. The classes ending on “Test” are unit tests and all those ending with “IT” are integration tests.

✓ lmsbackend (com.bfh.sharada) 35 sec 856 ms		
> ✓ AttendanceServiceTest	6 sec 481 ms	
> ✓ UserControllerQueryingIT	1 sec 286 ms	
> ✓ RoleServiceQueriesTest	17 ms	
> ✓ SectionControllerIT	370 ms	
> ✓ OptionServicePersistenceTest	93 ms	
> ✓ QuestionServiceQueryTest	12 ms	
> ✓ AnswerServiceQueryTest	38 ms	
> ✓ AttendanceControllerIT	1 sec 960 ms	
> ✓ ScheduledDateControllerIT	127 ms	
> ✓ AnswerRepositoryTest	31 ms	
> ✓ CourseServiceQueriesTest	30 ms	
> ✓ UserServicePersistenceTest	1 sec 158 ms	
> ✓ CertificateServiceQueryTest	1 sec 486 ms	
> ✓ CertificateControllerIT	1 sec 521 ms	
> ✓ UserCourseViewControllerIT	857 ms	
> ✓ BatchControllerIT	4 sec 347 ms	
> ✓ UserCourseViewQueryTest	324 ms	
> ✓ UserControllerManipulationIT	1 sec 908 ms	
> ✓ AssessmentServicePersistenceTest	269 ms	
> ✓ AnswerOptionServicePersistenceTest	41 ms	
> ✓ ImportServiceTest	394 ms	
> ✓ UserRepositoryTest	11 ms	

Figure 55: Automated tests: test classes 1

> ✓ SectionServicePersistenceTest	447 ms
> ✓ CourseTypeControllerIT	91 ms
> ✓ UserServiceQueriesTest	3 sec 314 ms
> ✓ AnswerOptionServiceQueryTest	9 ms
> ✓ AssessmentControllerIT	914 ms
> ✓ CourseTypeServiceTest	10 ms
> ✓ BatchRepositoryTest	24 ms
> ✓ BatchServicePersistenceTest	182 ms
> ✓ OptionServiceQueryTest	13 ms
> ✓ CurrentUserDetailsServiceTest	17 ms
> ✓ AttendanceCreatorTest	98 ms
> ✓ LoginControllerTest	182 ms
> ✓ AttendanceRepositoryTest	2 ms
> ✓ CourseServicePersistenceTest	83 ms
> ✓ QuestionServicePersistenceTest	66 ms
> ✓ AuthorizationServiceTest	6 ms
> ✓ AnswerServicePersistenceTest	2 sec 886 ms
> ✓ BatchStarterTest	49 ms
> ✓ AssessmentServiceQueryTest	51 ms
> ✓ BatchServiceQueriesTest	20 ms
> ✓ CourseControllerIT	4 sec 629 ms
> ✓ RoleRepositoryTest	2 ms

Figure 56: Automated tests: test classes 2

6.3.3.2 Automated execution gitlab runner

Whenever a new commit was made on GitLab the security scan and the tests were automatically execute. The following figure shows the last execution from the merge to the master branch.

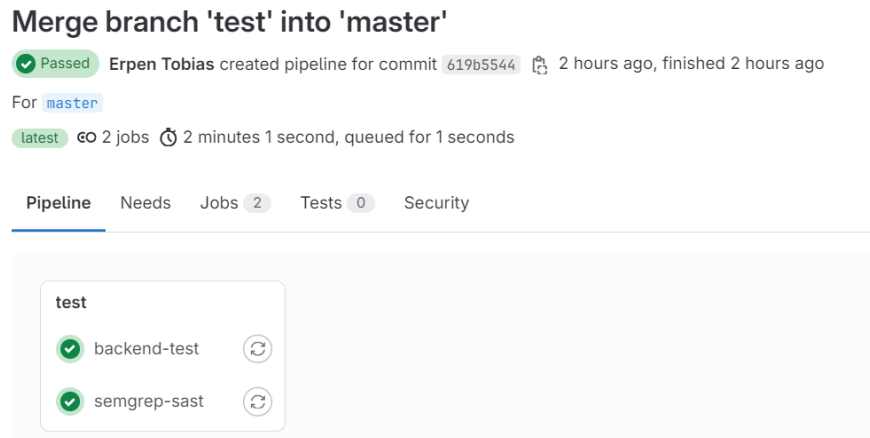


Figure 57: Atomated tests: pipeline backend

The next two figures show the results of the unit and integration tests for the last pipeline run describe both.

```
1119 [INFO] Results:
1120 [INFO]
1121 [INFO] Tests run: 184, Failures: 0, Errors: 0, Skipped: 0
1122 [INFO]
```

Figure 58: Automated tests: unit test results

```
1646 [INFO] Results:
1647 [INFO]
1648 [INFO] Tests run: 54, Failures: 0, Errors: 0, Skipped: 0
1649 [INFO]
1650 [INFO]
1651 [INFO] --- maven-failsafe-plugin:3.1.2:verify (default) @ LMS-Backend ---
1652 [INFO] -----
1653 [INFO] BUILD SUCCESS
1654 [INFO] -----
1655 [INFO] Total time: 01:52 min
1656 [INFO] Finished at: 2024-06-03T07:53:13Z
```

Figure 59: Automated tests: integration test results

6.3.4 Final Test run (acceptance tests)

This section summarises the results of the final test run and lists the feedback. Announcing the acceptance early was a critical success factor to keep the team focused on the final delivery, after which the main software development process would be stopped. The complete feedback from the final test runs can be found in the Appendix F: Test results. The final test runs were held by the project manager (PR03), thesis advisor (PR05) and the project partner (PR06). This was done to have project affiliated tester but still backbox tests to prevent code knowledge from muddying the test results to a positive result.

All testers got a test manual including a description of the test environment, the test scenario, all the testcases relevant for them and a standardised table to note their results. The test manual was derived from the test concept and contained mostly the same content. The following table shows the success rate of the different test cases:

Test Scenario	Test Case	Passed	Not Passed	Passed [%]
TS01	TC01	3	0	100
	TC02	3	0	100
	TC03	3	0	100
	TC04	3	0	100
TS02	TC05	3	0	100
	TC06	3	0	100
	TC07	3	0	100
	TC08	3	0	100
TS03	TC09	3	0	100
	TC10	3	0	100
TS04	TC11	3	0	100
	TC12	3	0	100
TS06	TC14	1	0	100

Table 49: Final test run results

The final test run was a success. All test cases but TC13 were successfully at its core. Some minor feedback was given and listed below. For TC13 this was the explorative testing and therefore not included in the final test run. With that are all relevant tests successfully.

Id	Test Case	Feedback	Evaluation
FB01	TC01	Created a new User – Anish Kamath without a Role. In the edit mode, tried mapping the User to Student role and revised the password to 4 digits. Getting an error message on saving “Some data send to the server is not correct” Screen shot attached below.	TRM03 (Medium) Does not occur often.
FB02	TC02	Edit button to be disabled for Courses as well as the lessons within the Course, which have been mapped to a batch already. Like in case of Batches, an eye icon should be applicable for Courses that cannot be edited. Currently getting a error message “Some data send to the server is not correct” instead of “Course is not editable” error message. Screen shot attached below.	TRM06 (Additional)
FB03	TC02	The other icons have hover-over text “Copy Course” and “Edit Course” so probably it would be better to have “Delete Course” rather than “Deletes Course” – more harmonious that way.	TRM05 (Very Low)
FB04	TC05	Question: Was not able to edit basic information about a course in progress.	TRM06 (Additional) Is intended

Id	Test Case	Feedback	Evaluation
FB05	TC07	After I take an assessment in the role of student, the assessment window closes up. As you can see (see Appendix F: Test results), Assessment 1 – History is closed. This is not what I would call a bug, but it is just awkward because the student needs to re-open this panel in order to see whether the assessment was successful or not. Because it is not a bug, I would not recommend fixing it	TRM05 (Very Low)
FB06	TC12	I then tried a course with a really long name, and the certificate looked like this. As you can see, the text is not formatted so nicely.	TRM04 (Low)

Table 50: Final test run feedback

The results look good and the most important ones were listed in the known bug's section.

7 Summary

This Chapter summarizes the achievements and shortcomings of the project. It also compares which goals were reached and which were not, including reasons why they were not reached. And in the end lists the project reviews and outlooks.

It must be mentioned at the outset, however, that the ultimate, overarching goal of the thesis was achieved, namely, that a tailor-made application was developed with high quality and put into production, and that the Project Partner (PR06) was able to carry out a soft launch of the application for real users, with real content!

7.1 Target/Actual Comparison

This Section briefly summarizes the key project management aspects, including the milestones, requirements, and working hours (effort).

7.1.1 Milestones

The following table lists all the milestones that were defined in the project planning. It shows whether and when we achieved them. The target was mostly set to the Thursday on the corresponding week of the milestone because this is the Development Teams (TA02) team workday. The delivery target for testing feedback was set to Wednesdays to discuss the feedback in the Project Team (TA01) on Thursday.

Id	Name	Target	Actual
MS01	Contract signed	15.03.2024	18.05.2024
MS02	Project Initialization finalized	22.03.2024	22.03.2024
MS03	Feature 1: Assessment online	11.04.2024	08.04.2024
MS04	Feature 2: Certification online	25.04.2024	15.04.2024
MS05	Feature 3: Smaller functionalities online	09.05.2024	06.05.2024
MS06	Testing feedback Feature 1 arrived	17.04.2024	11.04.2024
MS07	Testing feedback Feature 2 arrived	01.05.2024	18.04.2024
MS08	Testing feedback Feature 3 arrived	16.05.2024	09.04.2024
MS09	Testing complete	10.06.2024	03.06.2024
MS10	Documents final version delivered	13.06.2024	12.06.2024
MS11	Presentation held	14.06.2024	
MS12	Defence held	27.06.2024	
MS13	India visited	19.05.2024	19.05.2024

Table 51: Target/actual comparison milestones

The team was able to complete most milestones earlier than expected because Monday became the best deployment day due to the working days of the Project Team (TA01) members. Another reason was the very high willingness to test by our Project Partner (PR06) as well as the quality of the tests and their documentation.

At the time of this writing, the dates for the MS11 and MS12 are not reached, and they were not done by the finalization of the documentation. That is why they have no actual date. The MS13 is in last place because it was added later in the project. Once the travel dates were finalized and funding was secured, it was included. Thereafter everything went according to plan.

7.1.2 Requirements

This section lists all requirements described in the chapter 3 Requirements and whether and how they were achieved.

Overall, all requirements were achieved. Most of them are achieved as described in the chapter 3 but for some adjustments had to be made like for SR31, where the video length was extended with the blessing of the person responsible for the thesis module.

7.1.2.1 Functional Requirements

This section shows that the functional requirements were achieved that were posed in the beginning of the project and described in 3.2 Functional Requirements, with the results of the final test run as confirmation. If the tester (including the project partner (PR06) and thesis advisor (PR05)) deemed the test case successful it was served as evidence that the requirement was achieved.

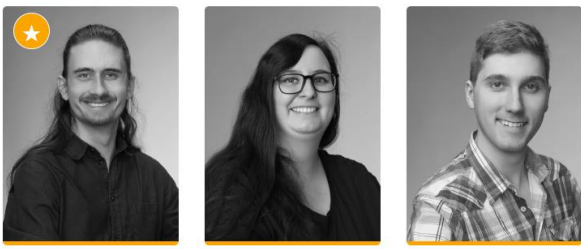
Id	Achieved	Prove
FR01	Yes	TC01
FR02	Yes	TC02
FR03	Yes	TC03
FR04	Yes	TC04
FR05	Yes	TC04, TC08
FR06	Yes	All sub requirements (SR01 to SR04) were achieved.
SR01	Yes	TC05
SR02	Yes	TC06
SR03	Yes	TC07
SR04	Yes	TC06
FR07	Yes	All sub requirements (SR05 and SR06) were achieved.
SR05	Yes	TC11
SR06	Yes	TC12

Table 52: Target/actual comparison functional requirements

7.1.2.2 Non-Functional Requirements

This section shows that the non-functional requirements were achieved that were posed in the beginning of the project and describen in 3.3 Non-Functional Requirements.

Id	Achieved	Prove
NR01	Yes.	All sub requirements (SR08 and SR13) were achieved.
SR08	Yes	Not all functions but functionalities are tested with unit tests. See 6.3.3 Automated Tests.
SR09	Yes	All endpoints of the backend are tested. See 6.3.3 Automated Tests.
SR10	Yes	See the results of the final 6.3.4 Final Test run.
SR11	Yes	See the section 6.3.1 Field-Testing.

Id	Achieved	Prove
SR12	Yes	See the section 6.3.4 Final Test run.
SR13	Yes	See the section 6.3.2 Explorative Tests and 7.4 Next Steps for the details of the known issues and possible new features.
NR02	Yes	All sub requirements (SRSR14 and SR15) were achieved.
SR14	Yes	See the section 7.1.1 Milestones for the exact dates.
SR15	Yes	See this was part of the testing and completed with the milestone Testing completed MS09 for the date see 7.1.1 Milestones.
NR03	Yes	All sub requirements (SR16 to SR23) were achieved.
SR16	Yes	See the Management Summary and the Chapter 1 Introduction.
SR17	Yes	See the chapter 2 Organization.
SR18	Yes	See the chapter 3 Requirements
SR19	Yes	See the chapters 4 Architecture and 5 Implementation.
SR20	Yes	See the chapter 4 Architecture.
SR21	Yes	See the chapter 5 Implementation.
SR22	Yes	See the chapter 6 Software Quality.
SR23	Yes	See the section 2.4 Time Estimate and chapter 7 Summary.
NR04	Yes	All sub requirements (SR24 and SR25) were achieved.
SR24	Yes	See the Appendix G: User Manual.
SR25	Yes.	See the Appendix G: User Manual.
NR05	Yes	All sub requirements (SR26 and SR27) were achieved.
SR26	Yes	See the Appendix G: User Manual.
SR27	Yes.	See the Appendix G: User Manual.
NR06	Yes	All sub requirements (SR28 to SR30) were achieved.
SR28	Yes	The Management Summary was used for this.
SR29	Yes	Arbeitsgruppe Student*innen  Tobias Erpen Alayne Larissa Lukas Vogel
SR30	Yes	The book entry was created and turned in via the BFH BookTool [20].
NR07	Yes	Tobias Erpen: ENDGÜLTIGE FREIGABE 31.05.2024 - 08:42 Silvia Gasenzer: ABSTRACT BESTÄTIGEN 31.05.2024 - 07:59 Tobias Erpen: AN SEKRETARIAT WEITERLEITEN 30.05.2024 - 19:34 Prof. Dr. Kenneth Ritley Please see the feedback I sent via email. 23.05.2024 - 12:29 Tobias Erpen: AN BETREUER*IN WEITERLEITEN 23.05.2024 - 11:40 START 18.03.2024 - 10:33

Id	Achieved	Prove
NR08	The video contains the necessary information and format.	All sub requirements (SR31 to SR34) were achieved.
SR31	Yes	The video is way longer than 90 seconds but this was approved by Anrig Bernhard responsible for the bachelor theses.
SR32	Yes	See the Appendix H: Video Script (5.1 BFH-Intro Sceen, 5.14 BFH-Outro Screen).
SR33	Yes	See the Appendix H: Video Script (4 Content)
SR34	Yes	See the Appendix H: Video Script.
NR09	Yes	All sub requirements (SR35 to SR37) were achieved.
SR35	Yes	It was created via the .pptx template and will be printed by the BFH to A1.
SR36	Yes	The file was 4.4 GB and turned in in the .pptx format.
SR37	Yes	The poster illustrates and describes the situation in India, our contribution and the future of the Sulabh app.

Table 53: Target/actual comparision non-functional requirements

7.1.3 Deliverables

This section lists all the deliverables that were required for this project to be successful as a bachelor's thesis. Because all the requirements were achieved the deliverables were all considered to be complete, and the table below lists only the deliverables and when they were turned in.

Id	Description	Turn in Date
D01	A working LMS enhanced by all the functional requirements	Last hotfix deployed 10.06.2024
D02	A project documentation i.e. the bachelor thesis document	12.06.2024
D03	User documentation	06.06.2024 (to Project Partner (PR06))
D04	System administrator documentation	06.06.2024 (to Project Partner (PR06))
D05	A book entry to be published on the BFHs website and in the bachelor thesis booklet	31.05.2024
D06	A video describing the project	12.06.2024
D07	A poster summarizing the project for the exhibition	25.05.2024

Table 54: Target/actual comparison deliverables

7.1.4 Agile Development

At the beginning of the project, the agile approach of the project was clearly articulated. This section shows some measures that were implemented to make the project more agile according to the Agile Manifesto [5]. The following table shows the measures and whether or how they were achieved.

Measure Id	Measure	Achieved and Prove
APM011	We have a minimal working version deployed at the start of the product.	Yes, there was a working version for the Project Partner (PR06) to test.
APM012	We will deploy a working version of the product whenever we have a new feature ready for user testing.	Yes, as soon as the team had a new version it could be deployed for the Project Partner (PR06), mostly a little early See 7.1.1 Milestones for the dates.
APM021	We are in constant exchange with the customer (see 2.2.1 Meetings) and will take feedback and implement small changes.	Yes, even if some meetings were cancelled, it was mostly because of offline discussions or e.g. in person in India.
APM022	This is a bachelor thesis. Therefore we have requirements that have to be met at the end of the project. To ensure the success of the bachelor thesis we can't accept major changes to them.	Yes, to the extent possible. For the detailed rundown see the 7.1.2.1 Functional Requirements to see how they were achieved.
APM031	We try to deploy in feature intervals (see APM012). If a feature takes too long, we find a way so split it. We will deploy at least every other month.	Yes. It was not necessary to split a feature in two. Either bugfixes or features closer were delivered every other week.
APM041	Since we don't work daily on the project we meet with the Project Partner (PR06) as often as reasonable: every week (see 2.2.1 Meetings). A WhatsApp channel ensures ad hoc requests are immediately fulfilled.	No, the Project Team (TA01) did not meet every week with the Project Partner. Sometimes if he was not available or no party had something to discuss, it was cancelled.
APM051	The Development Team (TA02) is motivated through their thesis, degree, that they work on a project that will be used and working for an NGO.	Yes, the Development Team was motivated for the most parts by these reasons.
APM052	The Project Partner (PR06) is motivated because they get a tool to enhance their mission.	Yes, as can be seen by all the testing he did and the huge involvement in preparing the India trip.
APM053	The Thesis Advisor (PR05) is motivated to work for a good cause and can together with the Head of Project Management (PR03) support the Development Team (TA02) with the environment they need.	Yes, the Thesis Advisor was motivated and attended all the meetings with great inputs and guidance, sometimes even if the project manager did not think of it.
APM061	The Development Team (TA02) has weekly meetings with the Thesis	No, the Development Team (TA02) did not meet every week with the Project Partner. Sometimes if he was

Measure Id	Measure	Achieved and Prove
	Advisor (PR05) and the Project Partner (PR06).	not available or no party had something to discuss, it was cancelled.
APM062	The Development Team (TA02) has a weekly in person workday.	No, the meeting was held but due to circumstances not always in person.
APM091	At the weekly in person meeting, we will make small code review of the most important things developed in the last week.	No, the code reviews were made regardless of whether we met in person, if someone had something to show.
APM092	The documentation will be proofread by multiple internal and external advisors.	No, this did not happen because the project advanced and things changed constantly and because of short comings on the project manager part.
APM101	Only implement things that fit in the narrative of the requirements.	Yes, many changes were that were not explicitly mentioned but aligned with the requirements.
APM102	Contest every change to evaluate whether it is really needed.	Yes, the Development Team (TA01) contested everything it had some good reason to believe it should not do it. And if it was not able to do it because of the time given, it rejected the changes and put it in a new Features list. See the 7.4 Next Steps section for the new Features.
APM111	The Development Team (TA02) is on their own with helpful correctional advice from the Thesis Advisor (PR06).	Yes, there was no external architect, not even the Project Partner (PR06), so the team itself made any needed changes to architecture and alike.
APM121	The Development Team (TA02) will hold a review and improvement section after each finished feature.	Yes, to an extent the team discussed it can enhance but some topics deserves more time in the next project.

Table 55: Target/actual comparision agile development

In summary, the AP07 and AP08 were not achieved because they were ruled out on the basis of the project's nature from the beginning. Half of the measures of AP09 were not achieved because of shortcomings. In AP04 and AP06 the standard that was set in the measure was not met, because it was set to absolute, but in the spirit of things these were achieved. The rest was achieved. That makes for a fairly agile project even if there is room for improvement. One such improvement would be to state the measures not in an absolute but still measurable manner.

7.1.5 Working hours

This Section lists the conclusion comparing the target and actual working hours listed in the section 2.4 Time Estimate.

The comparison over the whole project looks good. With an overtime of 0.96 % the project is well within bound and comes close to a perfect timing.

All members of the Development Team (TA02) achieved the necessary 360h that the bachelor's thesis is designed for. This also counts as a good result since the working times are close together and the minimum for the model is achieved.

Person	Time Estimated [h]	Time invested [h]
Project	1080	1090.50
Tobias Erpen	360	363.50
Alayne Hiltmann	360	362.75
Lukas Vogel	360	364.25

Table 56: Target/acutal comparison working time project

7.1.5.1 Tasks

This section compares the time of the different tasks. Since most tasks are somewhat in the range of the estimated time most tasks will not be listed in detail. There were overall too few hours planned for the implementation and too many for the testing. This can be ascribed mostly to the fact that the Thesis Advisor (PR05) and the Project Partner (PR06) were very passionate testers. Therefore, the Development Team (TA02) had time freed up that was invested into developing additional smaller parts and to give back to the Project Partner in this way.

Also, of not are the tasks named reserve. Every project phase had one and they were intended to absorb some of the overtime on the other task in the project phase. Thies tasks mostly have very few working hours reported but this was to be expected and intended.

The tasks listed in the table below were the ones that were of the most significance, and these are briefly described here.

- ▶ For TE02 it was expected that the Project Partner (PR06) would have more changes to the requirements, therefore more time was planned than needed.
- ▶ For TE06 considerable conceptional work was required, but the bugs (TE27) took less time to fix. This was due to extraction of one feedback into the Feature 3: Additional Features (PP4).
- ▶ The implementation of the assessment (TE10) was also more time intensive because the team implemented a more flexible approach that hopefully enables the app to be used for a longer time.
- ▶ The task TE31 did not take as long because a lot of bugs and issues were already fixed, and the project ended, so the remaining tasks were harshly prioritised.
- ▶ Lastly, the TE55 was added later in the project because of all the publicity work the Project Team (TA01) did during their stay in India.

Id	Activity	Effort Estimated [person-hours]	Time invested [h]
PP01	Project Initialization		
TE02	Requirements	20.00	4.00
TE06	Creating minimal version of Assessment	40.00	116.75
TE10	Development (Assessment)	70.00	105.00
TE27	Bug fixing minimal working version	60.00	32.25
TE31	Bug fixing on sight test	20.00	2.00
TE55	Representation & Publicity	0.00	66.00

Table 57: Target/actual comparison tasks

7.1.5.2 Project Phases

This section lists the working hour comparison of the different project phases and the meetings. As discussed in the section above the implementation phases took more time than expected and the testing was shortened because the Thesis Advisor (PR05) and the Project Partner (PR06) were very passionate testers. The overtime in the meetings was due to the Representation & Publicity task (TE55).

Id	Phase	Effort Estimated [person-hours]	Time invested [h]
PP01	Project Initialization	138.00	213.00
PP02	Feature 1: Assessment	130.00	173.00
PP03	Feature 2: Certification	60.00	52.25
PP04	Feature 3: Smaller functionalities	110.00	123.25
PP05	Testing	261.00	180.00
PP06	Additional Deliverables	110.00	103.00
PP07	Presentation and Demo	40.00	44.50
	Meetings	151.00	201.50

Table 58: Target/actual comparison working hours project phases

7.2 Publicity

The whole Project Team (TA01) is very thankful that the BFH Technology and Computer Science department provided funding for a one-week trip to India in which face-2-face meetings and field testing could be carried out. As an attempt to show gratitude for this investment, the Development Team (TA02) took active steps to publicize this trip and the application.

This section shows how and where the Development Team (TA02) was able to gather publicity for the app.

7.2.1 Presentations

The Development Team (TA02) held some presentations outside of the minimal presentation needed for the thesis module. These presentations gathered support for the project in a way that social media posts and quality work cannot.

Presentation to the Swiss General Consul and Swissnex on 13.05.2024

The first presentation was held in the Swiss consulate located in Bangalore, India. The Counsel General of Switzerland, Mr. Jonas Brunschwig, was present. Swissnex is an organisation that promotes cooperation between Swiss educational institutions and institutions and corporations abroad.

A small introduction from both sides was held and then a full demo of the tool in action was given. It is hoped that this project, the application, and cooperation between the BFH and Sharada Educational Trust can be used as a stepping stone for further projects and collaborations. For Sharada Educational Trust it is hoped to leverage Swissnex to connect with other NGOs that work in the same field. For the BFH and Swissnex, it is hoped there will be more cooperation in education. In fact, as a result of this meeting, there is an ongoing discussion about creating a shared module between the BFH and an Indian university.

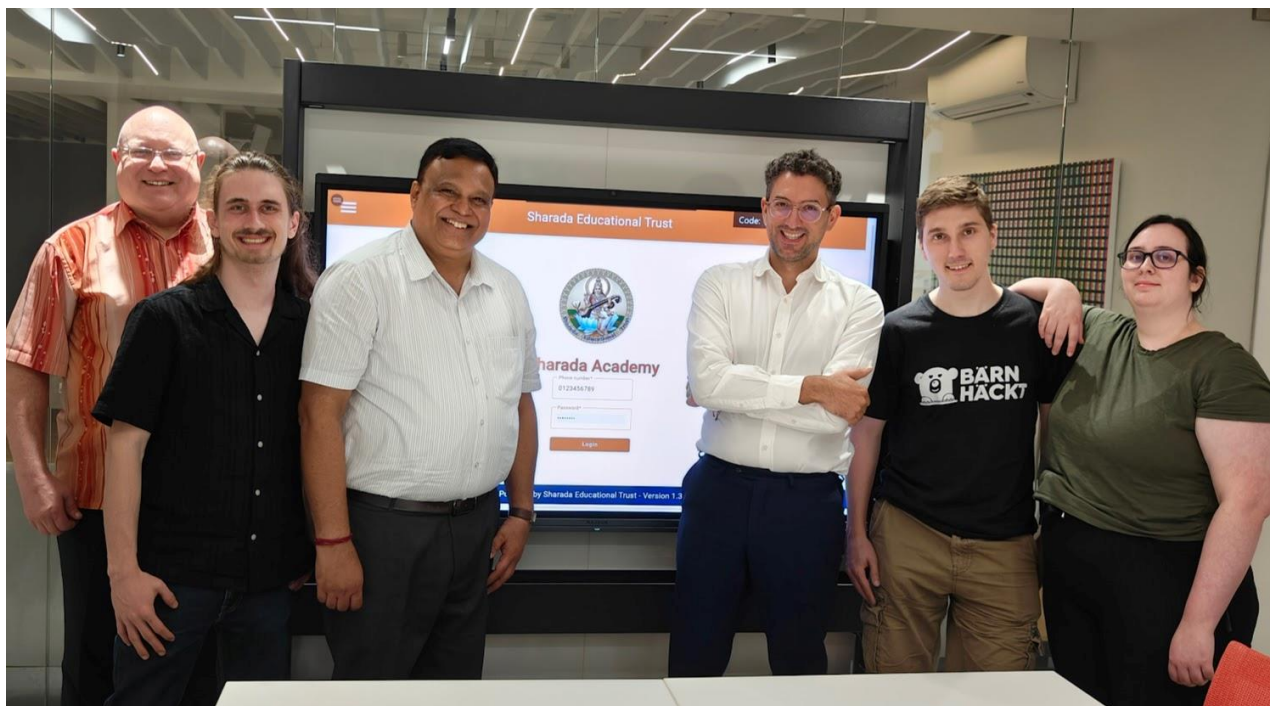


Figure 60: Presentation Swissnex

Presentation to Rural Government School on 15.05.2024

The second presentation was given during the visit to the rural Karnataka village of Cheeluru and the field testing there. A small explanation using their blackboard was given and then a run through the app from a student's point of view was carried out. This was followed by a testing and discussion part, where the students and teachers could use the app with the assistance from the Development Team (TA02). Also, feedback was shared and collected.

The feedback gathered was helpful and some small application functionality could be changed based on this, but other feedback will be left for a later date. The most important goal was to bring the app to the rural parts of India and to visit the students and teachers there. It was seen by the huge effort placed on the event by local authorities some local businesses representatives were there and most importantly teachers from neighbouring schools and a lot of children came during their summer holiday.



Figure 61: Project team with teachers and students in cheeluru

Presentation to InfoSys Center Director on 16.05.2024

The last presentation for the project in India was held during the visit to InfoSys. With over 350'000 employees InfoSys is the second largest global IT services company in the world. They operate the world's largest corporate training facility in Mysore, India, which uses 700 teachers, 3'500 support staff to teach 14'000 employees, all of whom live at the facility during their 6-month training. The project partner is an ex-InsoSys employee, and he used his contacts to arrange a personal tour of the facility. After a tour of their campus with a senior manager (Mr. Biligiri Ranga), the Project Team (TA01) had to honor of a personal meeting with the senior executive responsible for the facility, Mr. Satheesha B Nanjappa. In this meeting the app was demonstrated, and similarities and plans for the future of their InfoSys educational app as well as the one developed in this project were exchanged and compared.

The following image shows the project team with an independent teacher for local traditions and soft skills (Uma Nataraj, bottom left) and the guide of the tour Mr. Biligiri Ranga (top left). The image was taken in front of their main training building.



Figure 62: Project team in front of the education all at Infosys

Presentation to BFH students on 31.05.2024

To round things up the Development Team (TA02) held a presentation after their return from India to the class in the IT strategy module (BT14305). This presentation was educational in focus, so that the other BFH students could also benefit from the investment. The team highlighted their journey and what they did in India. This presentation was not focused on the application itself but more on what is possible with a bachelor's thesis at the BFH as well as critical success factors for working with a foreign stakeholder and rolling out a productive application.



Figure 63: Presentation IT strategy module

7.2.2 Local Newspaper

The soft launch of the app in Cheeluru at the rural government school drew a lot more attention than expected. This is shown by two local press articles released the day after the visit.

The Transcripts can be found in the Appendix I: News Paper Article Transcript. The content of the article differs a bit from what we expect in the West, with attention focussed on managers and administrators rather than with the Project Team (PA01) or students. In addition, the newspaper articles suggest that government schools are better off than the private schools. This is according to the Project Partner (PR06) not true. Most parents that can afford to do so send their students to private schools. Nevertheless, there are many passionate teachers at the government schools that try to improve the situation, and the app should help them. Even if the articles are not exactly what we would expect in the West, nevertheless they promoted the app, which helps a lot. The following two figures show the newspaper article in the local language Kanda.



Figure 64: Local news Prajavani



Figure 65: Local news Vijay Karnataka

7.2.3 Video

The video created by the project will help the Sharada Educational Trust to gain some support for their cause. In addition, it can generate publicity for the BFH computer science program when aspiring students in Bern see it. With the additional length of the video and the nature of it being about the trip to India it has the potential to draw in more internationally oriented students.

7.3 Lessons Learned

Overall, the team learned a lot in this project, the most important learnings focussing on the project management, the frontend development and the testing. Beside the learnings listed below corresponding to the topics above, each Development Team (TA02) member learned a lot of smaller but still very important lessons about the development in an international software project. The following summarize the key learnings:

Project Management

- ▶ Trust but verify. In Terms of project management, the team learned that it must check that work of others, especially when it comes to the requirements and the time planned for them. Trust is a good thing, but control (and feedback/requests for changes) is better.
- ▶ A single project schedule plan. An enhancement that could lead to better readability and communication is to combine the two GANTT diagrams. It gets crowded but it will help the two parts of the Development Team (TA02) to work together. For a bachelor's thesis it should also be used at the weekly meeting to open the meeting.
- ▶ Meetings. The team felt the meetings were done very well. A lot of ideas were exchanged, confusions lifted, and problems solved in the meetings in the Development Team (TA02) and the Project Team (TA01). All the meetings that had participants outside of the Development Team were protocolled just in case someone has a complaint.

Frontend Development

The team learned a lot in terms of frontend development, including the following.

- ▶ Coordination. Amongst the most important is to define the inputs in coordination with the backend, defining which are mandatory and in what format very tightly. Then to validate the input fields strictly to prevent a lot of confusing errors for the testers and a lot of time needed to fix it.
- ▶ Workflow. Another way to enforce correct data that was implemented later was to force a given workflow. That a user cannot see the content that should not be entered yet i.e. if some fields depend on the existence of some previous selections.
- ▶ Handling illegal actions. The third learning in the frontend development is to block and signalize illegal actions. If something can no longer be edited mark it as so.

But each of the points above stands in contrast with the philosophy to give a working version to the customer to see problems as soon as possible. Therefore, the last learning for the frontend is to make handle these in the order they are listed above. In this way it can be clearly communicated to the customer what is implemented, and they already have something to work with. It also enhances the ability to track what is done and in what state it is at a given moment.

Continuous Testing

- ▶ Bug tracking. During the continuous testing the team learned that a strict and enforced framework for reporting and communication is very important. An Excel sheet is nice, easy to setup and get going but in our case, it caused a lot of problems. For both tester and test manager (PR04) it was very hard to see the status. Sometimes it was hard to track whether there are duplicates or two in once and the information was not well readable. For the next time a harder restricted sheet or tool should be used. A kind of chat of each issue should be introduced. A lot of unneeded fields can be cut, and the status should be very well visible and easy to keep up to date by all parties.
- ▶ Target improvement. Another learning in testing would be to have a clear target for the testing, especially regarding the number of issues and kind of issues that can still be open at the end of the project. Clear targets for the automated tests and reasoning for that should be provided before the tests are written the have a clear goal in front.
- ▶ Field Testing. For the field tests next time we should ensure that not only the tool has a good structure for the test at hand, but also that the testing data provided by the Project Partner (PR06) is suited. This time we had the problem that some students lost themselves in the slides of the Project Partner (PR06) and did not test the app, the actual plan behind the field test. The team was lucky that the app is not too complicated to be run on the older phones of the rural population. But one is not always that lucky; therefore, next time the topic of compatibility with the hardware available should be raised early before the technology decisions.

7.4 Next Steps

After the completion of the bachelor thesis, the project will remain under the leadership of our project advisor. For further development a new engineer was found who can contribute to the application. After a handover, one option is for the developer to not just maintain the app but to add new features.

Just because the development phase of the project has ended does not mean that there are no future enhancements planned. Some wishes and features our customer wanted had to be defined as out of scope because of the time limit. The additional features elucidated during the project are summarized in the following table:

Id	Name	Description
AF01	Updatable batches	There are some critical parts on the Batch entity which would break the application if updated. Hence, the team did not create an endpoint to update batches when they have started. However, there are some fields which could be allowed, e.g. students could be added and removed from a batch.
AF02	Selectable location for user creation	For consistency and error prevention, the location of a user should be selectable.
AF03	Second usable batch id	At the moment, the batch id is simply a UUID which has completely random characters. There should be an id which states some information. Example: "SET-SE-B001"
AF04	Show contact information of students to their tutor	There are some scenarios where a tutor has to contact a student. At the moment, there is no way to get that kind of information through the app. The list of students with their phone numbers may be shown inside the tutor view of a course.
AF05	Lesson complete check	As some students will likely go directly to the assessments before studying the lesson material. There has to be at least a small obstacle for them to do this.

Table 59: Additional features

7.1 List of known issues

In every professional software application it is the responsibility of the development team to clearly communicate which aspects of the application are not working as intended or otherwise have weaknesses. In addition to the new features there is also a list of bugs that are known but there was no time to solve them yet. The following table lists them.

Id	Name	Description
KB01	User change to student	A user cannot be changed to student, cannot be changed to a pure student, because of a bug in the update method of the password. A user can be changed to be a student and have other roles and a user can be created as a pure student.
KB02	Saving Lesson	While adding a new Lesson getting error "Server error occurred" while saving. I could enter the details in 2nd attempt.
KB03	Uneditable fields are editable	Some fields in the course that are no longer editable after a batch started are not marked as such (and not greyed out).
KB04	Login without roles	If a user logs in in the frontend but has no roles then nothing happens. A error should be thrown.
KB05	Invisible last assessment	If an assessment is invisible the students can not continue the course because they can not complete the assessment.

Table 60: Known bugs

With the application at this point of development our project partner (PR06) will launch pilot projects where courses will be offered and taught. After the launch of the app at the government school in Cheeluru, 3 NGOs on behalf of government schools and 2 government schools have already contacted our partner saying that they also want to use the app. This will not be possible within a reasonable time frame because the pilot projects must first be successful. However, the goal of Sharada Education Trust [21] is to bring English and financial knowledge to rural government schools over all of India. With the interest the other organizations showing, the signs are good that vision will be fulfilled.

7.2 Personal Reviews

The following sections contain the personal reviews of the members of the development team and the project partner.

7.2.1 Ms. Alayne Hiltmann

“As the leading developer in the frontend, I was not only tasked with coding the functionalities and the lay outing of the user interface, but I also was allowed to design the look and feel for the application. Being able to have control not only over the implementation but also the design is quite a special experience. I am glad I got the opportunity to design a user interface which will hopefully be able to assist numerous students and teachers in their education.

Frontend development always seems to be a finicky process therefor I was grateful that I could focus almost all my attention solely into it, with the occasional help of Lukas and Tobias. Angular as a framework as well as the Material library and Bootstrap were not entirely new technologies for me. But within this project I got to deepen my understanding of them even more.

The split of responsibilities of our team was for me personally a great help in creating functional and running product. Having Tobias as our project manager helped me immensely in staying up to date with deadlines and additional tasks as well as him doing the heavy lifting of organizing and communicating with our partner and customers. Lukas has, both in our project 2 and in this bachelor thesis, helped me out in the frontend from implementing features to minor bugfixes. This close cooperation was very rewarding, and I hope we both gained some insight from it.

The experience overall in this project was incredible. From being able to work with a customer as far away as India and even having the opportunity to visit the country and interact directly with our end users. This experience has taught me a lot about working with outside customers and the often difficult process of communicating with each other.

From my viewpoint this was a successful project even with it's up and downs in between. I am very happy with the end result we have created.”

7.2.2 Mr. Lukas Vogel

“In my main role as backend developer in this project, I felt very comfortable. This is due to the distribution and execution of the team roles and also because of the consistent communication inside the development team. With Tobias as project manager, he was the main contact point for our customer and coordinated the work among us developers. Therefore, misunderstandings considering what to do next were very rare.

Developing with the technology stack we decided on was a big benefit for me because I already had some experience with it and did not have to invest much time in incorporating knowledge about it. Also, with the license for IntelliJ from the BFH I could work with a development environment which is very well suited for Java projects that I am familiar with.

With English as the language for the communication with our customer and for the deliverables we have to provide for the bachelor thesis, I faced some problems. As it is not my mother tongue, I noticed that I need considerably more time to create documentation, emails or just WhatsApp messages for our customer. In addition, the whole documentation has to go through multiple reviews where misspellings and wrong sentence structures are corrected. In order to create a readable and dynamic text, using diverse vocabulary is a crucial point. For me, this took considerably more time took more time to make points clear.

Scheduling time for project phases, features and tasks is hard. I experienced this during this project even though our project leader provided us with a well-arranged project plan. When reviewing the plan, everything seemed to make perfect sense but despite that, I was multiple times surprised how

fast the finish lines for some features arrived. In the future, I will take some time to think about the plan after certain events like new incoming features or unexpected time-consuming bugs. In Conclusion, the project is a great success. At the beginning, I could not imagine how well our final product will be working. I am very happy with the outcome.”

7.2.3 Mr. Tobias Erpen

“The communication within the Development Team (TA02) as well as in the Project Team (TA01) was very good. The meetings were for the most part productive also due to the splitting into roles. This helped to define the responsibilities of each person involved and therefore what they need to focus on.

My motivation for the project was higher than in most other projects I have taken part in. The fact that a working product was the goal (and it was achieved) and it serves a useful purpose were both very motivating. This helped to put minor disagreements and misunderstandings aside. It was also a very important part of the success of the project for me. Because every work is easier and most turn out better with motivation.

Nevertheless, I had a hard time adjusting to do thing not strictly in a Swiss style. It was a very good experience to see other way of doing things – especially when we were in India – but it caused a lot of friction. Sometimes understanding each other was hard because the Indians speak different than the Swiss regardless of the language. The most notable in this regard was that Indians don't say “no”. This was very difficult to handle. It was hard to understand when they actually disagree with me and when they are not because they will use a kind of yes one way or the other. I.e. when I asked a tester whether they like the product, they responded with yes but what do I do with this information after all they would not say no.

Even if I was not able to interpret there answer with certainty, I like where we got with the project. The project brought forward a product that will be used and brings value to many people. I was able to gather experience especially international experience that would be harder otherwise. I learned many things that I do not even know. So, for me personally, I am very happy with the result and what I can take from it into my own future.”

7.2.4 Feedback from Mr. Arvind Kamath

The following text is part of the letter of recommendation written by our project partner (PR06) Arvind Kamath.

“Throughout the duration of the project, your commitment to excellence was evident in every aspect. Your attention to detail, innovative ideas, and tireless effort were instrumental in achieving our goals and exceeding expectations. Your documents, comprising of the User Manual, User Testing Report and other Project related documentation, have been of exceptional quality, which are detailed yet simple to read and comprehend.

Furthermore, your ability to collaborate effectively with the team members, communicate ideas clearly, and problem-solve creatively has been invaluable to the project's progress. This was evident during the field tests conducted by you, at the Government Schools, during your visit to Bangalore from May 13th to May 18th 2024.

Your project's completion marks not only a significant achievement for you personally but also a testament to your capabilities as dedicated and talented individuals. Your passion for learning and your commitment to excellence shall serve as an inspiration to your peers.

On behalf of the entire team at Sharada Educational Trust, I would like to express our sincere appreciation for your hard work, dedication, and exceptional performance throughout this project. You have set a high standard for excellence, and we are confident that you will continue to achieve great things in the future.”

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10Glossary

API

Application Programming Interface

DDD

Domain-Driven Design

DTO

Data Transfer Object

LIFT

Locate Identify Flat Try to be Dry

LMS

Learning Management System

MS

Microsoft

NGO

Non-Government Organization

OS

Operating System

SOLID

A very common set of object-oriented software development principle. Called the first 5.

UVP

Unique Value Proposition

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12Declaration of Authorship

We hereby certify that we composed this work completely unaided, and without the use of any other sources or resources other than those specified in the bibliography. All text sections not of my authorship are cited as quotations and accompanied by an exact reference to their origin.

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Place, date: *Fahrni, 12.06.2024*

Signature: 

Appendix A: Use Cases

In this Appendix the main use cases are described in detail.

Id	UC01
Name	Create User
Related requirements	FR01
Actors	Admin (R01)
Trigger	A new user wants to join the platform
Precondition	-
Postcondition	The new user is saved and shown in the user overview
Steps	
1	The admin opens the user overview
2	The admin selects "new User"
3	The admin enters all the user's details
4	The admin submits the data
Exception Scenarios	
3 a	The e-mail is already used
3 b	The phone is already used
3 c	The password does not meet the requirements

Id	UC02
Name	Manage user
Related requirements	FR01
Actors	Admin (R01)
Trigger	Something about the user changed
Precondition	The user exists (UC01)
Postcondition	The changes to the user are saved and can be displayed
Steps	
1	The admin opens the user overview
2	The admin selects "edit" of the user in question
3	The admin changed the relevant details
4	The admin submits the data
Alternative Scenarios	
2.1.1	The admin selects "delete" of the user in question
2.1.2	The admin confirms the deletion
Exception Scenarios	
3 a	The e-mail is already used
3 b	The phone is already used
3 c	The password does not meet the requirements

Id	UC03
Name	Course creation
Related requirements	FR02
Actors	Admin (R01)
Trigger	They want to offer a new course or port one onto the system
Precondition	-
Postcondition	The new course is saved and shown in the course overview
Steps	
1	The admin opens the course overview
2	The admin selects “new course”
3	The admin adds the relevant details including the relevant sections
4	The admin submits the data
Alternative Scenarios	
2.3.1	The user wants to manage assessments. (includes UC10)

Id	UC04
Name	Manage course
Related requirements	FR02
Actors	Admin (R01) or course manager (R02)
Trigger	Something about the course changed
Precondition	The course exists (UC03)
Postcondition	The changes to the course are and can be displayed
Steps	
1	The user opens the course overview
2	The user selects “edit” of the course in question
3	The user changed the relevant details and may add new sections
4	The user submits the data
Alternative Scenarios	
2.1.1	The admin selects “delete” of the course in question
2.1.2	The admin confirms the deletion
2.3.1	The user wants to manage assessments. (includes UC10)
Exception Scenarios	
2.1.1 a	The course has one or more batches attached to it and can’t be deleted

Id UC05	
Name	Batch creation
Related requirements	FR03
Actors	Admin (R01)
Trigger	They want to create a new batch (run though) of a course
Precondition	Tutor, students and course for the batch exist
Postcondition	The new batch is saved and shown in the batch overview
Steps	
1	The admin opens the batch overview
2	The admin selects “new batch”
3	The admin selects the course
4	The admin adds the other details
5	The admin submits the data
Alternative Scenarios	
3.1.1	If it is not a “self-paced” course the admin selects the dates and selects “schedule dates” to automatically generate the dates

Id UC06	
Name	Manage batch
Related requirements	FR03
Actors	Admin (R01)
Trigger	Something about the batch changed
Precondition	The batch exists (UC05)
Postcondition	The changes to the course are and can be displayed
Steps	
1	The admin opens the batch overview
2	The admin selects “edit” of the batch in question
3	The admin changed the relevant details
4	The admin submits the data
Alternative Scenarios	
2.1.1	The admin selects “delete” of the course in question
2.1.2	The admin confirms the deletion
Exception Scenarios	
2.1.1 a	The batch has already started and can’t be deleted
4 a	The batch has already started and can’t be changed

UC07	
Name	List enrolled and tutoring courses
Related requirements	FR04, FR05
Actors	Tutor (R03) or Student (R04)
Trigger	The user wants to view their courses
Precondition	-
Postcondition	All courses where the user is enrolled or is tutoring are displayed
Steps	
1	The user opens the “my courses” overview

UC08	
Name	Detailed course
Related requirements	FR04, FR05
Actors	Tutor (R03) or Student (R04)
Trigger	The user wants to participate in a course
Precondition	The user has the “my course” overview open and at least one is displayed. Extends UC07
Postcondition	The detailed combination of course and batch are displayed
Steps	
1	The user opens the course in question

UC09	
Name	Manage attendance
Related requirements	FR05
Actors	Tutor (R04)
Trigger	The tutor wants to mark which students were in class
Precondition	The user has a course selected. Extends UC08
Postcondition	The detailed combination of course and batch are displayed
Steps	
1	The tutor selects “attendance management”
2	The tutor checks the students that were there.

UC10	
Id	
Name	Manage Assessment
Related requirements	SR01
Actors	Admin (R01) or course manager (R02)
Trigger	The course must be extended with an assessment or be changed
Precondition	The course in question is open.
Postcondition	The assessment was added and can be filled out by students
Steps	
1	The user adds a new section of the type “assessment” to the course or opens the assessment to be changed.
2	The user adds the questions, the available answers and selects which are right.
3	The user saves the changes.

UC11	
Id	
Name	Fill out Assessment
Related requirements	SR02, SR04
Actors	Student (R04)
Trigger	The tutor of a course tells the students to fill out the assessments
Precondition	The course in question is open and the end date is not reached. Extends UC08
Postcondition	The answers given are saved and assessment is corrected automatically, and the next part of the course is displayed if one exists.
Steps	
1	The student opens the assessment
2	The student answers all the question.
3	The student saves their results
Exception Scenarios	
2 a	The course’s end date is reached, and no more assessments can be filled out.
4 a	Not enough questions are answered correctly, and the student has to do it again.

UC12	
Name	View assessment results
Related requirements	SR03
Actors	Student (R04)
Trigger	The student wants to see how they did in the assessment
Precondition	The course in question is open and the assessment was sent. Extends UC08
Postcondition	The student sees their result
Steps	
1	The student opens the assessment in question.

UC13	
Name	Complete a Course
Related requirements	SR04
Actors	Student (R04)
Trigger	The student wants to finish a course to get a certificate
Precondition	The course in question is open and the end date is not reached. Extends UC11
Postcondition	The course is marked as complete for that student.
Steps	
1	The student fills out all the assessments successfully.
Exception Scenarios	
1 a	The course's end date is reached, and no more assessments can be filled out.

UC14	
Name	Get a certificate
Related requirements	SR05
Actors	Student (R04)
Trigger	The student wants to get a certificate for their work
Precondition	The course in question is marked as completed for that student. Extends UC13
Postcondition	The user can see their certificate with name, course and date.
Steps	
1	The student opens the "My Courses" overview.
2	The student finds the course they completed.
	The student requests the certificate via the newly displayed button.

Appendix B: Test Cases

In this Chapter the main test cases are described. The linking between the test and use cases can be found in 6.2.5 Test Cases

Id		TC01
Test method		TM04
Test scenario		TS01
Related use cases		UC01, UC02
Role		Admin (R01)
Description		Onboard a new student and tutor.
Precondition		The tool is started.
Steps	1	Login with the admin credentials
	2	Navigate via the hamburger navigation to the user overview.
	3	Navigate to the user creation view and create a student with the password in the format of ddMM, make sure it is active and save it.
	4	Create another inactive user without a role and save it.
	5	Edit the roleless user to be a tutor, change its name, activate it save the changes.
	6	Create another student, save it and then delete it.
Expected results		Three users were created. Only the first two are still available. The users are active, and the users can be displayed with the entered data. The only exception is the phone number which has no more spaces.

Id		TC02
Test method		TM04
Test scenario		TS01
Related use cases		UC03, UC04
Role		Admin (R01)
Description		Adding a new course
Precondition		The tool is started, and the admin logged in.
Steps	1	Navigate via the hamburger navigation to the course overview.
	2	Navigate to the course creation view and create a new course with at least two sections. Ensure it is active and save it.
	3	Create another minimal course, save it and then delete it.
Expected results		Two courses have been created of which only the first is still available. In the edit view are two courses as well as the other entered data displayed.

Id		TC03
Test method	TM04	
Test scenario	TS01	
Related use cases	UC05, UC06	
Role	Admin (R01)	
Description	Combining the new users with the new course	
Precondition	The tool is started, and the admin logged in. TC01 and TC02 were successful.	
Steps	1	Navigate via the hamburger navigation to the batch overview.
	2	Navigate to the batch creation view and create a new batch with the new course, the new tutor as the tutor and the new student as a student.
	3	Let the course start today and have at least two scheduled dates. Then save the new batch.
	4	Create another minimal batch that does not start today, save it and then delete it.
Expected results	Two courses have been created of which only the first is still available. In the edit view are two courses as well as the other entered data displayed.	

Id		TC04
Test method	TM04	
Test scenario	TS01	
Related use cases	UC07, UC08	
Role	Student (R04)	
Description	Viewing new course	
Precondition	The tool is started. TC03 was successful.	
Steps	1	Login as the new student.
	2	Navigate to my courses and inspect the course you are assigned to.
Expected results	The course is available in the list, the detail view opens. The displayed course combines the information of the new course and new batch. The active sections are visible.	

Id		TC05
Test method		TM04
Test scenario		TS02
Related use cases		UC04, UC10
Role		Admin (R04)
Description		Add an assessment to the new course
Precondition		The tool is started, and the admin is logged in. TS01 was successful.
Steps	1	Navigate via the hamburger navigation to the course overview.
	2	Edit the new course by changing its name and adding an assessment with two simple questions. Then save the changes.
Expected results		The course is available in the list and the assessment with the questions can be seen in the edit view.

Id		TC06
Test method		TM04
Test scenario		TS02
Related use cases		UC11
Role		Student (R04)
Description		Fill out the new assessment
Precondition		The tool is started, and the new student is logged in. TC05 was successful.
Steps	1	Navigate to the detailed overview of the new course.
	2	Find, open and answer the new assessment incorrectly.
	3	Save your answers.
Expected results		The answers are saved, and the correction of the assessment is displayed.

Id		TC07
Test method		TM04
Test scenario		TS02
Related use cases		UC11
Role		Student (R03)
Description		Retake an assessment
Precondition		The tool is started, and the new student is logged in. TC06 was successful.
Steps	1	Reopen the new assessment.
	2	Select the correct answer.
	3	Save your answers.
Expected results		The answers are saved, correction of the assessment is displayed and the next section is revealed.

Id		TC08
Test method		TM04
Test scenario		TS02
Related use cases		UC09
Role		Tutor (R03)
Description		Mark the student as attending
Precondition		The tool is started. TS01 was successful.
Steps	1	Login as the new tutor.
	2	Navigate to my courses, inspect the new course and open the attendance view.
	3	Mark the student as attending the first lesson.
Expected results		The student is marked as attending for the first lesson but not any others.

Id		TC09
Test method		TM04
Test scenario		TS03
Related use cases		-
Role		Student (R04)
Description		Asking a question
Precondition		The tool is started. Someone is logged in the talk.to dashboard.
Steps	1	Open the talk.to chat via the green button at the bottom left.
	2	Pose a new question to the admin.
Expected results		The admin can see the question in the talk.io dashboard

Id		TC10
Test method		TM04
Test scenario		TS03
Related use cases		-
Role		Admin (R01)
Description		Answering a question
Precondition		You are logged in the talk.to dashboard. TC09 was successful
Steps	1	Accept the chat opened via the TC10.
	2	Replay to the student via the dashboard chat.
	3	Mark the conversation as closed.
Expected results		The student can see the answer and the conversation is closed.

Id		TC11
Test method	TM04	
Test scenario	TS04	
Related use cases	-	
Role	Student (R04)	
Description	Complete a course	
Precondition	The tool is started, and the student is logged in. TC08 was successful	
Steps	1	Navigate to the new Course.
	2	Fill out the last assessment correctly.
	3	Save the answers.
Expected results	The answers are saved, correction displayed and the course is marked as complete.	

Id		TC12
Test method	TM04	
Test scenario	TS04	
Related use cases	UC14	
Role	Student (R04)	
Description	Generate a certificate	
Precondition	The tool is started, and the student is logged in. TC11 was successful	
Steps	1	Navigate to the “My Course” overview
	2	Find the new course in the list
	3	Get the certificate
Expected results	A PDF with the certificate opens in a new tab. The certificate has the right student’s name, course name and the date of today.	

Id		TC13
Test method	TM03	
Test scenario	TS05	
Related use cases	-	
Role	All roles but mostly admin (R01)	
Description	Free field testing in the application	
Precondition	The tool is started.	
Steps	1	Test a functionality
	2	Report to the development team the errors that were found.
Expected results	Depends on the functionality	

Id		TC14
Test method	TM01, TM02	
Test scenario	TS06	
Related use cases	-	
Role	All	
Description	Automated testing with unit and integration test	
Precondition	The backend project is loaded.	
Steps	1	The developer starts all unit tests
	2	The developer starts all integration tests
Expected results	All tests run successfully through.	

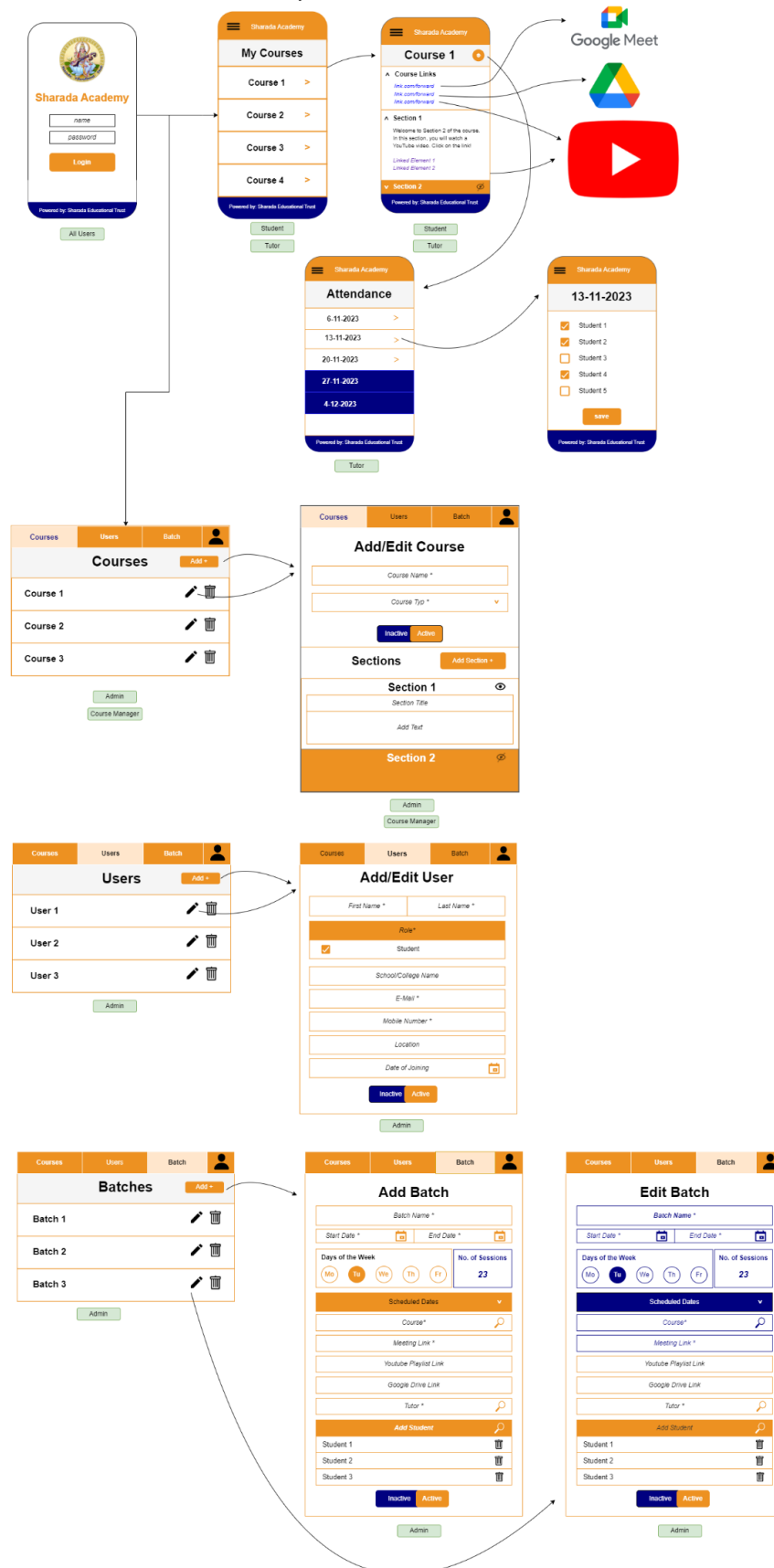
Appendix C: API Description

The whole YAML documentation generated by Swagger would cause 40 additional pages. Therefore, only a picture of the first few lines is shown here. To get the whole content, visit the URL (<https://env-1621132.jcloud.ik-server.com/api-docs> or <https://env-1621132.jcloud.ik-server.com/api-docs/swagger-ui/index.html> for the UI) described in 45 API Description.

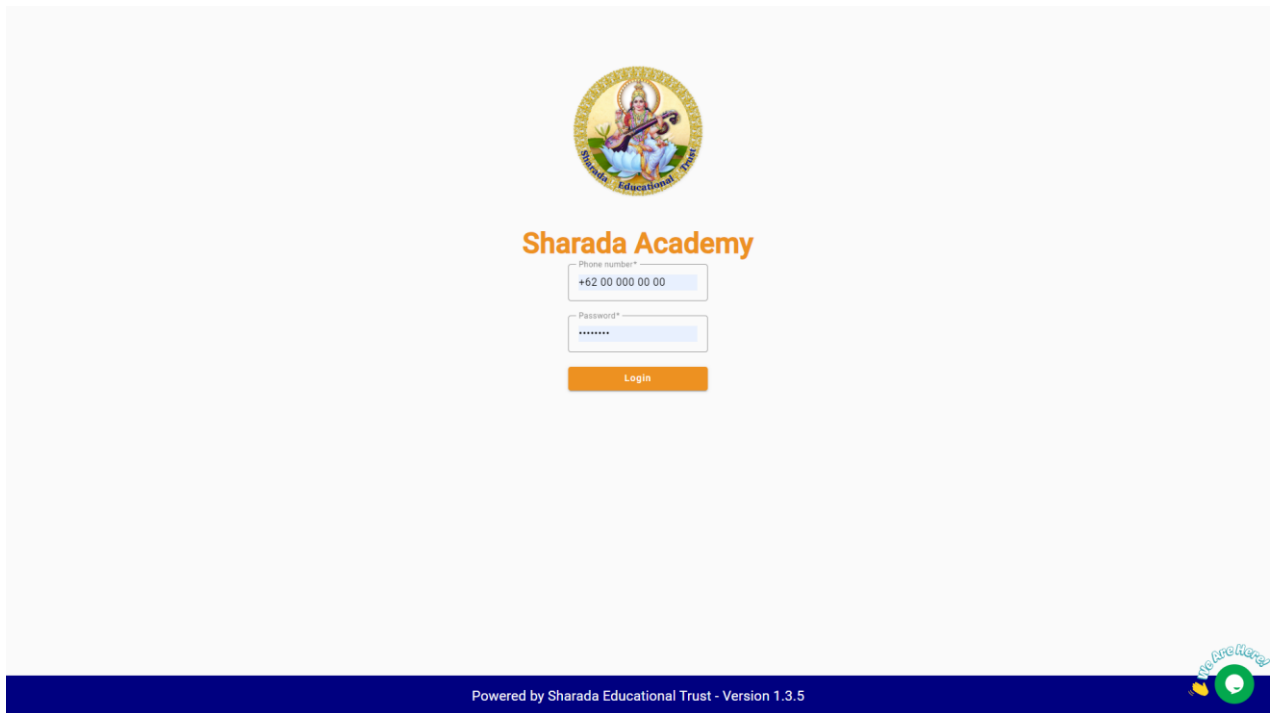
```
{
  "openapi": "3.0.1",
  "info": {
    "title": "OpenAPI definition",
    "version": "v0"
  },
  "servers": [
    {
      "url": "http://env-1621132.jcloud.ik-server.com",
      "description": "Generated server url"
    }
  ],
  "paths": {
    "/api/user/{id}": {
      "get": {
        "tags": [
          "user-controller"
        ],
        "summary": "Get a specific user.",
        "operationId": "getUser",
        "parameters": [
          {
            "name": "id",
            "in": "path",
            "description": "Id which users should be fetched.",
            "required": true,
            "schema": {
              "type": "string",
              "format": "uuid"
            }
          }
        ],
        "responses": {
          "404": {
            "description": "Requested user could not be found.",
            "content": {
              "application/json": {
                "schema": {
                  "$ref": "#/components/schemas/User"
                }
              }
            }
          },
          "200": {
            "description": "Successfully found user.",
            "content": {
              "application/json": {
                "schema": {
                  "$ref": "#/components/schemas/User"
                }
              }
            }
          }
        }
      },
      "put": {
        "tags": [
          "user-controller"
        ],
        "summary": "Update an existing user.",
        "operationId": "updateUser",
        "parameters": [
          {
            "name": "id",
            "in": "path",
            "description": "Id which user will be updated.",
            "required": true,
```

Appendix D: Screenshots of the frontend application

In this section screenshots of the application are provided. These show both the mobile application view as well as the desktop browser view.

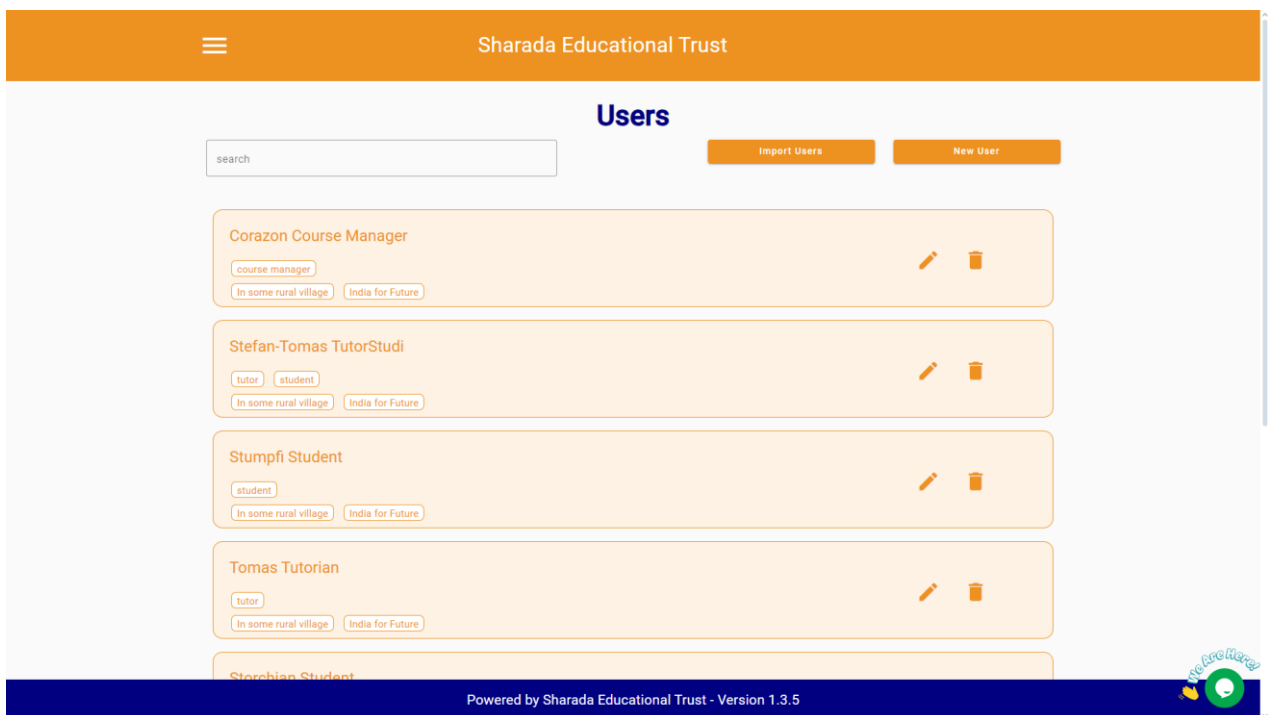


Screenshot login:



The screenshot shows the login page for Sharada Academy. At the top center is a circular logo featuring a deity playing a musical instrument. Below the logo, the text "Sharada Academy" is displayed in a bold, orange font. Underneath, there are two input fields: "Phone number*" with the placeholder "+62 00 000 00 00" and "Password*" with a masked password "*****". A blue "Login" button is positioned below these fields. At the bottom of the page, a dark blue footer contains the text "Powered by Sharada Educational Trust - Version 1.3.5" and a small circular icon with the text "We Are Here?".

User Overview Screenshot:



The screenshot displays the "Users" management interface of Sharada Educational Trust. The header is orange with a menu icon and the text "Sharada Educational Trust". Below the header, the title "Users" is centered. To the left of the user list is a search bar with the placeholder "search". To the right are two buttons: "Import Users" and "New User". The user list consists of five entries, each in a light orange box. Each entry includes a name, a role (in a pill), a location (in a pill), a country (in a pill), and edit/delete icons. The entries are: Corazon Course Manager (course manager, In some rural village, India for Future), Stefan-Tomas TutorStudi (tutor, student, In some rural village, India for Future), Stumpf Student (student, In some rural village, India for Future), Tomas Tutorian (tutor, In some rural village, India for Future), and Storchian Student (In some rural village, India for Future). The footer is dark blue with the text "Powered by Sharada Educational Trust - Version 1.3.5" and a small circular icon with the text "We Are Here?".

Name	Role	Location	Country	Actions
Corazon Course Manager	course manager	In some rural village	India for Future	Edit, Delete
Stefan-Tomas TutorStudi	tutor, student	In some rural village	India for Future	Edit, Delete
Stumpf Student	student	In some rural village	India for Future	Edit, Delete
Tomas Tutorian	tutor	In some rural village	India for Future	Edit, Delete
Storchian Student		In some rural village	India for Future	Edit, Delete

User Creation/Edit Screenshot:

Sharada Educational Trust

Edit User

First Name*
Tomas

Last Name*
Tutorian

Role

School/College Name
India for Future

E-mail
tutor@test.com

Mobile Number*
+62000000008

Update Password

Location
In some rural village

Join Date
24/11/2023

Last login
02/06/2024 11:11:35 915698

Active Inactive

Cancel Save

Powered by Sharada Educational Trust - Version 1.3.5

User Import Screenshot:

Sharada Educational Trust

Import Users

Drag and drop CSV or [Browse Files](#)

Users are imported as students

[Download user import file template](#)

Powered by Sharada Educational Trust - Version 1.3.5

Course Overview Screenshot:

The screenshot shows the 'Courses' overview page. At the top is an orange header with a menu icon and the text 'Sharada Educational Trust'. Below the header, the title 'Courses' is centered. A search bar with the placeholder 'search' is on the left, and a 'New Course' button is on the right. A list of courses is displayed in orange boxes: 'English 1', 'English 2', 'Finance 1', 'Finance 2', 'Testing Course', 'Plantes', and 'English 1'. Each course box has three icons on the right: a document, a pencil, and a trash can. At the bottom, a dark blue footer contains the text 'Powered by Sharada Educational Trust - Version 1.3.5' and a circular logo with the text 'We Are Here!'.

Course Creation/Edit Screenshot:

The screenshot shows the 'Edit Course' screen. At the top is an orange header with a menu icon and the text 'Sharada Educational Trust'. Below the header, the title 'Edit Course' is centered. A back arrow icon is on the left. The form includes a 'Course Name*' field with 'English 1' entered, and a 'Course Type*' dropdown menu with 'Tutor Led' selected. Below the form are 'Active' and 'Inactive' radio buttons, and 'Cancel' and 'Save' buttons. A horizontal line separates the form from the 'Lessons' section. The 'Lessons' section has two buttons: 'New Lesson' and 'New Assessment'. Below these are two lesson entries in orange boxes: 'English 1, Lesson 1' with the subtitle 'First lesson: IPSUM LOREM ...' and 'Quiz 1'. Each lesson entry has up/down arrow icons and a pencil icon. At the bottom, a dark blue footer contains the text 'Powered by Sharada Educational Trust - Version 1.3.5' and a circular logo with the text 'We Are Here!'.

Section Mockup:

Sections

New Sections

Section

Assessment

Section Title*

Section Text*

Visible

Invisible

New Section

Section 1



Sections

New Sections

Section

Assessment

Assessment Title*

Assessment Description*

What is a noun ?



Question

Visible

Invisible

New Assessment

Assessment 1

- Question 1

- Question 2



Mockup Course Creation:

Course Creation / Edit

Course info

Sections

New Text

New Assessment

Section 1

remove options
when open batch

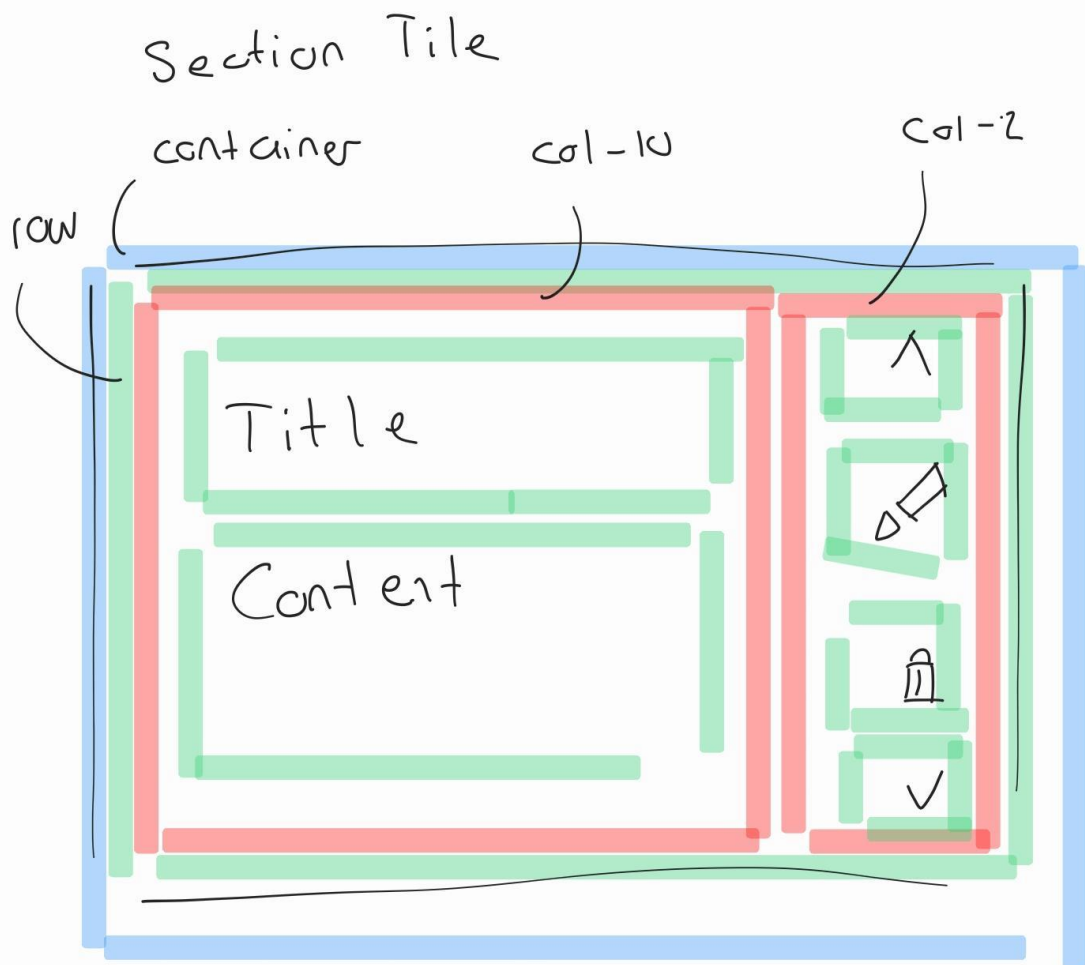
Assessment 1

Question

Question

Question

Mockup Section Tiles:



Mockup Lesson:

Text Section

A hand-drawn mockup of a 'Text Section' form. The form is enclosed in a blue border. Inside, there is a yellow border containing a green border. Within the green border, there is a button with a left-pointing arrow and the text 'Add / Edit Section'. Below this, there is a red border containing two input fields: 'title Input' and 'Content Description'. The 'Content Description' field is larger and has a small icon in the bottom right corner. The entire form is set against a light gray background.

Mockup Assessment Creation:

New / edit Assessment

Title		
Description		
		
Question		Correct Option
Option A	☐	☐
Option B	☐	☒
Option C	☐	☐
+		
Add Question		
Save		

Mockup Assessment Creation:

Add Assessment

Title

Description

Question

Option

Option

correct

☐

☐

+

Lesson Creation/Edit Screenshot:

The screenshot shows the 'Edit Lesson' interface. At the top is an orange header with a menu icon and the text 'Sharada Educational Trust'. Below the header is a white form area with a blue back arrow icon and the title 'Edit Lesson'. The form contains four input fields: 'Title*' with the value 'English 1, Lesson 1', 'Lesson material link*' with the value 'https://www.course.material.in/1', 'https://Lesson video link', and 'https://Additional resource link'. Below these is a 'Content*' field with the value 'First lesson: IPSUM LOREM'. At the bottom of the form are three buttons: 'Cancel', 'Visible' (highlighted in orange), and 'Invisible'. A 'Save' button is also present. The footer is a dark blue bar with the text 'Powered by Sharada Educational Trust - Version 1.3.5' and a circular logo on the right.

Sharada Educational Trust

Edit Lesson

Title*
English 1, Lesson 1

Lesson material link*
https://www.course.material.in/1

https://Lesson video link

https://Additional resource link

Content*
First lesson: IPSUM LOREM

Visible Invisible

Cancel Save

Powered by Sharada Educational Trust - Version 1.3.5

Assessment Creation/Edit Screenshot:

The screenshot shows the 'Edit Assessment Section' interface. At the top is an orange header with a menu icon and the text 'Sharada Educational Trust'. Below the header is a white form area with a blue back arrow icon and the title 'Edit Assessment Section'. The form contains two input fields: 'Title*' with the value 'Quiz 1' and 'Description*' with the value 'First Quiz for English 1'. Below these is a 'Visible' button (highlighted in orange) and an 'Invisible' button. The main section is a large white box with an orange border containing a 'Question*' field with the value 'What color has a fire department car?'. Below this are two 'Option*' fields: 'red' and 'blue'. To the right of each option is a 'Correct' checkbox; the 'red' checkbox is checked. Below the options is a blue '+' button. At the bottom of the form are three buttons: 'Cancel', 'Add Question' (highlighted in orange), and 'Save'. The footer is a dark blue bar with the text 'Powered by Sharada Educational Trust - Version 1.3.5' and a circular logo on the right.

Sharada Educational Trust

Edit Assessment Section

Title*
Quiz 1

Description*
First Quiz for English 1

Visible Invisible

Question*
What color has a fire department car?

Option*
red

Correct ☒

Option*
blue

Correct ☐

+

Add Question

Cancel Save

Powered by Sharada Educational Trust - Version 1.3.5

Batch Creation/Edit Screenshot:

The screenshot shows the 'Add Batch' form in the Sharada Educational Trust LMS. The form is titled 'Add Batch' and includes the following fields and controls:

- Course ***: English 1 (dropdown)
- Batch Type**: Tutor Led (dropdown)
- Batch Name ***: English 1, Batch 2023A (text input)
- Start Date *** and **End Date ***: Date pickers
- Days of the Week ***: Dropdown menu
- Schedule Dates**: Button
- Number of Sessions**: Input field with a minus sign
- Scheduled Dates**: Dropdown menu
- Meeting Link**: https://test.ch/classroomA (text input)
- Recordings Link**: https://www.lms.test.ch/recordings (text input)
- Additional Resources Link**: https://Additional Resources Link (text input)
- Tutor**: Tomas Tutorian (dropdown)
- Students** and **Added Students**: Input fields

Powered by Sharada Educational Trust - Version 1.3.5

Course Overview for Students/Tutors Screenshot:

The screenshot shows the 'My Courses' page in the Sharada Educational Trust LMS. The page is titled 'My Courses' and includes a search bar. The courses are listed in a table:

Course Name	Batch	Status	Action
English 1	English 1, Batch 2023A	Tutor Led, Completed	Download Certificate, >
English 1	English 1, Batch 2023C	Tutor Led	>
Finance 1	Finance 1, Batch 2024B	Tutor Led	>
English 1	English 1, Batch 2023A	Tutor Led	>

Powered by Sharada Educational Trust - Version 1.3.5

Course Detail View of a Tutor Screenshot:

The screenshot displays the 'Course Detail View' of a tutor interface for 'English 1'. The top navigation bar is orange with a menu icon on the left and the text 'Sharada Educational Trust' in the center. Below this, a grey header bar contains a back arrow icon, the course title 'English 1', and an 'Attendances' button. The main content area is divided into two sections. The first section, titled 'Course Links', contains two links: 'Classroom Link' with the URL test.ch/classroomA and 'Recordings Link' with the URL www.lms.test.ch/recordings. The second section, titled 'English 1, Lesson 1', contains a 'Quiz 1' titled 'First Quiz for English 1'. The quiz consists of two questions. The first question is 'What color has a fire department car?' with options 'red' (checked) and 'blue' (unchecked). The second question is 'What color has the Neptune?' with options 'red' (unchecked), 'blue' (checked), and 'yellow' (unchecked). The bottom of the screen features a dark blue footer with the text 'Powered by Sharada Educational Trust - Version 1.3.5' and a circular logo on the right.

Sharada Educational Trust

English 1

Attendances

Course Links

Classroom Link
test.ch/classroomA

Recordings Link
www.lms.test.ch/recordings

English 1, Lesson 1

Quiz 1
First Quiz for English 1

What color has a fire department car?

red ☒

blue ☐

What color has the Neptune?

red ☐

blue ☒

yellow ☐

Powered by Sharada Educational Trust - Version 1.3.5

Appendix E: Meeting protocols

This Appendix lists all the meeting protocols (minutes) for reference. Some of the Meeting Id were wrong in the versions sent to the participants otherwise nothing was changed to the protocol. All protocols but the one for M37 were sent to the participants and if not silently agreed upon the changes necessary were made.

Meeting notes from 20.02.2024

Topic	Testing feedback from Arvind
Meeting Id	M17
Present	Kenneth Ritley, Arvind Kamath, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic

Agenda Items

Id	Topic
AI001	All points recorded by Arvind were discussed or quietly accepted by all members. The important Points are listed in the following agenda items
AI002	The next feature that will be tackled is a very simple version of the assessment. It must only contain text questions and text answers. This version is done to check if the students use the website.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD001	Find a place and form for the static "About us" page.	Alayne	29.02.2024
TD002	Update the user creation to allow the password to be dd/MM for students and tutors but not admins.	Alayne (Frontend) Lukas (Backend)	29.02.2024
TD003	Update the error messages during user, batch and course creation in the frontend to more useful messages	Alayne	29.02.2024
TD004	Update the payment for the Hosting and inform Arvind ones it is up again.	Tobias	21.02.2024
TD005	Add a time stamp to the last login in the database.	Lukas	29.02.2024

Meeting notes from 21.02.2024

Topic	Payment Jelastic
Meeting Id	M33
Present	Kenneth Ritley, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic

Agenda Items

Id	Topic
AI003	Payment for Jelastic was initialized, the site is back up for testing and Arvind is informed. It remains to be checked how much the hosting effectively costs.

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 29.02.2024

Topic	The “About us” page and planning going forward, with the Gantt diagram.
Meeting Id	M18
Present	Arvind Kamath, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	Kenneth Ritley
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic
TD001	As proposed by Alayne the about us page will be added to the „hamburger“ menu.

Agenda Items

Id	Topic
AI004	The Gantt diagram was shared.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD006	Find out how to add the text for the Introduction written by Arvind to the sources.	Tobias	07.02.2024
TD007	Arvind will provide the first part of the spoken English course so that the team can get a better feel for it.	Arvind	07.02.2024
TD008	Provide the list of the tree most important additional features (requirements for this project) including the assessment functionality	Arvind	07.02.2024

Meeting notes from 07.03.2024

Topic	The Requirements were discussed
Meeting Id	M19
Present	Arvind Kamath, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	Kenneth Ritley
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic

Agenda Items

Id	Topic
AI005	The About us page was shared.
AI006	Requirements were shared. The different quizzes will be attached to the courses. Instead of an AI answering questions in the chat functionalities a FAQ will be added.
AI007	It was discussed that the field trip to India will most certainly take place in the CW 20 (13. May).

To-Dos

Id	To-Do	Responsible	Delivery Date
TD009	Finalize the requirements and send them to Arvind to get them approved.	Tobias, Arvind	14.03.2024
TD010	Provide the most frequently asked questions from students	Arvind	31.03.2024

Meeting notes from 14.03.2024

Topic	How good we are in the time plane considering we are three
Meeting Id	M04
Present	Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic

Agenda Items

Id	Topic
AI008	It was proposed that the user and administrator documentation could be made available online via the website or a WCMS. This would probably increase the usage of the documentations in comparison to a plain old Word document.
AI009	Work life balance was discussed and we came to the conclusion that no measures have to be taken yet but we keep an eye on it.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD011	Create dossier for the business trip to India including budget to see whether it gets approved and to what degree.	Ken	21.03.2024

Meeting notes from 14.03.2024

Topic	Future Plans
Meeting Id	M20
Present	Kenneth Ritley, Arvind Kamath, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic

Agenda Items

Id	Topic
AI010	Arvind will visit a school in Mumbai from the 17.04. If possible and the product has no show stoppers he can show the app to the children. Maybe gathering some test feedback.
AI011	The version with the mini assessment will be live on Monday the 18.03.

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 21.03.2024

Topic	Testing imagination
Meeting Id	M05
Present	Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic

Agenda Items

Id	Topic
AI012	The priorities of the testing and further development were discussed. We can to an agreement that we focus on the main functionality during development and user mostly explorative testing

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 21.03.2024

Topic	Testing feedback
Meeting Id	M21
Present	Kenneth Ritley, Arvind Kamath, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic

Agenda Items

Id	Topic
AI013	The most important questions and changes unearthed by the testing were discussed.
AI014	It was discussed that the Assessment and Certificate functionality will be the main targets and other functionalities are added to an optional features list. This list will be worked on if we have more time to fill at the end.

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 26.03.2024

Topic	Expert meeting
Meeting Id	M34
Present	Alain Joray, Kenneth Ritley, Alayne Hiltmann, Tobias Erpen
Absent	Lukas Vogel
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic

Agenda Items

Id	Topic
AI015	A round of introductions was held.
AI016	A presentation of the project and a demo of the current version of the product was held.
AI017	It was agreed that no second expert meeting will be held and that no documents need to be shared. Except for the final version of the document in PDF form.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD012	It was decided that the defense will be held on the 20.06.2024 in the SIPBB building. An invitation will be created and the room will be added once final.	Tobias	26.03.2024

Meeting notes from 28.03.2024

Topic	Catch up before easter
Meeting Id	M22
Present	Arvind Kamath, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	Kenneth Ritley
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic

Agenda Items

Id	Topic
AI018	It was discussed that we only show courses to the students that have already started and assessments can't be turned in after it has ended.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD013	Send the template for the certificate	Arvind	04.04.2024

Meeting notes from 04.04.2024

Topic	Organizing after easter and user documentation
Meeting Id	M07
Present	Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic

Agenda Items

Id	Topic
AI019	Ken provided the template for the contract so that Tobias can fill in the relevant information.
AI020	The user manual and a possible version inside the tool was presented and discussed. The resolution is to ask Arvind for what he likes the most.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD014	Ken sends the request for funding out today.	Ken	04.04.2024
TD015	Ken will prove read the project documentation and provide feedback.	Ken	08.04.2024

Meeting notes from 04.04.2024

Topic	Test results and fixing QNA format
Meeting Id	M23
Present	Arvind Kamath, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	Kenneth Ritley
Transcript Writer	Tobias Erpen

From Last Meeting

Id	Topic

Agenda Items

Id	Topic
AI021	Arvind presented some new testing results, and they were discussed. The Bugs are mentioned in the Bug report file (Bug 62-69).
AI022	It was discussed that the closing time of a course is not set to 6 months as falsely expected last time but is set to the end date of the batch.
AI023	It was discussed what items the team could bring to the photoshoot for the BFH Book to represent the project, Sharada Trust and southern India. The result is a flag of each country and some Indian sweets of which Arvind will provide the name.
AI024	It was fixed that the user manual/QNA will be done in Wordpress and added as a new section to the Sharada trust website. Tobias will provide the content in a rich text format and Arvind will upload it to the website.

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 11.04.2024

Topic	Test results and debriefing after demo
Meeting Id	M08
Present	Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI025	The test feedback from Ken was discussed and the necessary notes were made in the Testing feedback sheet.
AI026	Confirmation that the funding for the India trip will be available.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD016	Coordinate the apply for the stay in India and prepare the proposal for the group.	Ken, Tobias	15.04.2024

Meeting notes from 11.04.2024

Topic	Test results and two week outlook
Meeting Id	M24
Present	Arvind Kamath, Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI027	It was decided that the stay in India will be from 12.05.2024 to the 19.05.2024.
AI028	The test results from Arvind were discussed and the necessary notes were made in the testing feedback sheet.
AI029	Arvind will be in Mumbai on the 18.04.2024 and the meeting will not take place.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD017	Hold a demonstration that will be recorded and used as guidance to explain the tool to the students.	Tobias	25.04.2024

Meeting notes from 16.04.2024

Topic	Organization India field trip
Meeting Id	M35
Present	Kenneth Ritley, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI030	The cost calculation for the India field trip excluding flights was made. The result needs to be requested from the IT direction. The cost per person was determined to be about 963 CHF per person.
AI031	The general outline of the trip as it stands at the time of the meeting was shared.
AI032	Alain Jorai Expert (PR07) requested to shift the defense meeting. Some dates were presented but nothing was heard back. It was decided that we give him some more days to response otherwise Kenneth will seek contact with him instead of Tobias.
AI033	It was agreed that a briefing on Indian culture from Kenneth would be a good idea to prevent culture shock. It should include an emergency plan as well as gifts for the other parties.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD018	Organize meeting for Indian culture briefing.	Tobias	18.04.2024

Meeting notes from 18.04.2024

Topic	Demo for testing
Meeting Id	M25
Present	Arvind Kamath, Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI034	Arvind's test result and the requested answers were discussed.
AI035	A small demo was held to show the workflow of a student interacting with the app. This was done to help the testers test. It was agreed that another demo will be held on the 25.04.2024, that will be recorded and used to school the children and teachers. The recorded demo will use data as close to reality as possible.
AI036	The teachers and students how we will meet in India were introduced.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD018	Organize meeting for Indian culture briefing.	Tobias	18.04.2024
TD019	Arvind sends the course material and student names for the demo.	Arvind.	22.04.2024

Meeting notes from 25.04.2024

Topic	Demo for Students
Meeting Id	M26
Present	Arvind Kamath, Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI037	A demo of from the student point of view was held. This demo was recorded and will be cut to be used as a teaching tool.
AI038	The flights from Zürich to Bangaluru have been discussed and booked. We will fly on the 12.05.2024 and return on the 19.05.2024.
AI039	The hotel and travel options were discussed and Arvind will give a recommendation on the hotels/service apartments we will stay in.

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 02.05.2024

Topic	Demo for Tutors
Meeting Id	M27
Present	Arvind Kamath, Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI042	A demo with the tutor workflow was given. It was recorded and will be put onto the YouTube channel of Sharada. This video will be used for teaching/introduction purposes.
AI043	The test feedback from Arvind was discussed and some not necessary bug reports were discussed and closed.
	Einleitung indien

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 02.05.2024

Topic	Introduction to India
Meeting Id	M36
Present	Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI044	An introduction to Indian culture was given by Ken. The key points are on the slides: https://miro.com/app/board/uXjVKS_Slt8=?share_link_id=646798301023

To-Dos

Id	To-Do	Responsible	Delivery Date
TD020	All the members of the BFH how fly to India will request a visa.	All for their own	02.05.2024

Meeting notes from 09.05.2024

Topic	Preparing the ride to India
Meeting Id	M12
Present	Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI045	Plan for going to the Zurich airport: We meet on 04:30 in Bümpliz for Pickup. We meet on 05:45 in Olten for Pickup. We should arrive at the airport at 06:30. The flight takes off at 7:40.
AI046	Everyone should have their papers ready in physical and digital: • Passport • Voucher for the hotel • Visa • Tickets for the airplane

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 09.05.2024

Topic	Preparing the app and expectations for the India trip
Meeting Id	M28
Present	Arvind Kamath, Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI047	Smaller Problem found during testing by Arvind and the other tester in India were discussed.
AI048	From now on only fixes will be implemented. New changes, enhancement and new features won't be done for the time being.
AI049	For the stay in India, we won't upload a new version if not absolutely necessary.
AI050	The new catchy name for the tool is SULUB. The only place where that has to be changed is the About us.

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 15.05.2024

Topic	Field test in the rural village in India: Cheeluru
Meeting Id	M37
Present	Kenneth Ritley, Arvind Kamath, Alayne Hiltmann, Lukas Vogel, Tobias Erpen, many teachers from the government school and some children
Absent	
Transcript Writer	Lukas Vogel

Agenda Items

Id	Topic
AI051	Presentation of the different components of the app and how they work together
AI052	Questions & Answers section
AI053	Help new users to use the app
AI054	Gather feedback from the teachers

To-Dos

Id	To-Do	Responsible	Delivery Date
TD021	Teachers want that there is an obstacle for users who go directly to the assessment without reading the lesson material. Some sort of check, whether the material was read has to be implemented.	TBD	Out of scope during the thesis
TD022	Certificates could be downloaded without answering all questions right. This is by design because of a misunderstanding. The assessment is passed, when 80% of the questions were answered correctly. Change it to 100%	Lukas Vogel	20.05.2024
TD023	With some users, a server error occurred when they logged in and were redirected to the course overview, without seeing their courses. Debug and fix the bug.	Lukas Vogel Tobias Erpen	20.05.2024
TD024	The question was raised whether we know what the minimum OS system of the mobile device is. With an Android 7 Device, the app worked with some problems in displaying content. Check with different OS versions of Android and Apple whether the app works correctly. (Also with the platform where the lesson material is accessed)	TBD	Out of scope during the bachelor thesis
TD025	As the students are using mobile devices to use the app, it is crucial that the resource material is also optimized for this format	Arvind Kamath	Out of scope during the bachelor thesis

Personal feedback of the test

To demonstrate this app I recommend to have a test course ready with only 1 lesson and 1 assessment with 1 question. Inside the lesson, there is a link to the platform where all the resource materials are and

there is just the answer to the single question of the course and a short text which tells the student that he has to return to the app and answer the assessment. An example for the question could be: "In which year was Mahatma Gandhi born?" 1869. Inside the lesson material, the student has to be guided through as well. Tell him how long the lesson takes until he can return to the assessment and when he reaches the end, tell him again to return. Like that, the student is not unsure of what he has to do and is eager to process the information.

Also the structure of how we bring our app to the students can be optimized in my opinion. I would take the two following main steps:

1. Invite only tutors to the first tutorial. Explain what the app is meant to be and guide them through all tasks which the tutors have to do and also what the students have to do.
2. Invite tutors and students. Let the tutors themselves explain the app and what the students have to do to use it. Let the tutors guide their students through the example courses and only support when they have problems.

I think this way, tutors are more familiar with the app and are already a bit independent.

Meeting notes from 30.05.2024

Topic	Coordinating finalization of the project
Meeting Id	M15
Present	Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI51	Ken got the receipt for the airplane tickets.
AI52	The plan for the coming weeks was shared: • Until 02.06.2024 last changes to the code and final test runs are completed • 03.06.2024 is deployment to prod environment • The documentation is done by the 08.06.2024 • Last week (10.06.2024 until 13.06.2024): implement feedback to the documentation and presentation preparation
AI53	We will make a code review with an extern coder on the 06.06.2024.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD20	Setup a mail to get the funding from the international office in a ordered manner.	Tobias and Ken	03.06.2024

Meeting notes from 30.05.2024

Topic	Prepare final deployment and answer last questions
Meeting Id	M31
Present	Arvind Kamath, Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI54	The final test run was discussed, and it was revealed that the tester got the wrong test manual. Tobias will change that directly after the meeting.
AI55	The workload on the servers was discussed. Since there were no performance tests because they were not deemed relevant at the start of the project, no accurate answer can be given. The topic will be raised again once we have more user data.
AI56	The current version of the assessment concerning multiple possible answers was discussed. Currently it is implemented with checkboxes => multiple answers can be right and zero to all answers can be selected.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD021	A summary form the customers perspective will be done.	Arvind	06.06.2024

Meeting notes from 06.06.2024

Topic	External Code review
Meeting Id	M38
Present	Hugues Clavadetscher, Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI57	In introduction round of the project and the people how did not now the others was held.
AI58	There were a lot of concepts and tools that the team could have a look at in their spare time to improve their coding and designing skills. Among them are DDD, LIFT, nxgrx/store, Cypress.io and to review SOLID.
AI59	In the front end it would be better to group some components that build a unit into a module. This is to improve reusability and the overall readability of the structure.
AI60	It would be nice to separate the services and the front-end interaction in the form of the Snackbar. A solution would be to put a logger interface between the two.
AI61	It would be nice to not have the static data (that will be added on every new setup of a data source) in a Spring runner instead of in a SQL file.
AI62	It was noted that the backend was very well and comprehensively tested with automated tests.
AI63	Next time it would be nice to have certain endpoints with the /admin prefix to separate the functionalities or permission to execute an endpoint. i.e. a user can get their own user only and an admin can get all users. So, the GET /user/{id} would stay for the user and become GET /admin/user/{id} if you want to get any user.
AI64	Have some speaking minimal READMME.MD not just the generated.
AI65	The final verdict of the code review was: • GOOD PROJECT • GOOD WORK • EASY TO UNDERSTAND • Good collaboration - we can see the integration, naming is good, mapping between frontend and backend • Would be eager to have the students join his ongoing projects

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 06.06.2024

Topic	Coordinating finalization of the project
Meeting Id	M16
Present	Kenneth Ritley, Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI66	The missing results of Ken's the final test run can be treated as OK.
AI67	It would be quite easy to set up a second instance of the Shulabh app. This would be a good first task for the next person working on it.

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 06.06.2024

Topic	Coordinating finalization of the project
Meeting Id	M32
Present	Arvind Kamath, Kenneth Ritley, Shakir Sultanov Alayne Hiltmann, Lukas Vogel, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI68	Shakir Sultanov was introduced to the team and a little introduction about the app and situation with the app. He will start at the BFH at the beginning of August.
AI69	All available were informed that the test environment was taken down, the prod is taken up and tested by Arvind. He can login in.
AI70	Lukas did the change requested by Arvind to the user manual and will share them after the meeting.
AI71	Tobias offered to do the maintenance for showstopper until Shakir joins the BFH.
AI72	If Arvind needs a report from the DB, he can just ask Ken. Ken will be informed and will have access to the DB and the knowledge to get the report.

To-Dos

Id	To-Do	Responsible	Delivery Date

Meeting notes from 06.06.2024

Topic	Mail flights refunding
Meeting Id	M39
Present	Kenneth Ritley, Tobias Erpen
Absent	
Transcript Writer	Tobias Erpen

Agenda Items

Id	Topic
AI73	A walkthrough of the report for something in the DB was given to Ken.
AI74	It was exchanged how spend how much money in and for the India trip.

To-Dos

Id	To-Do	Responsible	Delivery Date
TD022	Create a mail to refund the flights for the students and send it all at the same time	Tobias	06.06.2024

Appendix F: Test results

In this Appendix the detailed test results are provided.

Test Results - Arvind

Meta Information			
Tester	Arvind Kamath		
Tested at	1 st June 2024		
Device used for Testing	MacBook Air		
Browser used for Testing	Google Chrome		
Test Scenario	Test Case	Passed	Notes
TS01	TC01	Yes, but with a comment	Created a new User – Anish Kamath without a Role. In the edit mode, tried mapping the User to Student role and revised the password to 4 digits. Getting an error message on saving “Some data send to the server is not correct” Screen shot attached below.
	TC02	Yes, but with a suggestion	However here is a suggestion: Edit button to be disabled for Courses as well as the lessons within the Course, which have been mapped to a batch already. Like in case of Batches, an eye icon should be applicable for Courses that cannot be edited. Currently getting a error message “Some data send to the server is not correct” instead of “Course is not editable” error message. Screen shot attached below.
	TC03	Yes	It is working fine after creating 2 new Users- one created as a student and the other as a Tutor.
	TC04	Yes	The student is able to view the Course mapped to the Batch, assigned to the student
TS02	TC05	Yes	Admin user is able to edit the Course not mapped to any Batch
	TC06	Yes	Student is able to answer the Assessment questions
	TC07	Yes	Student is able to open saved Assessment and answer the questions, subsequently
	TC08	Yes	Tutor can mark attendances of students only for the dates that have gone by or today’s date but not the future dates
TS03	TC09	Yes	Student (User) started a chat session with Admin and Admin received the chat on Tawk.To
	TC10	Yes	Admin responded to the chat which is seen by the student (User)

TS04	TC11	Yes	Student is able to move to the next Lesson after completing the Assessment and complete the course
	TC12	Yes	Student is able to generate the Certificate successfully.

Screenshots

TS01-TC01

Sharada Educational Trust

Edit User

First Name* Anish Last Name* Kamath

Role

- ☐ Admin
- ☐ Course Manager
- ☐ Tutor
- ☒ Student

School/Collage Name SET

E-Mail

Mobile Number* 8123496085

Update Password *****

Location BLR

Join Date 01/06/2024

Last login

Active Inactive

Cancel Save

Some data send to the server is not correct. OK

TS01-TC02

Edit Lesson

Title* Introduction to Sulabh Spoken English course

Lesson material link https://docs.google.com/presentation/d/1xFirZ0swha9lHYz1g1LEZR1fwlzbjE1/edit?usp=drive_link&ouid=110556192354025989821&rtopf=true&sd=true

Lesson video link <https://youtu.be/1Gz-Cw3HWz4>

Additional resource link https://youtu.be/9NP_mCmvFt0

Content*

- Please open the Course material PPT file in Slideshow mode or choose "Present on this Device" so that you can enjoy the animations
- Take the Online Spoken English course and learn to speak fluent English after completing the course with the help of a Tutor.

Visible Invisible

Cancel Save

Some data send to the server is not correct. OK

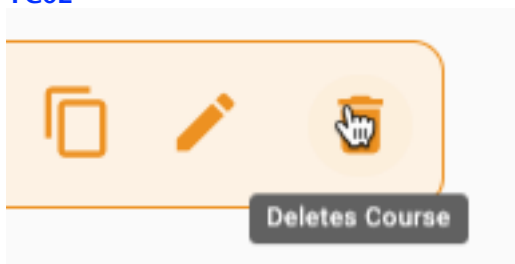
Test Results – Ken Ritley

Meta Information			
Tester	Ken Ritley		
Tested at	02.06.2024		
Device used for Testing	MacBook		
Browser used for Testing	Chrome		
Test Scenario	Test Case	Passed	Notes
TS01	TC01	OK	
	TC02	OK	See minor improvement
	TC03	OK	
	TC04	OK	
TS02	TC05	OK	Question: Was not able to edit basic information about a course in progress.
	TC06	OK	
	TC07	OK	See minor improvement
	TC08	OK	
TS03	TC09	OK	
	TC10	OK	
TS04	TC11	OK	
	TC12	OK	See note about certificate

Screenshots

Past your screenshots here and note which test cases it refers to.

TC02



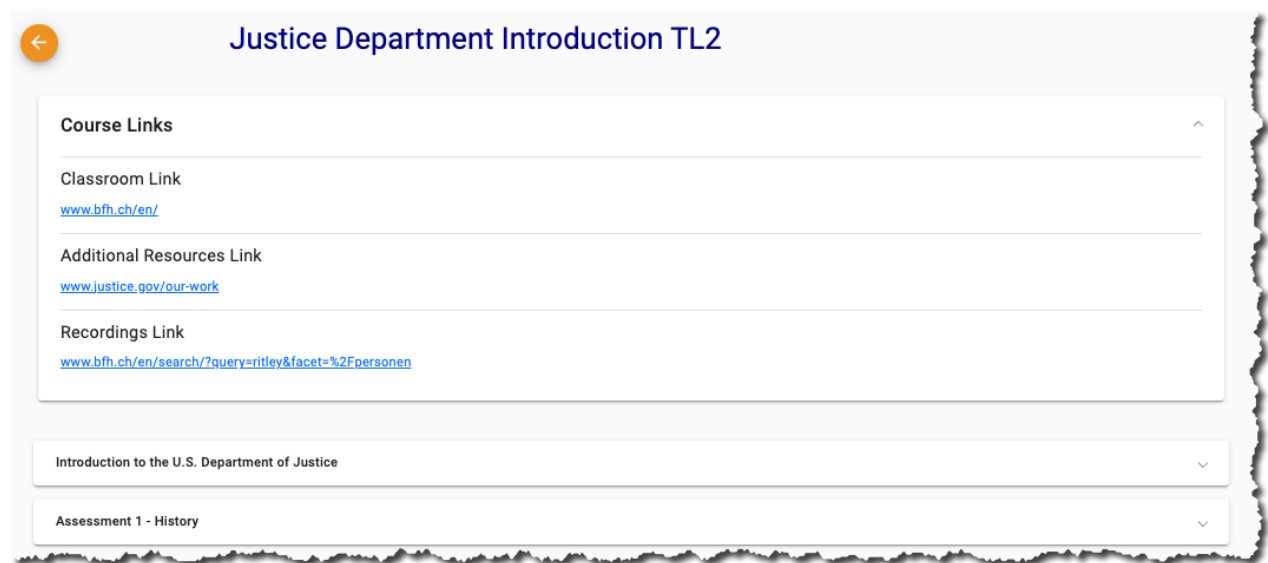
The other icons have hover-over text “Copy Course” and “Edit Course” so probably it would be better to have “Delete Course” rather than “Deletes Course” – more harmonious that way.

TC02

OK. Special note: I tested this in the role of Course Manager!

TC07

After I take an assessment in the role of student, the assessment window closes up:



As you can see, Assessment 1 – History is closed. This is not what I would call a bug, but it is just awkward because the student needs to re-open this panel in order to see whether the assessment was successful or not. Because it is not a bug, I would not recommend fixing it

TC12

OK. My first certificate looks like this:



I then tried a course with a really long name, and the certificate looked like this:



As you can see, the text is not formatted so nicely.

I don't think this is a showstopper . . . probably there should be a final "List of Known Issues"

Test Results – Tobias Erpen

Meta Information			
Tester	Tobias Erpen		
Tested at	03.06.2024		
Device used for Testing	Lenovo		
Browser used for Testing	Opera		
Test Scenario	Test Case	Passed	Notes
TS01	TC01	Yes	
	TC02	Yes	
	TC03	Yes	
	TC04	Yes	
TS02	TC05	Yes	
	TC06	Yes	
	TC07	Yes	
	TC08	Yes	
TS03	TC09	Yes	
	TC10	Yes	
TS04	TC11	Yes	
	TC12	Yes	
TS06	TC14	Yes	Tested via the GitLab pipelines

Appendix G: User Manual

What I would honestly suggest is REMOVING this document from the main Word document – since it is a standalone document. But then including it TOGETHER with the main document in the final PDF. On the Mac this is super easy, just drag and drop pages.



User Manual

Learning Management System for Rural India

Date: 29.03.2024
Version 1

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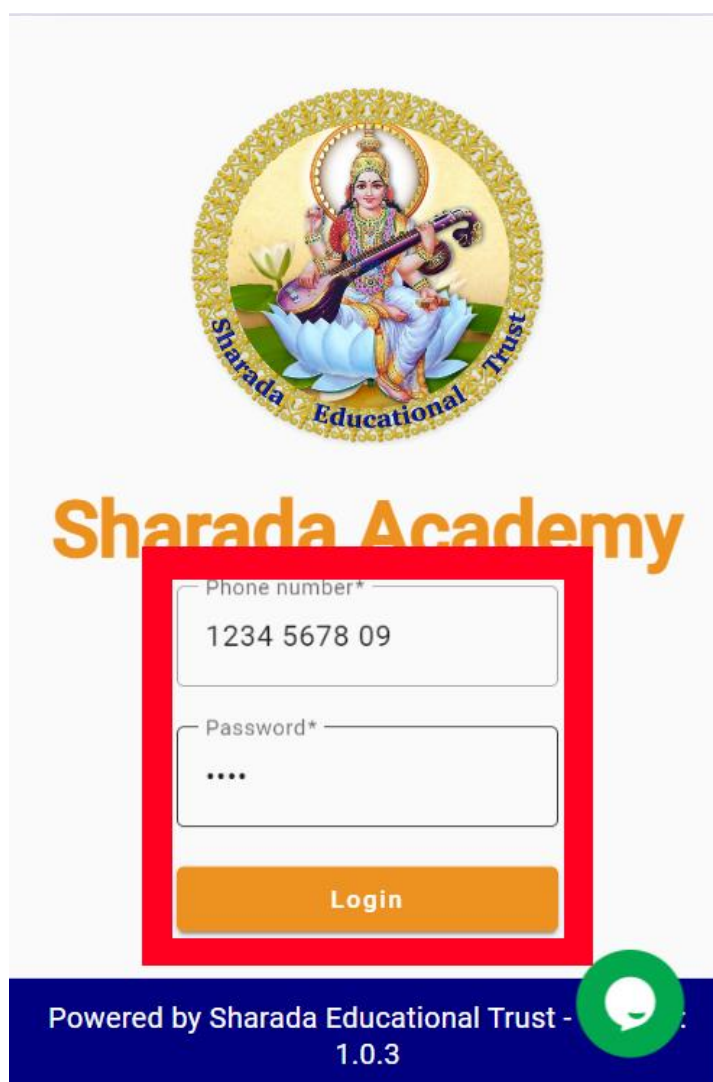
1 Introduction

This document shows and explains the various features a user can access. The features are sorted by the different roles.

Student Functionalities

2 How do I join the platform?

If you don't have an account an administrator needs to create one for you. Then you can login via the login page. Your login credentials should be your phone number and the default password is your birthdate in the format ddMM. For example, if you are born on the 26.05.2014 your password is 2605.

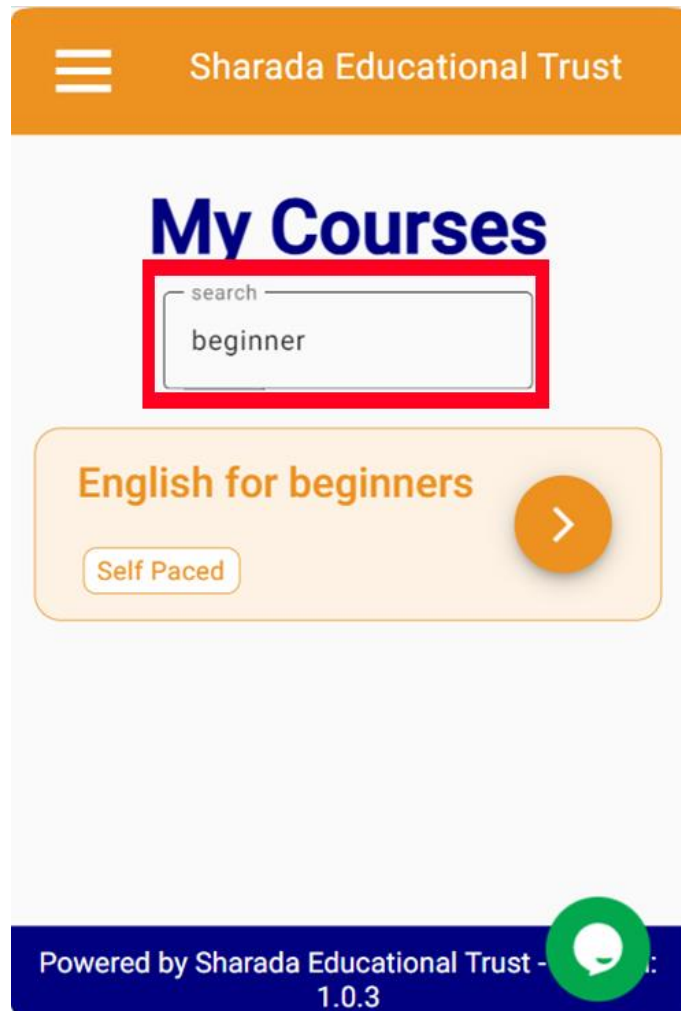


The image shows the login interface of the Sharada Academy mobile learning app (LMS). At the top, there is a circular logo featuring a Hindu deity (likely Sharada Devi) seated on a lotus, holding a veena. Below the logo, the text "Sharada Academy" is displayed in a large, orange, stylized font. Underneath the name, there is a login form with two input fields: "Phone number*" and "Password*". The phone number field contains the text "1234 5678 09". The password field contains four dots. Below these fields is an orange "Login" button. The entire login form is highlighted with a red rectangular border. At the bottom of the screen, there is a dark blue footer bar with the text "Powered by Sharada Educational Trust - 1.0.3" and a green circular chat icon on the right.

If this does not work, please contact the administrator with the chat functionality. For more information see "How do I contact the administrator?".

3 How do I join a course?

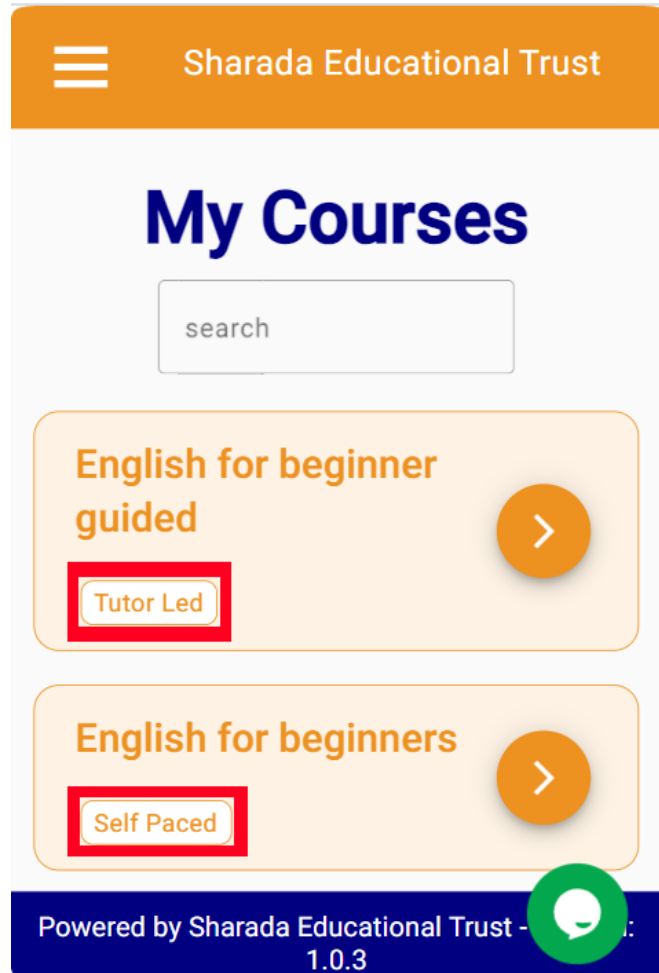
An administrator must add you to a batch so that you can join a course. Once the administrator has added you can find your courses on the “My courses” site. You can see all the courses you are in or search for the course name via the search bar at the top.



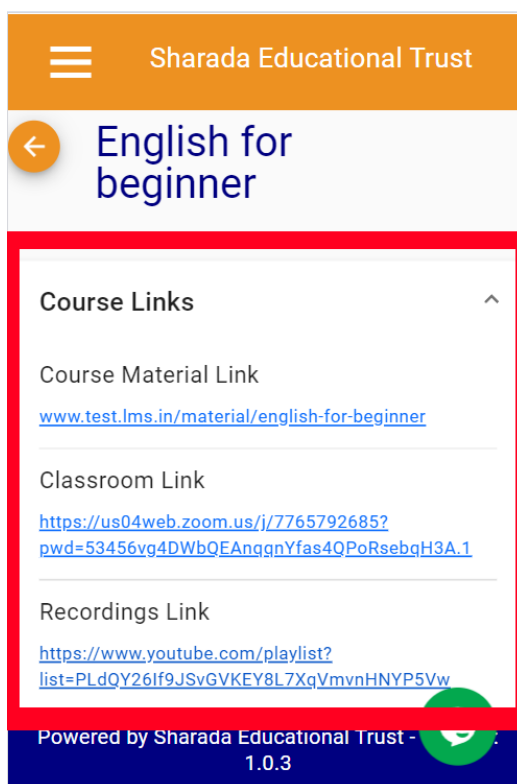
If you can't find a course, it is possible that the course has not started yet. It is also possible that the administrator did not put you on the course. For more information, please contact the administrator. For more information see “How do I contact the administrator.”

4 How do I participate in a course?

Once you have joined a course you can participate in it. There are two types of courses: “Self Paced” and “Tutor Led”. You can see the course type in the “My Courses” overview. In a “Tutor Led” course you meet on a regular basis with your tutor. They will teach you English and give you further information. In a “Self Paced” course you are given material but have to learn and make the tests on your own.



Once opened you can find helpful links in the top of the course. They contain additional material, a place to meet with your tutor and the old recordings, if such things exist for the course. Below you can find the sections, you already have access to.



5 How do I pass a course?

To progress in the course, you have to read the material given in the course. When you feel ready fill out the assessments. Once you answered every question of an assessment correctly, and you get access to the next part of the course. You can retry to pass an assessment as many times you want. When you passed all assessments, you have passed the course.

Important note: you can no longer finish a course once its end date is exceeded.

Assessment 1 - greetings

Show what you have learned

"Hello i am ..." is used to greet someone

Option 1
correct ☒

Option 2
false ☐

Goodbye is used to greet some one

Option 1
correct ☐

Option 2
false ☒

Save

5.1 What if I fail an assessment?

No worries. If you fail an assessment, you get feedback and can retry it. There are no more consequences.

5.2 What if I fail a course?

The only way to fail a course is to not complete all assessments in time. It is not possible to fail a course because you did fail one or more assessments too many times.

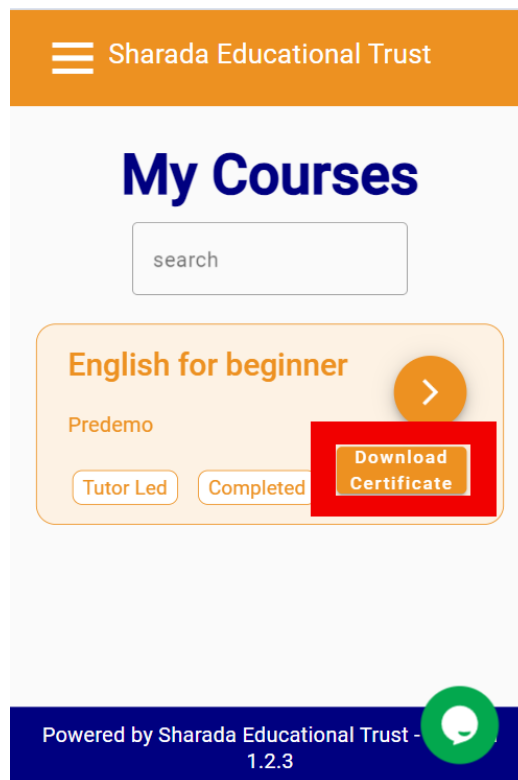
If you cannot finish all assessment in time, you can ask an administrator to take you on another batch. See how do I contact the administrator?

6 How do I get my certificate?

If you are using the app on the phone, you have to install a PDF reader before you can see your certificate. We recommend WPS Office (for Android click [here](#), for apple click [here](#)). If you have another PDF reader that is also fine.



Once you have passed a course you can find a button to get the certificate in the “My Course” overview.



7 How do I contact the administrator?

Should you have any questions or a problem you can't solve please contact an administrator. You can open a chat with an administrator via the green button at the bottom of the site. The administrator can answer questions immediately but if they are not available at the time they still get the message and will come back to you.



Sharada Academy

Phone number*
1234 5678 09

Password*

Login

Powered by Sharada Educational Trust
1.0.3



If you don't have started a chat yet you can start one via "New Conversation".



Hi there 🙋

Need help? Search our help center for answers or start a conversation:

Search for answers 

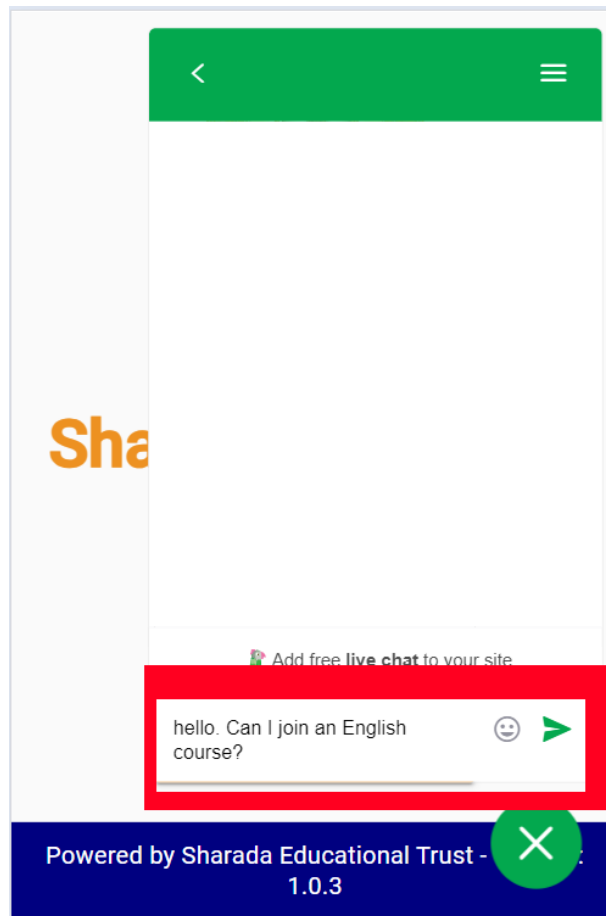
 New Conversation

 Add free live chat to your site

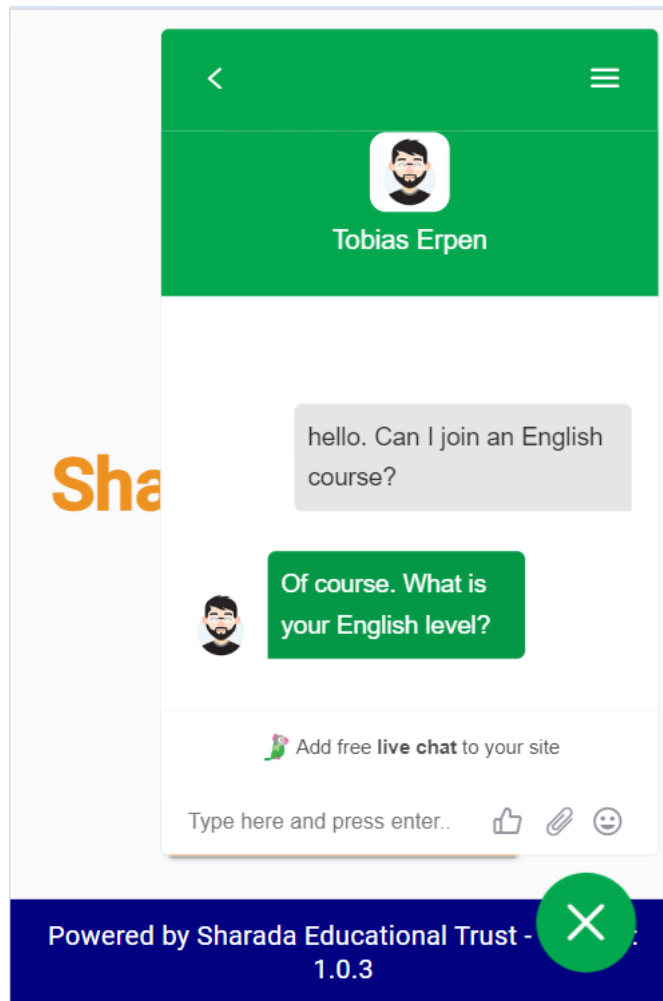
Powered by Sharada Educational Trust -
1.0.3



Just state your question or problem and send it via the arrow symbole.



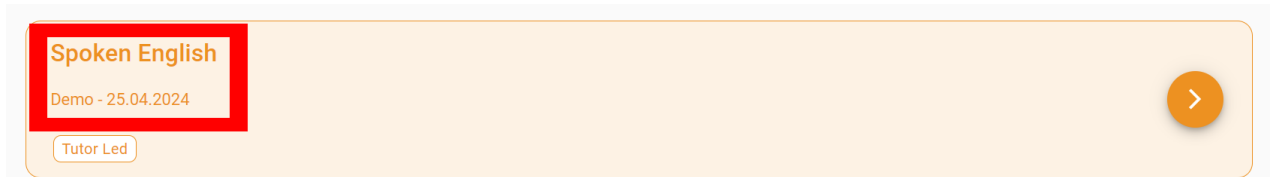
And the administrator will reach out to you as soon as possible.



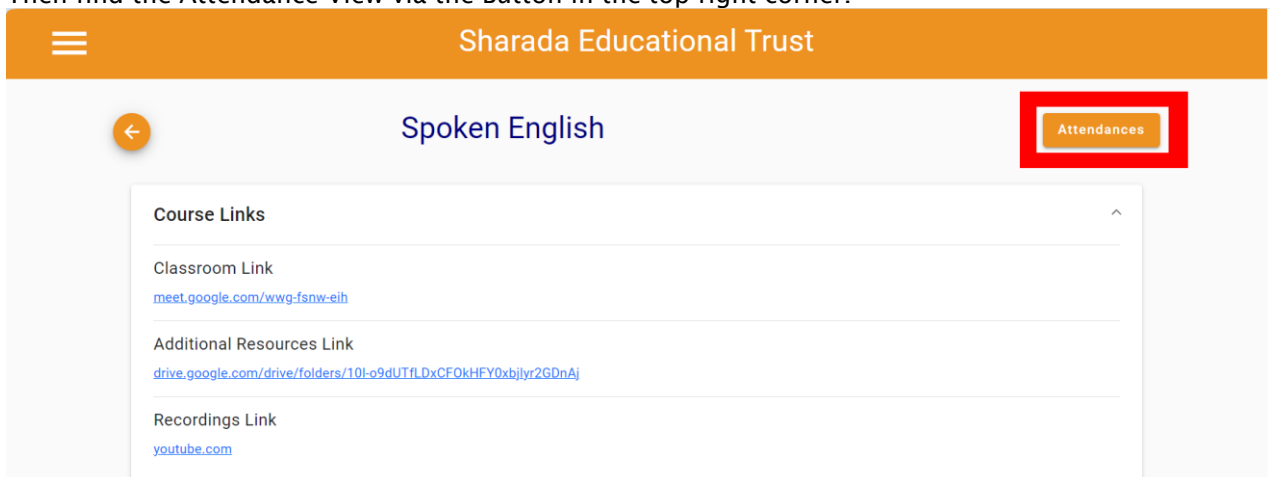
Tutor Functionalities

8 How do I keep track whether my students attended lessons?

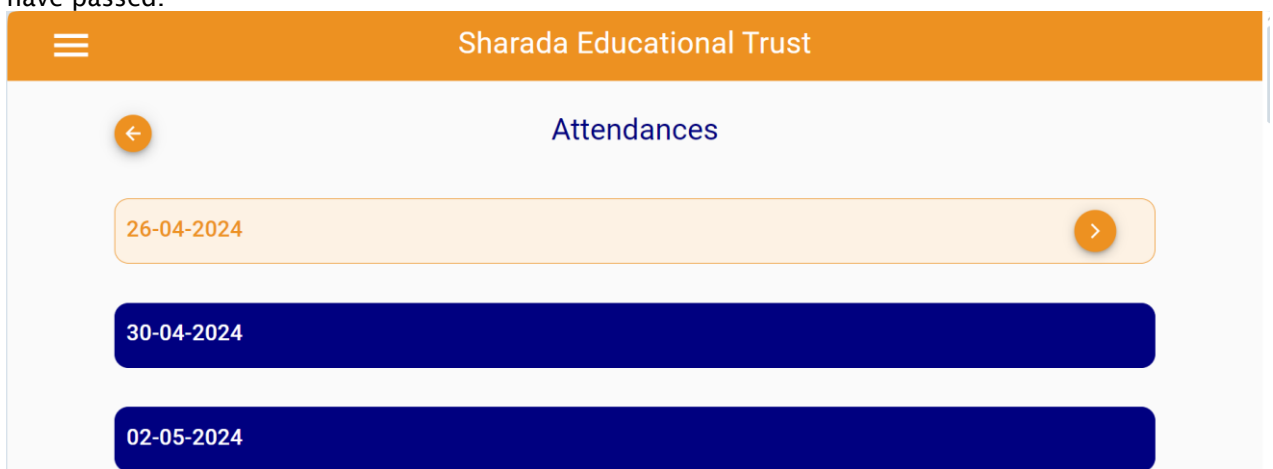
As a tutor you have the opportunity to track the attendance of your students. First open the course with the specific run through in question. You can see the the name of the course in the top of a Course tile and the batch right below. That is used to differentiate them if you teach the same course twice.



Then find the Attendance View via the Button in the top right corner.



You can see all the planed dates and for all the passed dates you can open them and check whether the students were there. The dates that are in the future are disabled and will be editable once they have passed.



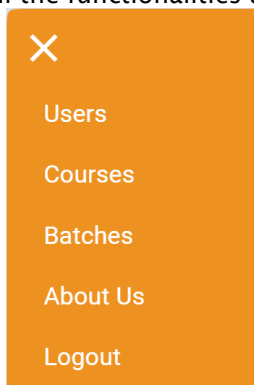
Administrator Functionalities

9 Where do I find the specific overviews?

In the top left corner of the page is an Icon that opens the menu.



When you open the menu, you can see all the functionalities and overviews you have access to.



If you cannot see the functionality, you are looking for ensure that you are logged in. If you still can't find it, you miss the necessary role. An admin can grant these roles. (see How do I contact the administrator?).

10 How do I setup a Course for my students from scratch?

To setup a Course for your students a few steps have to be followed:

1. You must add your students to the platform. (see How do I add my students to the platform?)
2. You have to setup a Course. This course will be reusable. (see How do I setup a Course?)
3. Create a new Batch that links the course to your students. (see How do I setup a Batch?)

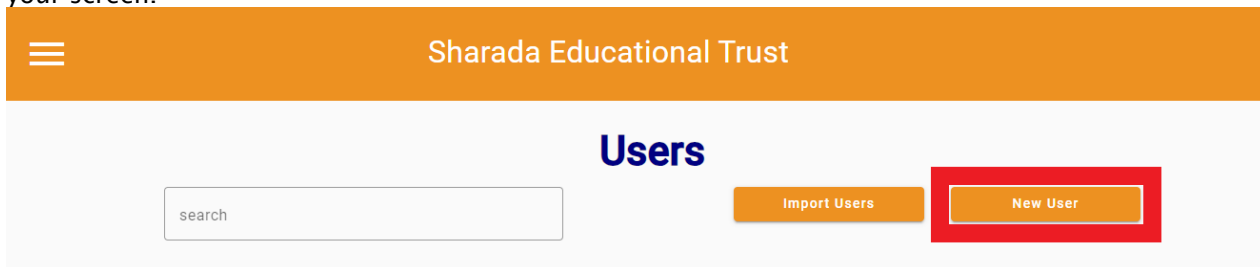
Thereafter and once the batch has started your students can find the course under "My Courses".

11 How do I add my students to the platform?

There is two ways to add students to the platform: one user at a time via the graphical interface (How do I add my students manually?) and a bulk of users via a list in a CSV file (see Is it possible to add a bunch of students at the same time?). Both are found in the User overview.

11.1 How do I add my students manually?

Navigate to the "User" overview. Then open the add view via the "New User" button on the right side of your screen.

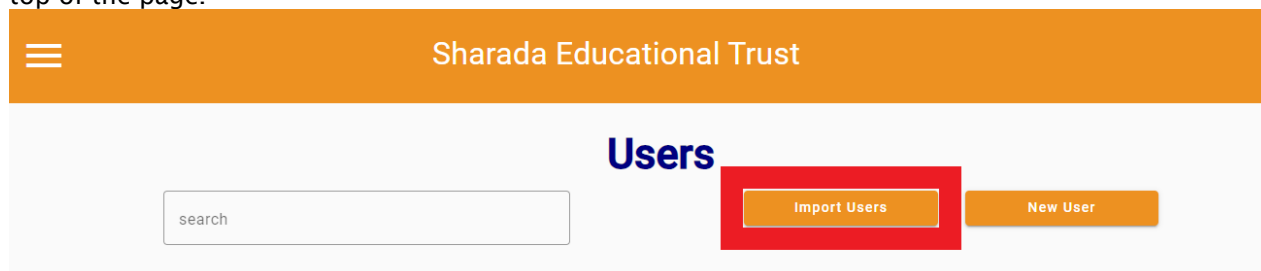


After a view should open where you have to enter the important information about the new user. The following table explains the different fields:

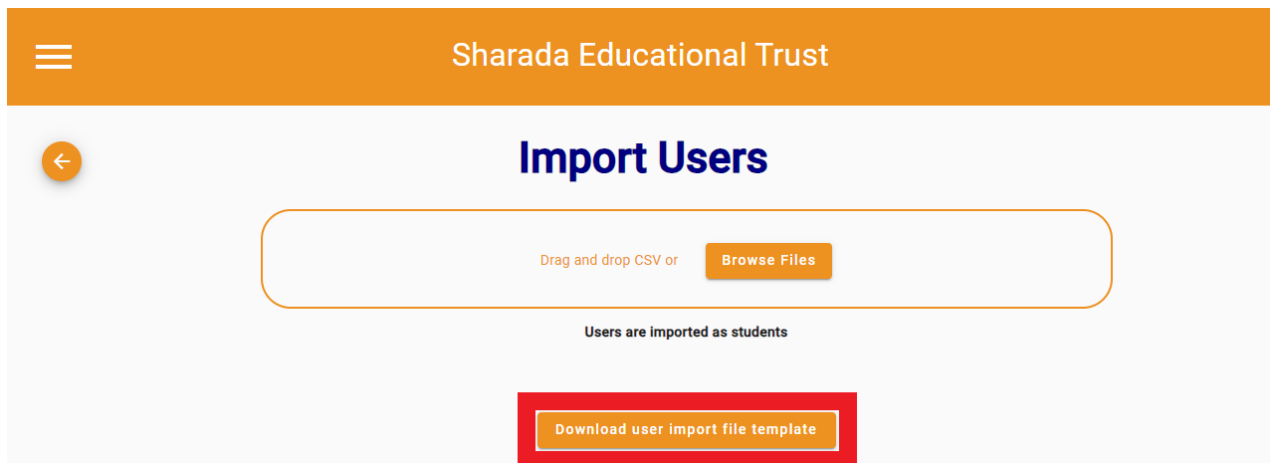
Field	Explanation
First Name*	Used to identify the user and for the certificates created.
Last Name*	Used to identify the user and for the certificates created.
Role	The roles can be folded open. A user can have multiple roles. They define what they can do on the system but also what they can see.
School/Collage Name	The school or college they attend. This is used to identify the user further.
e-mail	No two user can have the same e-mail address, but they can have no email address.
Mobile Number	The mobile number is used to login the user and must be unique across the system and it has to be 10 digits only. Meaning no two users can have the same number.
Password	the password for students must be a 4-digit code, for all others it must be a normal 8 character password.
Location	The location the user is in. This is used to identify the user further.
Join Date	The date the user was created. Automatically generated.
Last Login	Showes when the user last logged in. Automatically generated. If no date is present, they have not logged in yet.
Active/Inactive	If inactive the user can't login. This is used if the user is still needed but not at the given time.

11.2 Is it possible to add a bunch of students at the same time?

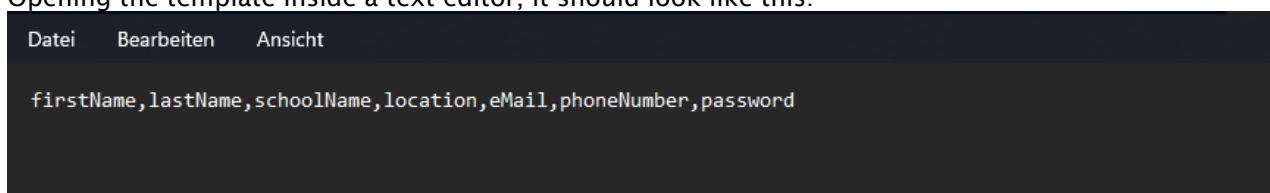
Yes, there is a way to register multiple users at the same time. However, this is only possible for students. To do this, navigate to the user overview page and click on the button "Import Users" at the top of the page.



To import multiple students at the same time, a CSV file is needed to be filled out. A template for that can be downloaded when clicking on the button at the lower end of the page.



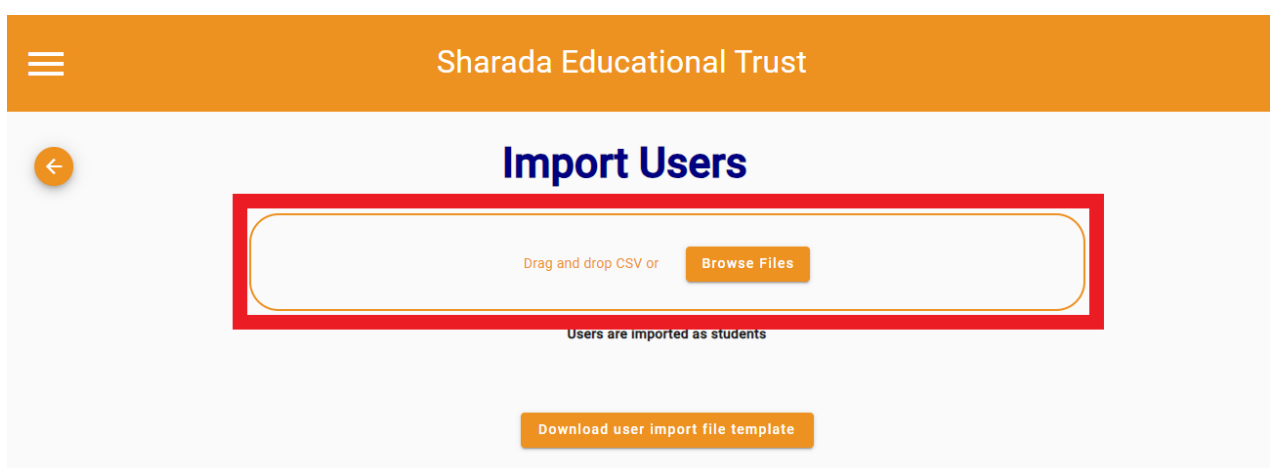
Opening the template inside a text editor, it should look like this:



Note: a CSV file can also be opened with Microsoft Excel but when saving, make sure, that you save it again as CSV before uploading to the application. As not all fields are required to create a new user, first name, last name, phone number and password are the only fields that must be filled out. An example could look like this:

```
firstName,lastName,schoolName,location,eMail,phoneNumber,password
max,musterman,,,,+62 00 000 11 00,1234
xam,namretsum,,,,+62000002200,1234
```

After filling out all the user information, you can either drag the file from the file explorer into the drop zone. Or click on "Browse Files" in order to upload the CSV.



When the creation completed successfully, you will be redirected to the user overview page, where you can check if the new users were created.

12What is the difference between a Course and a Batch?

A course contains the content for the students to learn and is reusable. A batch represents one run of a course containing the students, tutors as well as the relevant dates. That also means a batch is only used once: when its end date is reached it is over.

If you will a there can be many batches of one course, but a batch can only represent one course. And as soon as batch has started its corresponding course can no longer be edited/deleted to prohibit changes that would disrupt the integrity of the batches running and past.

12.1 Actions to take on Courses and Batches on the overview

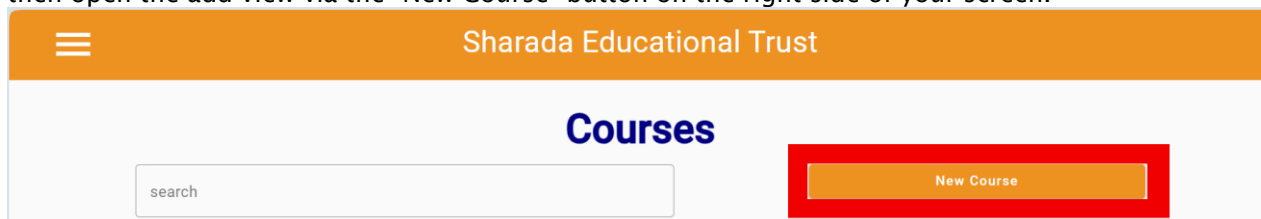
On a course or batch tile on the overview page, there are 3 icons which symbolises 3 different actions.



Icon	Name	Explanation
	Copy	This symbolises the action of duplicating a course or a batch. When clicked, you will be redirected to the creation page of the element with most of the information from the element, filled in the form. Then you are able to complete or adjust information and ultimately save it.
	Pencil	This symbolises the action of editing an existing element. When clicked, you will be redirected to the edit page of the given element. There you can update or add information to an existing element.
	Eye	This symbolises the action of getting detailed information on an element. This occurs on elements which do not have the possibility to get updates. Because the application architecture does not allow it.
	Garbage can	This symbolises the action of deleting an element. When it is blue, the action is no more possible, because the architecture does not allow it.

13 How do I setup a Course?

Navigate to the “User” overview. If you do not already have another course, you would like to copy then open the add view via the “New Course” button on the right side of your screen.



After a view should open where you have to enter the important information about the new course. The following table explains the different fields:

Field	Explanation
Course Name*	This is the name that will be displayed everywhere.
Course Type*	There are two types, which have an impact on the structural content of a course. “Self-Paced”: For a course the student has to do on their own and at their own speed. “Tutor Led”: For a course that is taught by a tutor with regular meeting.
Recordings Link	Only available for Self-Paces courses. This link if present will be added to the batches of that course. Is used to have only one link with all the recordings for the different batches of the course.
Active / Inactive	An active course is accessible for everyone who has some relation to it. An inactive course is not available for tutors or students. For them, the course is invisible and they can’t interact with. This may be needed when old or future courses are overflowing the overview.

Once this information is entered the course can be saved. Once saved Lessons and Attendances of the course can be added via the respective buttons. For more information see “What is important when creating a Lesson?” and “What is important when creating an Assessment?”. For more information on organizing the different Lessons and Assessments see “How do I organize the content of my course?”.

Note: For courses with no assessments no certificate will be generated.

Sharada Educational Trust

Edit Course

Course Name*

English for Beginner

Course Type*

Self Paced

Recordings link

https://www.youtube.com

Active

Inactive

Save

Cancel

Lessons

New Lesson

New Assessment

13.1 What is important when creating a Lesson?

The context for this question is Course lessons inside of the Sharada Academy tool. If you don't know how to create a lesson, see "How do I setup a Course?".

Sharada Educational Trust

Add New Lesson

Title*

Lesson 1

Lesson material link

https://sharada.com/material/lessons/1

Content*

Read the content in the lesson material.

Visible

Invisible

Save

Cancel

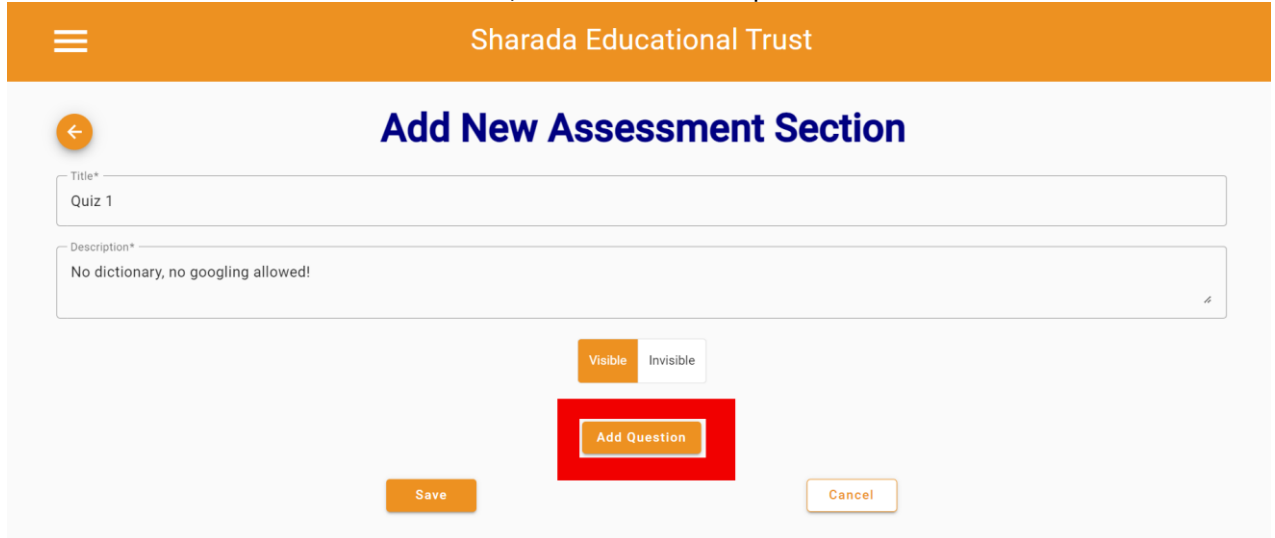
In a lesson there are the following fields that you can add information to:

Field	Explanation
Title*	This is the name that will be displayed to the students.
Lesson material link	This is a link to the lesson's material in another place i.e. google drive, a PDF, PPT or many more. This content will not be saved on in the Sharada Academy tool.
Content	The text/description of the lesson most often containing information and instructions.
Visible/Invisible	When a lesson is visible, students can see the lesson open it and work on it. When it is invisible, a student cannot see the lesson. This is handy for example, a tutor led course, where a course

manager and tutor wants to have complete control when students are allowed to enter the next lesson.
Note: If an assessment is set to “Invisible”, students cannot continue since the passing of an assessment is needed to precede.

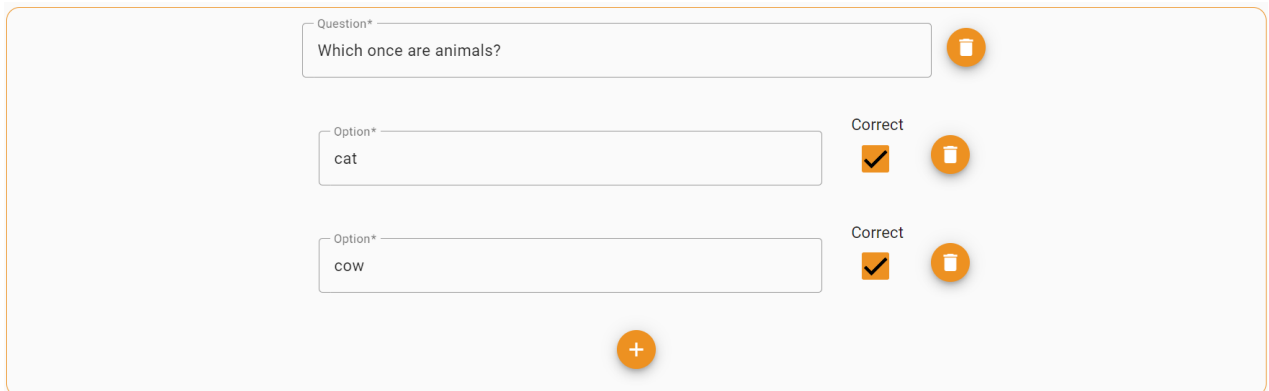
13.2 What is important when creating an Assessment?

The context for this question are Course assessments inside of the Sharada Academy tool. If you don't know how to create an assessment, see “How do I setup a Course?”.



Field	Explanation
Title*	This is the name that will be displayed to the students.
Description	The text/description of the assessments most often containing information and instructions.
Visible/Invisible	Indicates whether the Lesson is currently shown to the students.

There is some general information about the course and come all the questions. A new question can be added via the “Add Question” button. Then a new question is added.



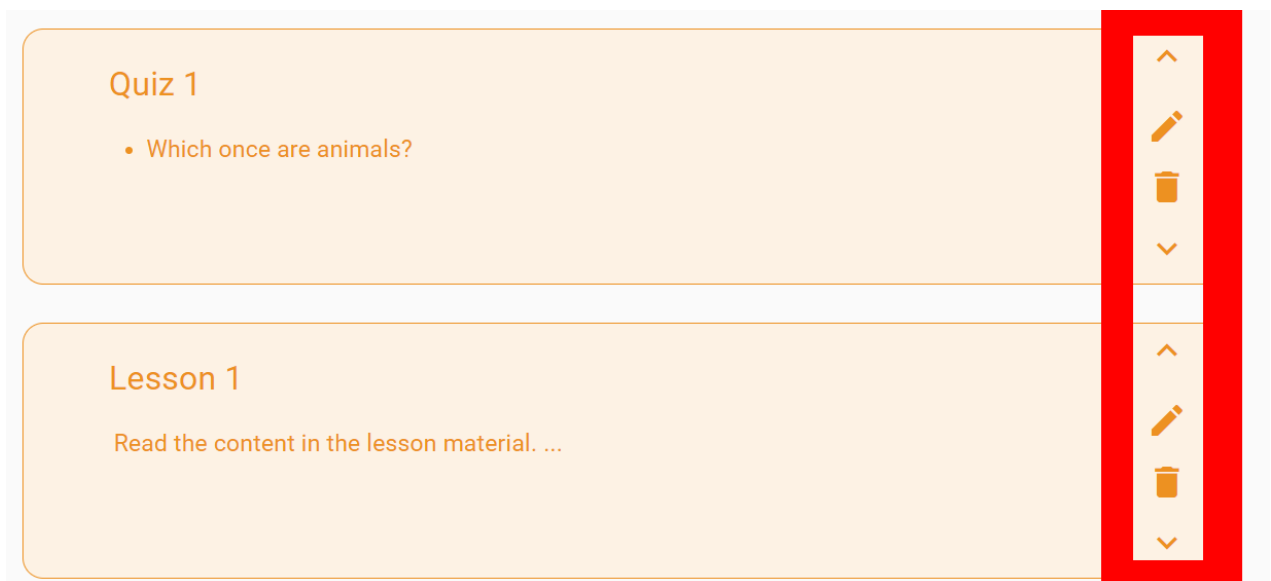
The following fields are important for a question:

Field	Explanation
Question*	The question text.

Option*	The text of the option. Via the "+" at the end a new option can be added.
Correct	Indicates whether the option is correct. A question can have multiple correct options. For a question to be answered correctly by a student all correct options must be selected and all incorrect options must not be selected.

13.3 How do I organize the content of my course?

If the order of the course changed or something changed, you can use the buttons to the right of a lesson or assessment.

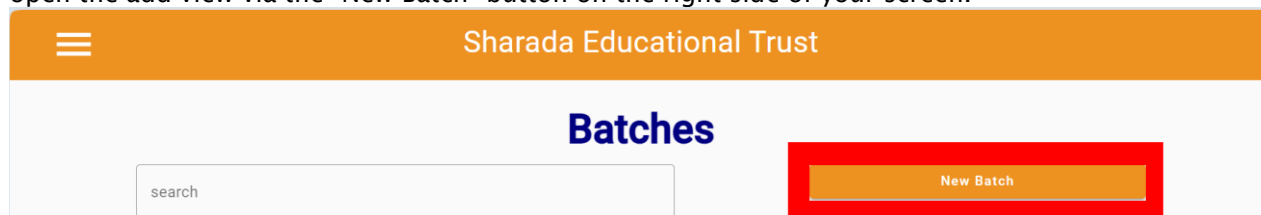


The screenshot displays a list of course items. The first item is 'Quiz 1' with a sub-item 'Which once are animals?'. The second item is 'Lesson 1' with a sub-item 'Read the content in the lesson material. ...'. To the right of each item is a vertical toolbar containing four icons: an upward arrow, a pencil, a trash can, and a downward arrow. A red rectangle highlights these toolbars for both items.

The arrows are used to change the order of the lessons and assessments. The pencil to edit and the trash to delete it.

14How do I setup a Batch?

Navigate to the “User” overview. If you do not already have another batch, you would like to copy then open the add view via the “New Batch” button on the right side of your screen.



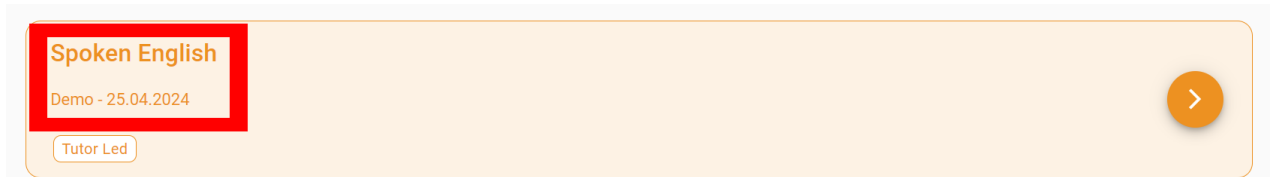
After a view should open where you have to enter the important information about the new batch. First you have to select the course and then the rest can be filled out according to the type of the course. The following table explains the different fields:

Field	Explanation
Course*	The content of the course will be displayed with the batch for the students.
Batch name*	Will be displayed and is used to identification the course.
Start Date	The date the batch starts. Is used schedule the dates in tutor lead courses. Also, students and tutors in the course will not see this batch in the “My Courses” view until the start date.
End Date	The date the batch ends. Is used schedule the dates in tutor lead courses. After the End Date has passed students will no longer be able to edit their assessments and complete the batch.
Recordings Link	A link displayed at the top of a course. This link is used to hold a reference to a Youtube playlist. This playlist is intended to hold all the recordings of past meetings. For Self-Paced Courses the link will automatically be added from the course if it has one but can be changed if need be.
Additional Resources Link	A link displayed at the top of a course. This link is used to hold a reference to a folder in a drive that is intended to hold additional material for students how are interested. Can be specific things that came up in a session and are not part of a course or a place to turn in some homework.
Classroom Link	A link displayed at the top of a course. This link is used to hold a reference to a continues meeting the students can regularly attend with there tutor. Only available for tutor lead courses.
Tutor	The tutor that will be leading the course.
Students	The students in the course.
Active / Inactive	It is the same with the courses. An active batch is accessible for everyone who has some relation to it. An inactive batch is not available for tutors or students. For them, the batch is invisible and they can’t interact with. This may be needed when old or future batches are overflowing the overview.

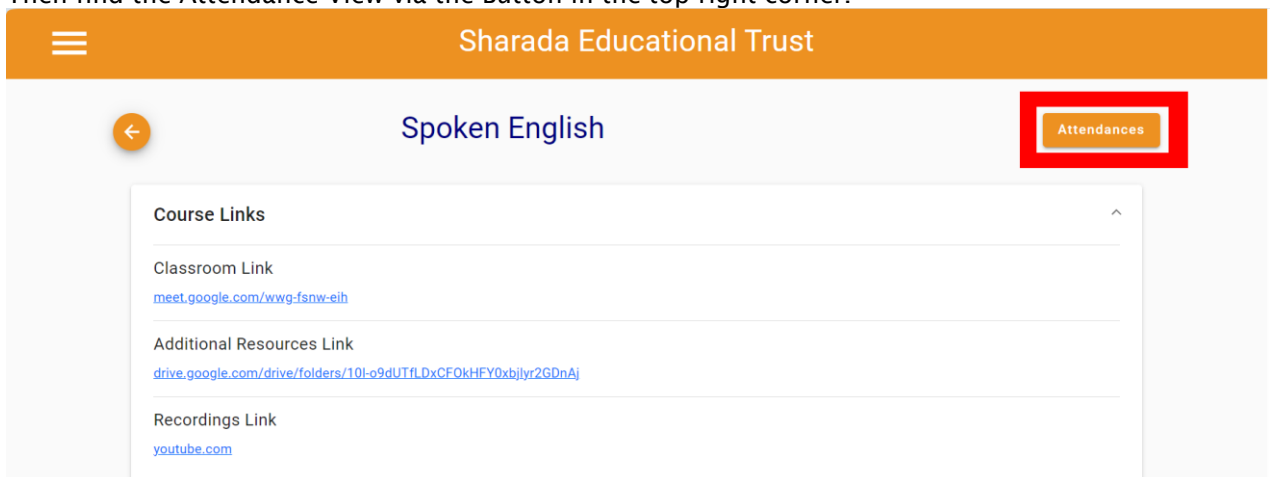
Tutor Functionalities

15 How do I keep track whether my students attended lessons?

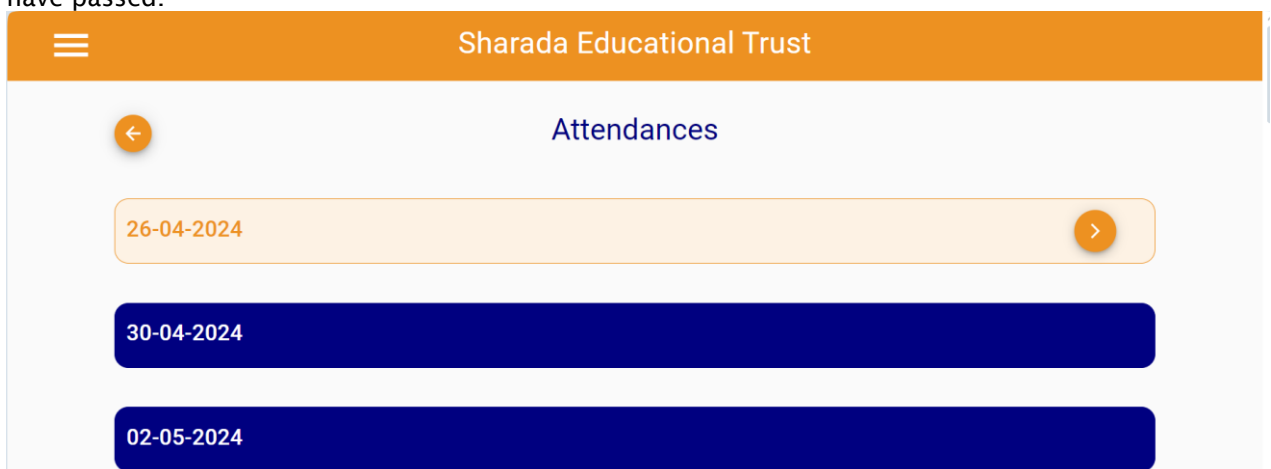
As a tutor you have the opportunity to track the attendance of your students. First open the course with the specific run through in question. You can see the the name of the course in the top of a Course tile and the batch right below. That is used to differentiate them if you teach the same course twice.



Then find the Attendance View via the Button in the top right corner.



You can see all the planed dates and for all the passed dates you can open them and check whether the students were there. The dates that are in the future are disabled and will be editable once they have passed.



Appendix H: Video Script



Script

Learning Management System for Rural India

Date:	31.05.2024
Version	1.0

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1 Introduction

This document contains the different scene descriptions for the video required for the bachelor thesis.

2 Requirements

The requirements for the video are given by the BFH and are summarized as follows:

- ▶ Use as good video and audio quality as possible
- ▶ 1 min to 5 min
- ▶ Video contains, situation, solution and result
- ▶ English
- ▶ Use the pre and post roll form BFH

Additional requirements/guidance to achieve the requirements by us:

- ▶ Use a voice over as much as possible to reduce background noise (which is prevalent in India)
- ▶ Only filmed by us and with the same phone, if possible, to ensure that the format stays the same.
- ▶ We target 3.5 min to have some room for unforeseen things.

3 Equipment

The following equipment is considered required and will be in use for most scenes. Therefore it won't be specifically mentioned.

- ▶ Phone Camera
- ▶ Laptop
- ▶ Cutting Program (Hitfilm)

4 Content

The following table shows the different main blocks (darker colours), given by the requirements, that make up the skeleton of the video. In lighter colors are the scene blocks, which contains all the scenes that contribute to the display of one concept or set of connected information (i.e. the information about Sharada).

Number	Block	Scenes	Estimated duration [sec]
1	Intro		30
1.1	BFH-Intro	BFH-Intro-Screen	5
1.2	Traven to India	At the BFH, At the airport, Traveling to Bangaluru	25
2	Situation		60
2.1	Contrast metropolitan and rural India	Metropolitan Bengaluru, Rural Village	30
2.2	Introduction Sharada Trust	Arvind's speech	30
3	Solution		50
3.1	Presentation of the App	Solving an assessment, Getting a Certificate	30
3.2	Display of a teaching session	Children being taught	20
4	Result		40
4.1	Result of the teachings	Expectations for the app, A student talking	40
5	End cut		30
5.1	Sharing a meal	Indian Meal	25
5.2	BFH-Outro	BFH-Outro-Screen	5
	<u>Sum</u>		<u>210</u>

5 Scenes

5.1 BFH-Intro-Screen

Aspect	Description
Place	Generated
Content	▶ Title and Participants
Scene Description	The image is shown on the whole screen for the duration
Camera position	-
Sound	No sound
Lighting	No lighting
Additional Equipment	-

5.2 At the BFH

Aspect	Description
Place	BFH Quellgasse
Content	▶ Students coding the app ▶ Getting ready to go to India
Scene Description	The students sit around in the BFH, coding. With the sentence “ So then we are ready. Let’s go and take our work to Bangaluru. ” they get up and leave.
Camera position	At about head hight, while sitting.
Sound	No sound
Lighting	As good natural light as possible
Additional Equipment	-

5.3 At the Airport

Aspect	Description
Place	Zürich Airport
Content	▶ The Team moving to the airplane
Scene Description	The students walking to the airplane ready to board.
Camera position	Normal head height, while standing
Sound	Sound will be muted, Explain the project and why we go to India.
Lighting	The best we can get.
Additional Equipment	-

5.4 Travel to Bangaluru

Aspect	Description
Place	Arrival at the hotel
Content	▶ The travel through the city
Scene Description	The taxi ride or the arrival at the hotel.
Camera position	From bottom to top.
Sound	Sound will be muted
Lighting	The best we can get.
Additional Equipment	-

5.5 Metropolitan Bangaluru

Aspect	Description
Place	The Info-Sys Campus
Content	▶ The IT-side of the India society
Scene Description	A longshot of the campus.
Camera position	Left to right
Sound	Will be muted, explain that the city and it is very much into English
Lighting	As good as we get
Additional Equipment	-

5.6 Rural Village

Aspect	Description
Place	The school we run some tests in
Content	▶ The rural side of india
Scene Description	<i>See as we are there</i>
Camera position	<i>See as we are there</i>
Sound	No sound, explain that the rural village often lack English
Lighting	As good as we get
Additional Equipment	-

5.7 Arvind's speech

Aspect	Description
Place	Not specific, preferably calm
Content	▶ Present the Sharada Trust
Scene Description	The image is shown on the whole screen for the duration.
Camera position	Frontal on the face
Sound	Arvind's voice, less background noise if possible
Lighting	With good daylight
Additional Equipment	-
Actions	-

5.8 Solving an Assessment

Aspect	Description
Place	Generated, Screen cast
Content	▶ Showing how to use the app.
Scene Description	The app shows the assessment and then a congratulation section.
Camera position	-
Sound	Off: explanation of the tool
Lighting	No lighting
Additional Equipment	-

5.9 Getting a Certificate

Aspect	Description
Place	Generated, screen cast
Content	▶ Show the result of students work
Scene Description	The app shows the image of the certificate
Camera position	-
Sound	Off: explanation what they can get from the app
Lighting	No lighting
Additional Equipment	-

5.10 Children being taught

Aspect	Description
Place	The school we run some tests in
Content	▶ Showing the processes in a school
Scene Description	<i>See as we are there</i>
Camera position	<i>See as we are there</i>
Sound	No sound; explanation in the off
Lighting	As good as we get
Additional Equipment	-

5.11 Expectations for the app

Aspect	Description
Place	Not specific, preferably calm (same as in 5.7)
Content	▶ Show the potential of the app
Scene Description	The image is shown on the whole screen for the duration
Camera position	-
Sound	No sound
Lighting	No lighting
Additional Equipment	-

5.12 A student taking

Aspect	Description
Place	Generated
Content	▶ Title and Participants
Scene Description	The image is shown on the whole screen for the duration.
Camera position	Frontal on the face
Sound	Arvind's voice, less background noise if possible
Lighting	With good daylight
Additional Equipment	-

5.13 Indian Meal

Aspect	Description
Place	The whole team eating together
Content	▶ Show the end of the project
Scene Description	A stable view of the eating, proseded with a tost.
Camera position	Stable from not too far away.
Sound	If possible, with original sound
Lighting	Dimmer lighting to simulate a dinner table
Additional Equipment	-

5.14 BFH-Outro-Screen

Aspect	Description
Place	Generated
Content	▶ Title, Participants and images
Scene Description	The image is shown on the whole screen for the duration
Camera position	-
Sound	No sound
Lighting	No lighting
Additional Equipment	-
Actions	-

6 Background voice Text

6.1 Scene 5.2 to 5.4

So, with a product **ready** for show, we set out to India, to **present** it to and **meet** our audience in person.

As we got there, we realized that we landed in a vastly **different world**.

6.2 Scene 5.5

We found ourselves in a **bustling metropolis**, that expands faster than we could grasp initially. Littered with opportunity and growth. But India has another side as well.

6.3 Scene 5.6

A big junk of rural India is **left behind** the fast-evolving cities. There infrastructure and education are not exactly getting worse but the gap to the richer cities **grows** with them. And there our partner "Sharada Educational Trust" comes in. But I'll let them introduce themselves.

6.4 Scene 5.8 and Scene 5.9

The SULABH app as it was named offers the opportunity to bring education to the rural parts of India via the teacher and girls that made it to the collage. It is easy to use, which is necessary given that a lot of the users rarely interact with computers and smart phones. It offers a way to bring education of any kind to the students and lets them complete assessments feel a sense of progress.

In the end it even provides a certificate to show off to a potential employer or just feel good about.

With that basis we look into the future with great confidence for the Sulabh app.

6.5 Scene 5.13

With that, our part in this journey as well as in this project ends and we give it to the Sharada Educational Trust to spread the education from here.

Appendix I: News Paper Article Transcript

Prajavani - Ramanagara edition - Thursday 16th May 2024:

An English learning app was launched on Wed 15th May 2024 at Government Higher Primary School-Cheeluru in Harohalli Taluk
Emphasis on quality education in Government Schools

At Government Higher Primary School Cheeluru, Harohalli taluk, Sharada Educational Trust along with Bern University of Applied Sciences of Switzerland, launched the Sulabh App, which will help the Government school children learn English.

Dignitaries present at the function, who launched the App: BEO – Ramappa; Sharada Educational Trust - Arvind Kamath; Bern University Professor – Ken

BEO Ramappa said - “English is necessary for everyone in their day-today life and it is essential to make it easy for children to learn. Government schools are also providing good education compared to private schools. We should appreciate the Foreign University for taking the initiative to develop the app, which will help children to learn English. The school children should utilize this app.”

Other dignitaries present during this launch are: SDMC - Muniraju; CRP - Anand and Shivananjappa School faculty and students were also present at the launch event.

Vijay Karnataka - Ramanagara edition - Thursday 16th May 2024:

At Harohalli- BEO Ramappa said “Government is taking many steps and planning programs to improve quality of education in Government schools.”

On Wednesday in Harohalli taluk Govt Higher Primary School - Cheeluru, Sharada Educational Trust along with Bern University - Switzerland launched the Sulabh App, which will help rural school children learn English effectively.

These days people don't turn to Government schools thinking the quality of Government schools are not up to mark as private schools. However, now the situation is improving and Government schools are much better than Private schools.

In Government schools, quality education is provided which helps in building good society and at the same time basic necessities like toilets, drinking water, good building infrastructure are also provided in Government schools. And children are getting involved in sports too.

Dignitaries presented at the launch event - Arvind Kamath of Sharada Educational Trust; Ken from Bern University, Muniraju - President of SDMC, Anand - CRP and Shivananjappa. School children and faculty members were present during the event.

Appendix J: Contract

Agreement

between

Bern University of Applied Sciences, Engineering and Information Technology (BH+IT)
represented by
Kenneth Ritly
Hoheweg 80, 2502 Biel/Bienne, Switzerland

hereinafter referred to as 'BFH'

and

Sharada Educational Trust
Sowmya Springs, Orwan Madhava Rao Road, Basavanagudi, Bengaluru, Karnataka, India

hereinafter referred to as 'the trust'

and

Tobias Erpen
Hafenweg 8, 3400 Burgdorf, Switzerland

and

Alayne Hiltmann
Sonneggstrasse 12, 4600 Olten, Switzerland

and

Lukas Vogel
Obere Murggen 123, 3617 Fährni, Switzerland

Tobias Erpen, Alayne Hiltmann, Lukas Vogel hereinafter referred to as 'the students'

hereinafter jointly referred to as 'the parties'

regarding student project thesis in the field of
Learning Management System for Rural India

1. Preamble

BFH offers companies the opportunity to have issues of interest investigated by students within the framework of a project thesis or a Bachelor thesis.
The current agreement regulates the rights and obligations of the parties within the framework of this type of project thesis or Bachelor thesis.

2. Subject Matter of the Agreement

The Students will conduct a Bachelor thesis in the field of Learning Management System for Rural India at BFH, in collaboration with the Trust, according to the project description outlined in Appendix A (hereinafter referred to as "the Project"). The Project will involve the development of software and related artifacts, including software documentation and end-user documentation, tailored to the Trust's specified needs within its field of business.

The Trust contributes its expertise by defining current issues for investigation and actively engaging in the iterative refinement of the Project's features and the management of software defects.

BFH provides the infrastructure and specialist supervision required for the Project, ensuring an environment that supports the academic and developmental objectives of the thesis work.

3. Project leaders

BFH: Kenneth Ridley, Hüheweg 80, 2502 Biel/Bienne, Switzerland, +41 31 848 54 72, kenneth.ridley@bfh.ch
Trust: Arvind Kamath, Sowmya Springs, Dewan Madhava Rau Road, Basavanagudi, Bengaluru, Karnataka, India, +91 9019911604, arvind@sharadatrust.org
Student: Tobias Erpen, Hüheweg 8, 5400 Burgdorf, Switzerland, +41 79 134 95 53, tobias.erpen@student.bfh.ch

4. Project Results and Intellectual Property Rights

4.1 The students retain all rights, title, and interest in and to the source code and related artifacts of the software developed as part of the Sherada-Academy-LMS ("Software"), excluding end-user documentation. Also excluded are any intellectual property of the trust or BFH, such as logos, graphics, and fonts.

4.2 The students are entitled to use the Software and related artifacts for commercial purposes under the terms set forth in this Agreement.

5. License to Modify and Use

5.1 The Trust and BFH are each granted a non-exclusive, royalty-free license to modify, use, exercise, and display the source code and related artifacts of the Software for non-commercial purposes.

5.2 Modifications must be for internal, operational, and non-commercial purposes only and may not be sold, sublicensed, or otherwise transferred to any third parties.

5.3 Both the Trust and BFH agree to maintain any proprietary notices on all copies of the source code and related artifacts, including any modifications thereof.

6. Guarantee and Liability

6.1 The parties use the greatest possible care and the current state of the art for the purpose of carrying out the project. They endeavour to achieve the intended project outcomes.

6.2 Research and teaching imply the acceptance of unpredictable results. As a result, the parties assume no guarantee for achievement of the intended project results. In particular, the parties do not guarantee that the project results will be usable and error-free or will be able to be exploited commercially or on an industrial scale.

6.3 The parties assume no liability for slight breaches of duty of care on the part of their project members (own employees, graduates, students, etc.). They also refuse to accept any liability for indirect damage.

6.4 The parties endeavour not to breach any third-party rights as a result of applying or exploiting the project results. However, they refuse to accept any liability for damage caused as the result of breaching third-party rights.

6.5 If a device or machine is developed as part of the student's work, it is a prototype for which BFH excludes any warranty and liability to the extent permitted by law. By accepting the prototype and using it in any way, the recipient assumes full responsibility for its use and all resulting consequences, including consequential damage (such as indirect damage, bodily injury, loss of use and profit).

7. Scientific Publications

The results of the project can be published in compliance with the confidentiality agreement in accordance with clause 9.

The parties obtain mutual approval prior to publication. Consultation should for example ensure that the Bachelor thesis, other publications or possible protection of intellectual property is not impaired in any way. It is not permissible to refuse consent without giving an appropriate reason. As a matter of principle, the results remain confidential until this point in time.

8. Plagiarism detection

The BFH uses plagiarism detection software (e.g. Turnitin) to check student work. Student work is uploaded to the corresponding database, where it is checked for matches with other deposited texts. BFH reserves the right to upload student work to a plagiarism detection database. The disclosure of work or parts thereof is only made individually to requesting bodies.

9. Confidential Information

The parties mutually undertake to keep confidential any confidential information and materials ("Confidential Information") provided to them by the other parties during the term of the contract, even if such information has not been expressly designated as secret or confidential. The duty of confidentiality shall apply regardless of whether Confidential Information is provided in writing, orally, electronically or in the form of equipment, samples, specimens or products. The duty of confidentiality shall expire two (2) years after termination of this Agreement.

The parties agree not to use the Confidential Information for any purpose other than the student work contemplated in this Agreement.

The confidentiality obligation does not apply to information and materials that:

- was already known to the public or third parties before it was disclosed by either party, or which becomes publicly available during the term of the confidentiality obligation agreed herein without misconduct by either party;
- have been made known to a party by a third party with power of disposal
- were already in the possession of the Receiving Party prior to the Collaboration or were developed without the use of the Confidential Information or Materials, provided that this can be proven in writing.

The legal and contractual provisions on data protection must be complied with.

The Receiving Party shall take all necessary precautions, both technical and organisational, to keep Confidential Information and materials safe and secret. The Recipient shall impose on all persons to whom it provides access to the Confidential Information and Materials for the purpose of fulfilling its contractual obligations the duty of secrecy and confidentiality in accordance with this Confidentiality Agreement.

If the Recipient becomes aware that it is required to disclose Confidential Information or Materials by law, court order or official process, it shall promptly notify the Donor in writing.

The Recipient shall promptly notify the Donor in writing if it discovers that the Confidential Information or Materials have been used or disclosed without permission.

When using online tools, the recipient is prohibited from entering confidential information.

10. Term and Right of Termination

- 10.1 This Agreement shall commence on the date of the last signature below and continue in effect until terminates with the expected completion date of the project on 13.05.2024.
- 10.2 The provisions of clauses 4-8 and 11 shall continue to apply after termination of the agreement. The duty of confidentiality shall continue to apply even if the main contract is not concluded or is terminated prematurely.

10.3 Prior to the expiration of the term, the agreement can only be cancelled in the case of serious breaches of the agreement. Before cancellation takes place, the party wishing to cancel must in any event give the other party the opportunity to comment and to make amends within 30 days.

10.4 Upon termination, all rights and licenses granted under this Agreement shall revert to the respective Parties, subject to any rights and obligations that have accrued prior to termination.

11. Other Arrangements

11.1 Modifications and Amendments to the Agreement

The current agreement regulates the contractual relationship between the contractual parties solely in relation to the project. No ancillary agreements in this regard exist and any earlier agreements in this regard end herewith. Modifications and amendments to this agreement must be made in written form and require the legal signatures of all parties to the agreement.

11.2 Partial Invalidity

Should individual provisions of this agreement be inoperative or fulfilment becomes impossible, the validity of the remaining provisions will not be affected. In this event, the contractual parties undertake to replace the inoperative provision by a permissible operative agreement which comes closest to the original purpose in terms of content. The same applies in the event of a contractual gap.

11.3 Announcement of Collaboration

The contractual parties agree that general information regarding the type of collaboration (collaboration territory, identity of the contractual parties) can be made public.

11.4 Rights

As a result of this contractual relationship, the contractual parties acquire no rights from the other party except those which are expressly granted in this agreement.

11.5 Transfer to Third Parties

Rights and obligations from this agreement may not be transferred to third parties without the written permission of the contractual parties. Permission must not be refused inappropriately.

11.6 Support

The contractual parties are to promptly provide the necessary mutual support within reasonable limits to enable the contractual parties to exercise the rights they have acquired as a result of this agreement. In particular, they provide the necessary signatures for the acquisition of rights to the intellectual property.

11.7 Applicable Law and Place of Jurisdiction

This contract is governed by Swiss law (excluding conflict of laws and the Vienna Sales Convention), with the exception of those aspects of intellectual property law that are subject to mandatory foreign law. The exclusive place of jurisdiction is Bern. The parties shall endeavour to settle any disputes arising from the performance of a contract amicably in the first instance.

11.8 Appendices

The appendices form an integral component of this agreement.

BFH


Kenneth Ritley, Professor

Date: 18.05.2024

Trust


Arvind Kamath, Managing Trustee Sharada Educational Trust

Date: 18/05/2024

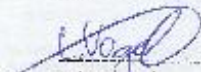
Students


Tobias Erpen

Date: 18.05.2024


Mayne Hiltmann

Date: 18.05.2024


Lukas Vogel

Date: 18.05.2024

12. Appendix A: Project description



Berner Fachhochschule
Hochschule für angewandte Wissenschaften
Berner Hochschule für Applied Sciences

Bachelor Thesis

to: Lukas Vogel
Alexander Fritsch-Hermann
Tobias Eppen

Course: Computer science

Supervisor: Rainer Hölz

Learning Management System for Rural India

BACKGROUND

Shiksha Educational Trust is a non-profit organization dedicated to improving the lives of underprivileged and rural children and women through education. Their focus areas include expanding literacy, teaching Spoken English, conducting self-paced training programs, supporting school infrastructure among others.

They intend to use modern tools such as Moodle, or their equivalents. But they lack something like a LMS or a LMS (Learning Management System) or Moodle to continue this work.

PREVIOUS WORK

In the Project 8 course a "universally usable product" was released into production that included only the basic features:

- User management (admins, teachers, admin) including their security
- Course management (setting up course curriculum and uploading course content)
- E-Classroom management (assigning students to classes)

PROPOSAL

For the Bachelor Thesis the goal is to implement all the significant features that will provide real business value, including:

- Attendance management (for students and teachers)
- Assessments (quizzes and homework)
- Generating certificates for both students and teachers
- And explore other innovative features, that will stimulate learning and reward good learning behavior

J. Edgar